

Connecting Score Matching, Maximum Likelihood, and Expectation-Maximization in Mixed Linear Regression

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Abstract

We study variance-preserving diffusion of the response in mixed linear regression (MLR) with unknown mixing weights. Our analysis separates the statistical guarantees of score matching from the loss geometry and optimization signal at a fixed diffusion noise level. The KL divergence links the denoising score matching objective integrated over the diffusion path with the likelihood and a terminal discrepancy. Under mild regularity conditions and terminal schedule, the resulting estimator converges up to the ground truth parameters of MLR, and its scaled error converges to the Gaussian limit of the maximum-likelihood estimator. At a fixed scale of the diffusion noise level, we derive a decomposition linking the score matching loss to cross-entropy and Expectation-Maximization (EM) operators. This decomposition yields an EM-related low-noise gradient expansion with additional correction terms of latent variance. In the high-noise limit, we further characterize gradient descent on this limiting loss under isotropic covariance. Along fixed high signal-to-noise ratio rays, the score matching imbalance gradient and the latent-variance term tend to zero pointwise. Numerical experiments illustrate our theoretical findings and statistical guarantees.

1 Introduction

For a response law with several modes, its mean can fall in a region of low probability and fail to describe a plausible outcome (Bishop, 1994; Han et al., 2022). Diffusion models instead learn scores for Gaussian-smoothed versions of the response law (Ho et al., 2020; Song et al., 2021b). Conditional variants retain the covariates while adding noise to the response. Batzolis et al. (2021) justified conditional denoising score estimation, while Fu et al. (2024) and Tang et al. (2025) established conditional-score and distribution-estimation guarantees under their respective assumptions. Building on this distributional perspective, we study parameter recovery in a structured latent model, its connection to latent-variable optimization, and the variation of parameter information and gradient signal with diffusion noise level and signal strength. We study these questions in mixed linear regression (MLR), a classical latent model (Quandt, 1972; De Veaux, 1989; Zilber & Nadler, 2023). Under Gaussian diffusion of the response, the conditional score and the posterior of the latent sign remain explicit. We denote the covariate by $\mathbf{x} \in \mathbb{R}^d$ and the response by $y_0 \in \mathbb{R}$ in the MLR model. The ground truth regression parameters θ^* represent the separation between components, ν^* characterizes the imbalance of the ground truth mixing weights $\pi^* = (\pi^*(1), \pi^*(2)) := ((1 + \tanh \nu^*)/2, (1 - \tanh \nu^*)/2)$, and the conditional law for the response is

$$y_0 \mid \mathbf{x} \sim \pi^*(1) \mathcal{N}(\langle \theta^*, \mathbf{x} \rangle, 1) + \pi^*(2) \mathcal{N}(-\langle \theta^*, \mathbf{x} \rangle, 1). \quad (1)$$

Questions: For this MLR model, under what conditions does the path-integrated score matching estimator recover the regression parameters and mixing weights?

How is the score matching loss related to Expectation-Maximization, and how do the noise level and signal strength affect the optimization dynamics and parameter information retained by score matching?

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Score matching and mixture learning. Score matching was introduced for unnormalized models, connected to maximum likelihood under perturbation, and linked to denoising objectives (Hyvärinen, 2005; Lyu, 2009; Vincent, 2011). Analyses based on variational methods and maximum likelihood connected diffusion score objectives to likelihood under their respective reverse models and weightings (Huang et al., 2021; Song et al., 2021a). Results on learning densities and distributions with scores further related an integrated score objective to density and parameter estimation (Li & Yan, 2024; Chewi et al., 2025a), while results for Gaussian mixtures provided end-to-end distribution-learning guarantees for diffusion and score methods (Chen et al., 2025; Gatmiry et al., 2025). Shah et al. (2023) analyzed balanced spherical Gaussian mixtures. Their large-noise proof used a spectral iteration, while their small-noise proof compared a gradient step with the M-step of Expectation-Maximization (EM). Our conditional Gaussian components have centers that move with $\mathbf{x}$, and both their regression parameters and mixing weights are unknown. Prior work studied MLR through identifiability, spectral and moment methods, convex formulations, alternating minimization, approximate message passing, and EM (Hennig, 2000; Chaganty & Liang, 2013; Zhong et al., 2016; Yi et al., 2014; Chen et al., 2018; Tan & Venkataramanan, 2023; Balakrishnan et al., 2017; Klusowski et al., 2019; Kwon et al., 2021; 2024). Zilber & Nadler (2023) studied estimation of unknown imbalanced MLR, and Luo & Hashemi (2025; 2026) characterized the coupled EM operators for the regression parameters and the imbalance of mixing weights. Ozkara et al. (2026) used a related conditional diffusion method to estimate client-specific weights in a two-component Gaussian mixture at one noise level with fixed component means. Our setting estimates both the covariate-dependent centers and the imbalance of mixing weights.

Score blindness and the role of noise. At a high diffusion noise level, component separation can remove the information about the mixing weights from the score as established by Wenliang & Kanagawa (2020) in one-dimensional Gaussian examples with two mixtures where the Fisher divergence between them vanished as the separation grew. Zhang et al. (2022) proposed a mixture Fisher divergence to mitigate this failure, while Koehler et al. (2023) related score matching efficiency to isoperimetry. The corresponding behavior of population EM in Luo & Hashemi (2024) showed that the EM update for the mixing weights was independent of the incoming imbalance of mixing weights in the noiseless setting but remained informative about the ground-truth mixing imbalance through the alignment of the current and ground-truth regression parameters. For finite Gaussian mixtures with known mixture weights and a known shared covariance, Qin & Risteski (2024) proved a polynomial asymptotic bound on the statistical complexity of fitting the component means with a generalized score matching objective that used continuous tempering and was related to annealed score matching. For mixing weights of a two-Gaussian family with known component parameters, Lütke Schwienhorst et al. (2026) showed that suitable horizon tuning removed separation dependence from their finite-sample error bound, while Dennehy et al. (2026) showed that mode-overlapping intermediate scales of the diffusion noise level could retain information about the mixing weights.

1.1 Technical Overview and Contributions

The technical development combines the relative-entropy and likelihood bridges by following the de Bruijn identity (Stam, 1959), Expectation-Maximization update (Dempster et al., 1977) with the coupled EM operators (Weinberger & Bresler, 2022; Luo & Hashemi, 2025), and standard M-estimation arguments (van der Vaart, 1998; Chewi et al., 2025b; Lütke Schwienhorst et al., 2026). The main contributions are as follows.

- **Path-Integrated Score Matching: Maximum-Likelihood Asymptotics** (Section 3). Lemmas A.15 and A.16 establish population and empirical likelihood bridges for which terminal discrepancies form the only parameter-dependent gap between the path-integrated objective and likelihood. Proposition 3.1 shows that the terminal population KL divergence preserves identifiability, and Main Theorem 3.2 proves that controlling the terminal discrepancy transfers consistency and asymptotic normality with inverse-Fisher covariance from maximum likelihood to the path-integrated Score Matching estimator.
- **Fixed-Scale Score Matching: Cross-Entropy and EM Decompositions** (Section 4). Propositions 4.1 and 4.2 identify the diffused MLR Expectation-Maximization (EM) operators and express the cross-entropy gradients as EM residuals, while Main Theorem 4.3 shows that the fixed-scale loss

admits equivalent decompositions of cross-entropy and EM operators. Corollaries 4.4 and 4.5, together with Proposition 4.6, translate these decompositions into exact gradient identities, showing that the score matching gradient field involves EM operators, operator derivatives, and latent variance terms. Equation (15) and Propositions 4.7 and 4.8 then characterize the two endpoint regimes: the low-noise gradient limit retains this EM-related correction structure, whereas the high-noise limiting objective depends on the candidate parameters and the ground truth parameters.

- **High-Noise Gradient Dynamics and Score Matching Blindness** (Section 5). Main Theorem 5.1 gives an $\mathcal{O}(K^{-2})$ loss bound when the effective target is zero, where K denotes the number of gradient-descent iterations, and an $\mathcal{O}(\log(1/\epsilon))$ bound for generic initializations when the effective target is nonzero, with only a measure-zero exceptional set attracted to a saddle. Proposition 5.2 formalizes pointwise blindness to the mixing imbalance: as the ground-truth signal diverges along a fixed ray, the gradient in the imbalance parameter vanishes at fixed scale of the diffusion noise level.

Organization. In Section 2, we formulate the MLR model, score matching objectives, and associated EM operators. In Section 3, we show how a likelihood bridge transfers maximum-likelihood asymptotics to the path-integrated Score Matching estimator. In Section 4, we derive exact cross-entropy and EM decompositions and use them to characterize the fixed-scale loss and gradients across the low- and high-noise regimes. In Section 5, we analyze high-noise gradient dynamics and pointwise score matching blindness. In Section 6, we consolidate the corresponding numerical implications. In Appendix A, we develop the auxiliary identities and bounds used in the main proofs for results of Sections 3–5 in Appendices B–D. In Appendix E, we record the experimental details for the numerical results to verify the theoretical findings of Sections 3–5.

2 Problem Setup

In this section, we specify the mixed linear regression (MLR) model and its variance-preserving diffusion of the response. We then define the population and empirical score matching objectives. We also introduce the cross-entropy and EM operators and the notations required for the estimation and asymptotic analysis. See the detailed derivations of the equations below in Appendix A.

2.1 Mixed Linear Regression and Variance-Preserving Diffusion

In the two-component mixed linear regression (MLR) model, the response y_0 is generated from the regression parameter θ and additive noise ε_0 according to

$$y_0 = (-1)^{z+1} \langle \theta, \mathbf{x} \rangle + \varepsilon_0, \quad (2)$$

where $\mathbb{E}[\mathbf{x}\mathbf{x}^\top] = \Sigma \succ 0$ and $\varepsilon_0 \sim \mathcal{N}(0, 1)$. The latent variable $z \in \{1, 2\}$ has probabilities $\mathbb{P}(z = 1 \mid \pi) = \pi(1)$ and $\mathbb{P}(z = 2 \mid \pi) = \pi(2)$. The mixing weights $\pi = (\pi(1), \pi(2))$ are positive and sum to one. We impose the independence assumptions $(z; \pi) \perp\!\!\!\perp \theta$, $\mathbf{x} \perp\!\!\!\perp (z; \theta, \pi)$, and $\varepsilon_0 \perp\!\!\!\perp (\mathbf{x}, z; \theta, \pi)$. The half log-odds parameter $\nu := (\ln \pi(1) - \ln \pi(2))/2$ characterizes the mixing imbalance and satisfies $\tanh \nu = \pi(1) - \pi(2)$, $\pi(1) = (1 + \tanh \nu)/2$, and $\pi(2) = (1 - \tanh \nu)/2$. The variance-preserving diffusion process satisfies the stochastic differential equation (SDE)

$$dy_t = -\frac{\beta_t}{2} y_t dt + \sqrt{\beta_t} dB_t, \quad (3)$$

where $\beta_t > 0$ is the noise schedule and B_t is a standard Brownian motion satisfying $\mathbb{E}[B_t] = 0$ and $\mathbb{E}[B_t B_s] = \min(t, s)$. Noise schedules used or analyzed in the literature include the constant choice $\beta_t = 2$ (Chen et al., 2023), the linear choice $\beta_t = a + bt$ (Ho et al., 2020), and the power-law family $\beta_t = (a + bt)^\rho$ with $\rho \geq 1$ (Gao et al., 2025; Gao & Zhu, 2025). The retained-signal factor $\bar{\alpha}_t := \exp\left(-\int_0^t \beta_\tau d\tau\right)$ encodes cumulative attenuation along the diffusion path, and its square root $\sqrt{\bar{\alpha}_t}$ controls the decay of the signal from y_0 . The corresponding forward transition density is $p(y_t|y_0) = \mathcal{N}(y_t; y_0\sqrt{\bar{\alpha}_t}, 1 - \bar{\alpha}_t)$, so the response variable y_t at time t admits the representation

$$y_t = y_0\sqrt{\bar{\alpha}_t} + \xi_t\sqrt{1 - \bar{\alpha}_t} = (-1)^{z+1} \langle \theta_t, \mathbf{x} \rangle + \varepsilon_t, \quad (4)$$

where $\xi_t \sim \mathcal{N}(0, 1)$ and $\varepsilon_t \sim \mathcal{N}(0, 1)$ are additive Gaussian noises independent of $\mathbf{x}$ and z , while $\theta_t := \theta\sqrt{\bar{\alpha}_t}$ and $\mu_t := \langle \theta_t, \mathbf{x} \rangle$ provide concise notation for the scaled parameters at time t . At $t = 0$, these definitions give $\theta_0 \equiv \theta$ and $\mu_0 \equiv \langle \theta, \mathbf{x} \rangle$. These identities imply $\mu_t = \mu_0\sqrt{\bar{\alpha}_t}$ and $d\mu_t/dt = -(\beta_t/2)\mu_t$. For the ground-truth data-generating parameters $\theta^* \in \mathbb{R}^d$ and $\pi^* \equiv (\pi^*(1), \pi^*(2))$, we have $\nu^* := (\ln \pi^*(1) - \ln \pi^*(2))/2$, $\theta_t^* := \theta^*\sqrt{\bar{\alpha}_t}$, and $\mu_t^* := \langle \theta_t^*, \mathbf{x} \rangle$.

2.2 Score Function and Score Matching Loss

With the scaled regression parameters $\theta_t \equiv \theta\sqrt{\bar{\alpha}_t}$, mixing weights π , and shorthand $\mu_t \equiv \mu_0\sqrt{\bar{\alpha}_t} = \langle \theta_t, \mathbf{x} \rangle\sqrt{\bar{\alpha}_t}$, the Stein score function with respect to y_t is

$$s_{\theta_t, \nu}(y_t, \mathbf{x}) := \nabla \log p(y_t | \mathbf{x}; \theta_t, \pi) = -y_t + \tanh(\mu_t y_t + \nu)\mu_t. \quad (5)$$

The score matching loss $\mathcal{L}_t(\theta, \nu)$ is the covariate expectation of the relative Fisher information $\text{FI}(p(y_t | \mathbf{x}; \theta_t^*, \pi^*) \parallel p(y_t | \mathbf{x}; \theta_t, \pi)) := \mathbb{E}_{y_t | \mathbf{x}; \theta_t^*, \pi^*} \|s_{\theta_t, \nu}(y_t, \mathbf{x}) - s_{\theta_t^*, \nu^*}(y_t, \mathbf{x})\|^2$ between the model score with parameters θ, π and the true score with ground-truth parameters θ^*, π^* under the same normalization of $1/(2\bar{\alpha}_t)$.

$$\mathcal{L}_t(\theta, \nu) := \frac{1}{2\bar{\alpha}_t} \mathbb{E}_{y_t, \mathbf{x} | \theta^*, \pi^*} \|s_{\theta_t, \nu}(y_t, \mathbf{x}) - s_{\theta_t^*, \nu^*}(y_t, \mathbf{x})\|^2 = \frac{1}{2\bar{\alpha}_t} \mathbb{E}_{\mathbf{x}} \text{FI}(p(y_t | \mathbf{x}; \theta_t^*, \pi^*) \parallel p(y_t | \mathbf{x}; \theta_t, \pi)). \quad (6)$$

The auxiliary loss $\mathcal{J}_t(\theta, \nu)$ compares the model score with the score of the forward transition. The Gaussian transition is $p(y_t | y_0) = \mathcal{N}(y_t; y_0\sqrt{\bar{\alpha}_t}, 1 - \bar{\alpha}_t)$. Its score satisfies $\nabla \log p(y_t | y_0) = -\frac{y_t - y_0\sqrt{\bar{\alpha}_t}}{1 - \bar{\alpha}_t} \stackrel{d}{=} -\frac{\xi}{\sqrt{1 - \bar{\alpha}_t}}$, where $\xi \stackrel{d}{=} \xi_t \sim \mathcal{N}(0, 1)$ is independent of $(y_0, \mathbf{x})$. The equivalent representation $y_t \stackrel{d}{=} y_0\sqrt{\bar{\alpha}_t} + \xi\sqrt{1 - \bar{\alpha}_t}$ gives

$$\begin{aligned} \mathcal{J}_t(\theta, \nu) &:= \frac{1}{2\bar{\alpha}_t} \mathbb{E}_{y_t, y_0, \mathbf{x} | \theta^*, \pi^*} [\|s_{\theta_t, \nu}(y_t, \mathbf{x})\|^2 - 2\langle s_{\theta_t, \nu}(y_t, \mathbf{x}), \nabla \log p(y_t | y_0) \rangle] \\ &= \frac{1}{2\bar{\alpha}_t} \mathbb{E}_{y_0, \mathbf{x} | \theta^*, \pi^*} \mathbb{E}_{\xi} \left[\left\| s_{\theta_t, \nu}(y_0\sqrt{\bar{\alpha}_t} + \xi\sqrt{1 - \bar{\alpha}_t}, \mathbf{x}) \right\|^2 + 2 \left\langle s_{\theta_t, \nu}(y_0\sqrt{\bar{\alpha}_t} + \xi\sqrt{1 - \bar{\alpha}_t}, \mathbf{x}), \frac{\xi}{\sqrt{1 - \bar{\alpha}_t}} \right\rangle \right]. \end{aligned} \quad (7)$$

For i.i.d. observations $\{(y_0^{(i)}, \mathbf{x}^{(i)})\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} p(y_0, \mathbf{x} | \theta^*, \pi^*)$, we define $\mathbb{P}_n := n^{-1} \sum_{i=1}^n \delta_{(y_0^{(i)}, \mathbf{x}^{(i)})}$ and $\mathbb{E}_{\mathbb{P}_n}[f(y_0, \mathbf{x})] := n^{-1} \sum_{i=1}^n f(y_0^{(i)}, \mathbf{x}^{(i)})$. Thus $\mathbb{E}_{\mathbb{P}_n}$ averages over the observed pairs. The inner expectation in $\mathbb{E}_{\mathbb{P}_n} \mathbb{E}_{y_t | y_0}[\cdot]$ averages over forward-diffusion noise. The finite-sample loss is given by

$$\begin{aligned} \mathcal{J}_t^n(\theta, \nu) &:= \frac{1}{2\bar{\alpha}_t} \mathbb{E}_{\mathbb{P}_n} \mathbb{E}_{y_t | y_0} [\|s_{\theta_t, \nu}(y_t, \mathbf{x})\|^2 - 2\langle s_{\theta_t, \nu}(y_t, \mathbf{x}), \nabla \log p(y_t | y_0) \rangle] \\ &= \frac{1}{2\bar{\alpha}_t} \mathbb{E}_{\mathbb{P}_n} \mathbb{E}_{\xi} \left[\left\| s_{\theta_t, \nu}(y_0\sqrt{\bar{\alpha}_t} + \xi\sqrt{1 - \bar{\alpha}_t}, \mathbf{x}) \right\|^2 + 2 \left\langle s_{\theta_t, \nu}(y_0\sqrt{\bar{\alpha}_t} + \xi\sqrt{1 - \bar{\alpha}_t}, \mathbf{x}), \frac{\xi}{\sqrt{1 - \bar{\alpha}_t}} \right\rangle \right]. \end{aligned} \quad (8)$$

2.3 Negative Log-Likelihood and Expectation-Maximization Operators

The unit-variance negative log-likelihood and the associated population EM operators are given by the following equations, see detailed derivations in Appendix A and Proposition 3 and Appendix A of (Luo & Hashemi, 2025). The negative log-likelihood is

$$F(y_t, \mu_t, \nu) := -\ln p(y_t | \mathbf{x}; \theta_t, \pi) = \frac{1}{2}(y_t^2 + \mu_t^2) - \ln \frac{\cosh(\mu_t y_t + \nu)}{\cosh(\nu)} + \frac{1}{2} \ln(2\pi). \quad (9)$$

We use $\mathbb{E}_*[\cdot] := \mathbb{E}_{y_t, \mathbf{x} | \theta^*, \pi^*}[\cdot]$ for expectation under the ground-truth law. The cross-entropy $\mathcal{H}(\theta_t, \nu) := \mathbb{E}_*[-\ln p(y_t | \mathbf{x}; \theta_t, \pi)] = \mathbb{E}_*[F(y_t, \mu_t, \nu)]$ is the expected negative log-likelihood. The population Expectation-Maximization (EM) operators for θ_t and the imbalance of mixing weights $\tanh \nu \equiv \pi(1) - \pi(2)$ are

$$M(\theta_t, \nu) = \Sigma^{-1} \mathbb{E}_*[\mathbf{x} y_t \tanh(\mu_t y_t + \nu)], \quad N(\theta_t, \nu) = \mathbb{E}_*[\tanh(\mu_t y_t + \nu)]. \quad (10)$$

2.4 Notations on Metrics, Asymptotics, and Estimation Objectives

For probability measures P, Q on a common measurable space, we define $\text{KL}(P \parallel Q) := \int \ln(dP/dQ) dP$ when $P \ll Q$ and $\text{TV}(P, Q) := \sup_A |P(A) - Q(A)|$ over measurable sets. For probability measures P, Q with finite second moments, the Wasserstein distance is defined as $W_2^2(P, Q) := \inf_{\gamma \in \Gamma(P, Q)} \int \|u - v\|_2^2 \gamma(du, dv)$ over their set of couplings $\Gamma(P, Q)$. We write $X_n = \mathcal{O}_P(a_n)$ when X_n/a_n is stochastically bounded and $X_n = o_P(a_n)$ when $X_n/a_n \xrightarrow{\mathbb{P}} 0$. The model parameters $\vartheta := (\theta, \nu)$ and the ground-truth values $\vartheta^* := (\theta^*, \nu^*)$ are introduced for the convenience of notation. T_n denotes a deterministic terminal horizon for the total number of samples n , and $\bar{\alpha}_{T_n}$ is the retained-signal factor at the terminal horizon. Using $\mathbb{P}_n$ and $\mathcal{J}_t^n$ defined above, the empirical negative log-likelihood is defined as $\hat{\mathcal{R}}_n^{\text{MLE}}(\vartheta) := -\mathbb{E}_{\mathbb{P}_n}[\ln p(y_0 \mid \mathbf{x}; \theta, \pi)]$, and the path-integrated score matching objective is denoted by $\hat{\mathcal{R}}_{n, T_n}^{\text{SM}}(\vartheta) := \int_{\bar{\alpha}_{T_n}}^{\bar{\alpha}_0} \mathcal{J}_t^n(\theta, \nu) d\bar{\alpha}_t$. On the event that this objective has a nonempty argmin, let $\hat{\vartheta}_{n, T_n}^{\text{SM}} := (\hat{\theta}_{n, T_n}^{\text{SM}}, \hat{\nu}_{n, T_n}^{\text{SM}})$ be a sample-measurable exact minimizer; on the complementary event, set $\hat{\vartheta}_{n, T_n}^{\text{SM}} := (\mathbf{0}, 0)$. This convention defines the score matching estimator on the entire sample space, and Lemma B.2 supplies the measurable selection used under the conditions of Main Theorem 3.2. The empirical conditional terminal discrepancy is defined as $\Delta_n(\vartheta) := \mathbb{E}_{\mathbb{P}_n}[\text{KL}(p(y_{T_n} \mid y_0) \parallel p(y_{T_n} \mid \mathbf{x}; \theta_{T_n}, \pi))] \geq 0$. The constant term $c(\bar{\alpha}_{T_n}) := \frac{1}{2}(\ln \frac{1 - \bar{\alpha}_{T_n}}{\bar{\alpha}_{T_n}} + \ln(2\pi) + 1)$ only depends on the retained-signal factor $\bar{\alpha}_{T_n}$ at the terminal horizon.

3 Path-Integrated Score Matching: Maximum-Likelihood Asymptotics

In this section, we show how integrating score matching along the diffusion path recovers the asymptotic guarantees of maximum likelihood. We first derive population and empirical likelihood bridges that isolate the terminal discrepancy separating the integrated objective from likelihood and identify the terminal population KL divergence's global minimizers. We then control this discrepancy to transfer consistency up to joint sign and asymptotic normality with inverse-Fisher covariance to the path-integrated estimator in Main Theorem 3.2, with the supporting arguments deferred to Appendices A and B. by the two-law relative analogue of de Bruijn identity (Stam, 1959) for entropy and Fisher information under the variance-preserving Ornstein–Uhlenbeck flow (see the derivation of Lemma A.4 in Appendix A).

$$\int_{\bar{\alpha}_T}^{\bar{\alpha}_0} \mathcal{L}_t(\theta, \nu) d\bar{\alpha}_t = \text{KL}(p(y_0, \mathbf{x} \mid \theta^*, \pi^*) \parallel p(y_0, \mathbf{x} \mid \theta, \pi)) - \text{KL}(p(y_T, \mathbf{x} \mid \theta_T^*, \pi^*) \parallel p(y_T, \mathbf{x} \mid \theta_T, \pi)). \quad (11)$$

Its finite-sample counterpart involves the empirical objectives $\hat{\mathcal{R}}_{n, T_n}^{\text{SM}}(\vartheta)$, $\hat{\mathcal{R}}_n^{\text{MLE}}(\vartheta)$, and $\Delta_n(\vartheta)$, which correspond to the left-hand side and the first and second terms on the right-hand side of the equation above, respectively, and differ from them only by the constant $c(\bar{\alpha}_{T_n})$ (see Lemmas A.15 and A.16 in Appendix A).

$$\hat{\mathcal{R}}_{n, T_n}^{\text{SM}}(\vartheta) = \hat{\mathcal{R}}_n^{\text{MLE}}(\vartheta) - \Delta_n(\vartheta) - c(\bar{\alpha}_{T_n}). \quad (12)$$

We can bound the terminal population KL divergence from the true law to the model law by the Wasserstein geometry of the discrete mixing measures supported on at most two points $D \equiv D(\mathbf{x}; \theta, \pi) \equiv \pi(1)\delta_{\langle \theta, \mathbf{x} \rangle} + \pi(2)\delta_{-\langle \theta, \mathbf{x} \rangle}$ and $D^* \equiv D(\mathbf{x}; \theta^*, \pi^*) \equiv \pi^*(1)\delta_{\langle \theta^*, \mathbf{x} \rangle} + \pi^*(2)\delta_{-\langle \theta^*, \mathbf{x} \rangle}$ (see Lemmas A.6, A.7, A.13 in Appendix A).

$$\text{KL}(p(y_T, \mathbf{x} \mid \theta_T^*, \pi^*) \parallel p(y_T, \mathbf{x} \mid \theta_T, \pi)) \leq \frac{\bar{\alpha}_T}{2(1 - \bar{\alpha}_T)} \mathbb{E}_{\mathbf{x}} W_2^2(D, D^*), \quad (13)$$

Similarly, the finite-sample counterpart $\Delta_n(\vartheta)$ decays in the rate of $\frac{\bar{\alpha}_{T_n}}{2(1 - \bar{\alpha}_{T_n})}$ as $n \rightarrow \infty$ (see Lemma A.8 in Appendix A). If $\ln \frac{1 - \bar{\alpha}_{T_n}}{\bar{\alpha}_{T_n}} - \ln n \rightarrow \infty$, then $\frac{\bar{\alpha}_{T_n}}{2(1 - \bar{\alpha}_{T_n})} = o(n^{-1})$ and $\Delta_n(\vartheta)$ is $o_P(n^{-1})$ when ϑ is bounded.

The next proposition complements that the ground truth parameters are the global minimizers of the terminal population KL divergence.

Proposition 3.1 (Global Minimizers of the Terminal KL Divergence). *The retained-signal level satisfies $0 < \bar{\alpha}_T \leq 1$, the ground-truth parameters satisfy $\theta^* \neq \mathbf{0}$ and $\nu^* \in \mathbb{R}$, and $\mathbf{x}$ has a density that is positive almost everywhere on $\mathbb{R}^d$ and finite second moments. For the terminal population KL divergence $\text{KL}_T(\theta, \nu) := \text{KL}(p(y_T, \mathbf{x} \mid \theta_T^*, \pi^*) \parallel p(y_T, \mathbf{x} \mid \theta_T, \pi))$, the set of global minimizers is $\arg \min_{(\theta, \nu) \in \mathbb{R}^d \times \mathbb{R}} \text{KL}_T(\theta, \nu) = \{(\theta^*, \nu^*), (-\theta^*, -\nu^*)\}$.*

Proof sketch. $\text{KL}_T(\theta, \nu) = 0$ only if $p(\cdot \mid \mathbf{x}; \theta_T^*, \pi^*) = p(\cdot \mid \mathbf{x}; \theta_T, \pi)$ almost everywhere by Pinsker's inequality (Cover & Thomas, 2006) $\text{KL}_T(\theta, \nu) \geq 2\mathbb{E}_{\mathbf{x}} \text{TV}^2(p(\cdot \mid \mathbf{x}; \theta_T^*, \pi^*), p(\cdot \mid \mathbf{x}; \theta_T, \pi))$. Therefore, $(\theta, \nu) \in \{(\theta^*, \nu^*), (-\theta^*, -\nu^*)\}$ when $\text{KL}_T(\theta, \nu) = 0$ by the positive-density assumption on $\mathbf{x}$, while $\mathbb{E}_{\mathbf{x}} W_2^2(D, D^*) = 0$ has the same two solutions. Thus the lower bound on the terminal KL divergence and the upper bound on the expected discrete Wasserstein cost have the same global zero set, proving the stated minimizer set.

The next theorem adapts the similar argument of asymptotic normality in Chewi et al. (2025a, Assumption 3, Theorem 3.1 on page 25) to MLR but without the regularity condition on Lipschitz continuity of the log-probability density. Proposition 3.1 suggests that the sign identification is needed for this model.

Main theorem 3.2 (Consistency and Asymptotic Normality of Score Matching Estimator). *The ground-truth parameters $(\theta^*, \nu^*) \in \mathbb{R}^d \times \mathbb{R}$ satisfy $\theta^* \neq \mathbf{0}$. The deterministic terminal-horizon sequence $\{T_n\}_{n \geq 1}$ satisfies $\ln \frac{1 - \bar{\alpha}_{T_n}}{\bar{\alpha}_{T_n}} - \ln n \rightarrow \infty$, and the covariate $\mathbf{x}$ has a density that is positive almost everywhere and satisfies $\mathbb{E}\|\mathbf{x}\|^2 < \infty$. The estimator $\hat{\vartheta}_{n, T_n}^{\text{SM}} \equiv (\hat{\theta}_{n, T_n}^{\text{SM}}, \hat{\nu}_{n, T_n}^{\text{SM}})$ is consistent up to a joint sign change. For the alignment map $s_n \in \arg \min_{s \in \{+1, -1\}} \left\| s \begin{pmatrix} \hat{\theta}_{n, T_n}^{\text{SM}} \\ \hat{\nu}_{n, T_n}^{\text{SM}} \end{pmatrix} - (\theta^*, \nu^*) \right\|$, as $n \rightarrow \infty$,*

$$s_n \begin{pmatrix} \hat{\theta}_{n, T_n}^{\text{SM}} \\ \hat{\nu}_{n, T_n}^{\text{SM}} \end{pmatrix} \xrightarrow{\mathbb{P}} (\theta^*, \nu^*).$$

If $\mathbb{E}\|\mathbf{x}\|^4 < \infty$, then the aligned Score Matching estimator is asymptotically normal. As $n \rightarrow \infty$,

$$\sqrt{n} \left[s_n \begin{pmatrix} \hat{\theta}_{n, T_n}^{\text{SM}} \\ \hat{\nu}_{n, T_n}^{\text{SM}} \end{pmatrix} - (\theta^*, \nu^*) \right] \xrightarrow{d} \mathcal{N}(\mathbf{0}, I(\theta^*, \nu^*)^{-1}),$$

where I denotes the Fisher information matrix.

Proof sketch (Bridge between Score Matching and MLE). Lemma B.2 shows that the empirical score matching objective $\hat{\mathcal{R}}_{n, T_n}^{\text{SM}}(\vartheta)$ admits $\hat{\vartheta}_{n, T_n}^{\text{SM}}$ with probability tending to one, and the global minimizer $\hat{\vartheta}_{n, T_n}^{\text{SM}}$ is attained asymptotically almost surely. See the detailed proof of Main Theorem 3.2 in Appendix B.

Step 1. Boundedness and excess-risk control. On the attainment event, the empirical bridge (12) bounds the empirical likelihood excess risk by the terminal discrepancy such that $\hat{\mathcal{R}}_{n, T_n}^{\text{MLE}}(\hat{\vartheta}_{n, T_n}^{\text{SM}}) - \hat{\mathcal{R}}_{n, T_n}^{\text{MLE}}(\vartheta^*) = \hat{\mathcal{R}}_{n, T_n}^{\text{SM}}(\hat{\vartheta}_{n, T_n}^{\text{SM}}) - \hat{\mathcal{R}}_{n, T_n}^{\text{SM}}(\vartheta^*) + \Delta_n(\hat{\vartheta}_{n, T_n}^{\text{SM}}) - \Delta_n(\vartheta^*) \leq \Delta_n(\hat{\vartheta}_{n, T_n}^{\text{SM}})$. The transition-kernel KL bound and a coercive likelihood bound give $\|\hat{\vartheta}_{n, T_n}^{\text{SM}}\| = \mathcal{O}_P(1)$. The terminal-scale condition then gives $\Delta_n(\hat{\vartheta}_{n, T_n}^{\text{SM}}) = o_P(n^{-1})$.

Step 2. Consistency up to joint sign. The population likelihood excess risk is a conditional KL divergence whose zero set is $\{\vartheta^*, -\vartheta^*\}$. The attainment result, the empirical bridge, and Step 1 make the estimator an $o_P(n^{-1})$ -near maximizer of the empirical likelihood. Theorem 5.14 of van der Vaart (1998, p. 48) then gives consistency up to joint sign and $\|\hat{\vartheta}_{n, T_n}^{\text{SM}}\| = \mathcal{O}_P(1)$.

Step 3. Asymptotic normality. After joint sign alignment, the estimator is consistent and remains an $o_P(n^{-1})$ -near maximizer. Under the fourth-moment condition, the population log-likelihood has a nonsingular quadratic expansion with Hessian $-I(\theta^*, \nu^*)$, and the local log-likelihood admits a square-integrable Lipschitz envelope. These properties verify the hypotheses of Theorem 5.23 of van der Vaart (1998, p. 53), which gives the stated Gaussian limit with covariance $I(\theta^*, \nu^*)^{-1}$.

Remark (On Consistency and Asymptotic Normality). Lemma A.2 shows that $p(y_0 \mid \mathbf{x}; \theta, \pi) \equiv p(y_0 \mid \mathbf{x}; -\theta, \pi^{\text{swap}})$, where $\pi^{\text{swap}} = (\pi(2), \pi(1))$. The likelihood is invariant under $(\theta, \nu) \mapsto (-\theta, -\nu)$, and both estimators $\hat{\mathcal{R}}_{n, T_n}^{\text{MLE}}$ and $\hat{\mathcal{R}}_{n, T_n}^{\text{SM}}$ inherit this invariance. The alignment map $s_n \in \{\pm 1\}$ selects the estimator nearest to the ground truth, after which $s_n \hat{\vartheta}_{n, T_n}^{\text{SM}} \xrightarrow{\mathbb{P}} \vartheta^*$. After sign alignment, the local M-estimator expansion yields the $n^{-1/2}$ rate with limiting covariance $I(\theta^*, \nu^*)^{-1}$, which matches the asymptotic variance of the MLE.

4 Fixed-Scale Score Matching: Cross-Entropy and EM Decompositions

In this section, we establish the connection between the EM operators M, N for regression parameters and the imbalance of mixing weights, and the cross entropy $\mathcal{H}(\theta_t, \nu) \equiv \mathbb{E}_*[-\log p(y_t \mid \mathbf{x}; \theta_t, \pi)] = \mathbb{E}_*[F(y_t, \mu_t, \nu)]$ and EM-latent-variance decompositions of the loss, together with gradient identities that retain operator-

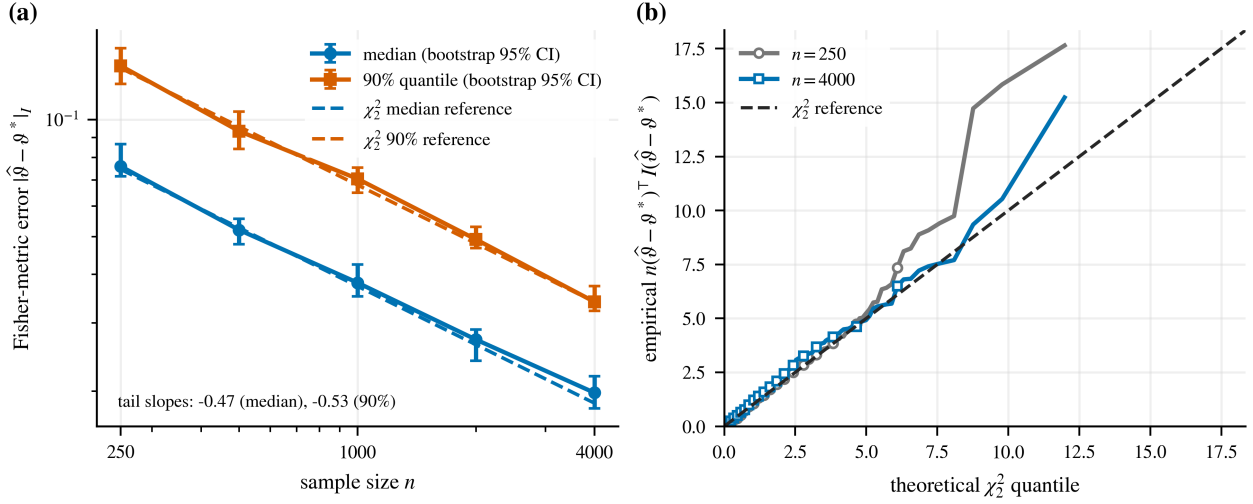


Figure 1: **Illustration of the finite-sample estimation results in Theorem 3.2.** The experiment uses a two-parameter Gaussian 2MLR instance, the theorem’s joint sign alignment, and a terminal sequence whose terminal-scale condition is verified in Appendix E. Panel (a) reports the median and 90% quantile of the error after joint sign alignment in the Fisher metric, together with 95% bootstrap intervals and the corresponding limiting quantiles of $\sqrt{\chi^2_2/n}$. Over the three largest sample sizes, the fitted slopes are -0.471 and -0.526 , close to the predicted $-1/2$ in the rate of $n^{-1/2}$. Panel (b) compares the squared, n -scaled Fisher error with χ^2_2 at the smallest and largest sample sizes. Coverage of the joint 95% Fisher ellipse ranges from 93.5% to 96.5%, equals 95.0% at the largest sample size, and has a corresponding covariance discrepancy of 0.070. These diagnostics provide finite-sample illustrations under the alignment and terminal-scale conditions of Main Theorem 3.2. Appendix E provides additional calibration details.

Jacobian and posterior-variance corrections. We then use these identities to characterize the loss and its gradients at the low- and high-noise endpoints. Appendix C contains the proofs of Main Theorem 4.3, Propositions 4.1–4.2 and 4.6–4.8, and Corollaries 4.4–4.5.

By applying the established Lemma A.17 for the derivatives of the negative log-likelihood $F(y_t, \mu_t, \nu) \equiv -\ln p(y_t \mid \mathbf{x}; \theta_t, \pi)$, we can derive the following proposition for the EM update operators M, N for the regression parameters θ_t and the imbalance parameter ν , and further formulate the relation between the derivatives of the EM update operators and derivatives of the cross entropy $\mathcal{H}(\theta_t, \nu) = \mathbb{E}_*[F(y_t, \mu_t, \nu)]$.

Proposition 4.1 (Expectation-Maximization Rules). *The population EM operators for the regression parameter θ_t and imbalance of mixing weights $\tanh \nu$ are*

$$M(\theta_t, \nu) = \Sigma^{-1} \mathbb{E}_*[\mathbf{x} y_t \tanh(\mu_t y_t + \nu)], \quad N(\theta_t, \nu) = \mathbb{E}_*[\tanh(\mu_t y_t + \nu)].$$

In addition to defining the EM updates, these operators satisfy the scalar identity

$$\langle \Sigma M(\theta_t, \nu), \theta_t \rangle = \mathbb{E}_*[\mu_t y_t \tanh(\mu_t y_t + \nu)] = \mathbb{E}_*[y_t^2 - y_t \partial_{y_t} F(y_t, \mu_t, \nu)],$$

while differentiation with respect to ν yields $\Sigma \nabla_\nu M(\theta_t, \nu) = \nabla_{\theta_t} N(\theta_t, \nu) = \mathbb{E}_[\mathbf{x} y_t \operatorname{sech}^2(\mu_t y_t + \nu)]$.*

Proposition 4.2 (Cross-Entropy and EM Update Rules). *The cross-entropy $\mathcal{H}(\theta_t, \nu)$ satisfies*

$$\nabla_{\theta_t} \mathcal{H}(\theta_t, \nu) = \Sigma(\theta_t - M(\theta_t, \nu)), \quad \nabla_\nu \mathcal{H}(\theta_t, \nu) = \tanh \nu - N(\theta_t, \nu).$$

From the established Proposition 4.1 and Proposition 4.2, we can derive the following theorem for the information theoretical decomposition of the score matching loss, and show its connection with the EM update operators M, N and derivatives of the cross entropy $\mathcal{H}(\theta_t, \nu)$.

Main theorem 4.3 (Information Decomposition of the Score Matching Loss). *The expected squared activation is $\mathcal{A}(\theta_t, \nu) := \frac{1}{2} \mathbb{E}_*[\mu_t^2 \tanh^2(\mu_t y_t + \nu)]$ and its ground-truth counterpart is $\mathcal{A}(\theta_t^*, \nu^*) := \frac{1}{2} \mathbb{E}_*[(\mu_t^*)^2 \tanh^2(\mu_t^* y_t + \nu^*)]$. With these definitions, the score matching loss satisfies*

$$\bar{\alpha}_t \mathcal{L}_t(\theta, \nu) = \langle \nabla_{\theta_t} \mathcal{H}(\theta_t, \nu), \theta_t \rangle - \mathcal{A}(\theta_t, \nu) + \mathcal{A}(\theta_t^*, \nu^*).$$

Defining the latent variance $V(\theta_t, \nu) \equiv \frac{1}{2} \mathbb{E}_[\mu_t^2 \text{sech}^2(\mu_t y_t + \nu)]$ gives an equivalent representation.*

$$\bar{\alpha}_t \mathcal{L}_t(\theta, \nu) = \|\theta_t\|_{\Sigma}^2/2 - \langle \Sigma M(\theta_t, \nu), \theta_t \rangle + \|\theta_t^*\|_{\Sigma}^2/2 + V(\theta_t, \nu) - V(\theta_t^*, \nu^*).$$

Proof sketch. Lemmas A.2 and A.3 express both scores as negative y_t -derivatives of the negative log-likelihood $F(y_t, \mu_t, \nu) \equiv -\ln p(y_t | \mathbf{x}; \theta_t, \pi)$. Expanding their squared discrepancy and integrating the mixed and squared ground-truth-score terms by parts with vanishing boundary terms yields an identity involving the first two derivatives. The derivative formulas and the conditional second moment of y_t then reduce this identity to quadratic and sech^2 terms. Proposition 4.1 identifies the remaining mixed moment and gives the representation with EM operators and latent variance. Proposition 4.2, together with $\text{sech}^2(u) = 1 - \tanh^2(u)$, converts it into the representation with the cross entropy $\mathcal{H}(\theta_t, \nu)$ and expected squared activation.

Corollary 4.4 (Gradients of the Score Matching Loss via Cross-Entropy and Expected Squared Activation). *The gradients of the score matching loss $\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)$ with respect to the regression parameter θ_t and the mixing imbalance ν are as follows.*

$$\begin{aligned} \nabla_{\theta_t} [\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)] &= \nabla_{\theta_t} \mathcal{H}(\theta_t, \nu) + \nabla_{\theta_t}^2 \mathcal{H}(\theta_t, \nu) \theta_t - \nabla_{\theta_t} \mathcal{A}(\theta_t, \nu), \\ \nabla_{\nu} [\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)] &= \langle \nabla_{\theta_t \nu}^2 \mathcal{H}(\theta_t, \nu), \theta_t \rangle - \nabla_{\nu} \mathcal{A}(\theta_t, \nu). \end{aligned}$$

Corollary 4.5 (Gradients of the Score Matching Loss via EM Operators and Latent Variance). *The gradients of the score matching loss $\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)$ with respect to θ_t and ν are as follows.*

$$\begin{aligned} \nabla_{\theta_t} [\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)] &= \Sigma(\theta_t - M(\theta_t, \nu)) - (\nabla_{\theta_t} M(\theta_t, \nu))^{\top} \Sigma \theta_t + \nabla_{\theta_t} V(\theta_t, \nu), \\ \nabla_{\nu} [\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)] &= -\langle \nabla_{\theta_t} N(\theta_t, \nu), \theta_t \rangle + \nabla_{\nu} V(\theta_t, \nu). \end{aligned}$$

The normalized EM operators are $\bar{M}_t(\theta, \nu) := M(\theta_t, \nu)/\sqrt{\bar{\alpha}_t}$, $\bar{N}_t(\theta, \nu) := (N(\theta_t, \nu) - (1 - \bar{\alpha}_t) \tanh \nu)/\bar{\alpha}_t$. These definitions imply $\theta - \bar{M}_t(\theta, \nu) = (\theta_t - M(\theta_t, \nu))/\sqrt{\bar{\alpha}_t}$, $\tanh \nu - \bar{N}_t(\theta, \nu) = (\tanh \nu - N(\theta_t, \nu))/\bar{\alpha}_t$, and $\Sigma \nabla_{\nu} \bar{M}_t(\theta, \nu) = \nabla_{\theta} \bar{N}_t(\theta, \nu)$, which yield the following gradient identities with $V_t(\theta, \nu) := V(\theta_t, \nu)/\bar{\alpha}_t$.

Proposition 4.6 (Gradients of the score matching Loss). *The gradients of the score matching loss $\mathcal{L}_t(\theta, \nu)$ are as follows:*

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_t(\theta, \nu) &= \Sigma(\theta - \bar{M}_t(\theta, \nu)) - (\nabla_{\theta} \bar{M}_t(\theta, \nu))^{\top} \Sigma \theta + \nabla_{\theta} V_t(\theta, \nu), \\ \nabla_{\nu} \mathcal{L}_t(\theta, \nu) &= -\langle \nabla_{\theta} \bar{N}_t(\theta, \nu), \theta \rangle + \nabla_{\nu} V_t(\theta, \nu). \end{aligned}$$

Remark. As $\bar{\alpha}_t \rightarrow 1$, we have $\theta_t \equiv \theta \sqrt{\bar{\alpha}_t} \rightarrow \theta$, so $\bar{M}_t(\theta, \nu) \rightarrow M(\theta, \nu)$, $\bar{N}_t(\theta, \nu) \rightarrow N(\theta, \nu)$, and $V_t(\theta, \nu) \rightarrow V(\theta, \nu)$. Substitution into Proposition 4.6 gives

$$\begin{aligned} \lim_{\bar{\alpha}_t \rightarrow 1} \nabla_{\theta} \mathcal{L}_t(\theta, \nu) &= \Sigma(\theta - M(\theta, \nu)) - (\nabla_{\theta} M(\theta, \nu))^{\top} \Sigma \theta + \nabla_{\theta} V(\theta, \nu), \\ \lim_{\bar{\alpha}_t \rightarrow 1} \nabla_{\nu} \mathcal{L}_t(\theta, \nu) &= -\langle \nabla_{\theta} N(\theta, \nu), \theta \rangle + \nabla_{\nu} V(\theta, \nu). \end{aligned} \tag{14}$$

Proposition 4.7 (Limit at Low Noise and Expectation-Maximization). *If $\mathbb{E}[\|\mathbf{x}\|^4] < \infty$, the gradients of the score matching loss $\mathcal{L}_t(\theta, \nu)$ satisfy the following expansions as $\bar{\alpha}_t \rightarrow 1$.*

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_t(\theta, \nu) &= \Sigma(\theta - M(\theta, \nu)) - (\nabla_{\theta} M(\theta, \nu))^{\top} \Sigma \theta + \nabla_{\theta} V(\theta, \nu) + \mathcal{O}(1 - \bar{\alpha}_t), \\ \nabla_{\nu} \mathcal{L}_t(\theta, \nu) &= -\langle \nabla_{\theta} N(\theta, \nu), \theta \rangle + \nabla_{\nu} V(\theta, \nu) + \mathcal{O}(1 - \bar{\alpha}_t). \end{aligned}$$

Remark (low-noise loss as $\bar{\alpha}_t \rightarrow 1$). For fixed θ, θ^* , under the independence assumptions of Section 2.1 and $\mathbb{E}[\|\mathbf{x}\|^2] < \infty$, with $\mu_0 := \langle \theta, \mathbf{x} \rangle$ and $\mu_0^* := \langle \theta^*, \mathbf{x} \rangle$, the normalized conditional squared score discrepancy in

Lemma A.10 is bounded uniformly in $\bar{\alpha}_t$ by $(|\mu_0| + |\mu_0^*|)^2$, which is integrable under the stated second-moment condition. Dominated convergence over $\mathbf{x}$ therefore gives the low-noise limit

$$\lim_{\bar{\alpha}_t \rightarrow 1} \mathcal{L}_t(\theta, \nu) = \frac{1}{2} \mathbb{E}_{\mathbf{x}} \mathbb{E}_{y_0 | \mathbf{x}; \theta^*, \pi^*} \left[(\mu_0 \tanh(\mu_0 y_0 + \nu) - \mu_0^* \tanh(\mu_0^* y_0 + \nu^*))^2 \right]. \quad (15)$$

Thus the objective limit itself requires only a second moment, whereas Proposition 4.7 uses the stronger fourth-moment condition to control differentiation under the covariate expectation and obtain the corresponding gradient expansion with EM operators, latent variance, and an $\mathcal{O}(1 - \bar{\alpha}_t)$ remainder.

Proposition 4.8 (Effective Coefficient Limit at High Noise). *If $\mathbb{E}[\|\mathbf{x}\|^2] < \infty$, the effective coefficient $c^* := \theta^* \tanh \nu^*$ determines the limiting objective through $\lim_{\bar{\alpha}_t \rightarrow 0} \mathcal{L}_t(\theta, \nu) = \frac{1}{2} \|\theta \tanh \nu - c^*\|_{\Sigma}^2$. The condition $\mathbb{E}[\|\mathbf{x}\|^4] < \infty$ suffices for the $\mathcal{O}(\bar{\alpha}_t)$ gradient expansions as $\bar{\alpha}_t \rightarrow 0$*

$$\nabla_{\theta} \mathcal{L}_t(\theta, \nu) = \nabla_{\theta} \lim_{\bar{\alpha}_t \rightarrow 0} \mathcal{L}_t(\theta, \nu) + \mathcal{O}(\bar{\alpha}_t), \quad \nabla_{\nu} \mathcal{L}_t(\theta, \nu) = \nabla_{\nu} \lim_{\bar{\alpha}_t \rightarrow 0} \mathcal{L}_t(\theta, \nu) + \mathcal{O}(\bar{\alpha}_t).$$

Remark. Proposition 4.8 characterizes only the leading high-noise objective, whose gradient dynamics will be discussed in Section 5; its $\mathcal{O}(\bar{\alpha}_t)$ gradient remainder is checked componentwise in Figure 2(b). On the balanced slice $\nu^* = \nu = 0$, that leading objective is flat in θ , so the proposition supplies no preferred direction. We report a separate next-order directional diagnostic, illustrated in Figure 5, which compares finite-noise score-matching updates with power steps on the spiked fourth-order covariance in Appendix E.

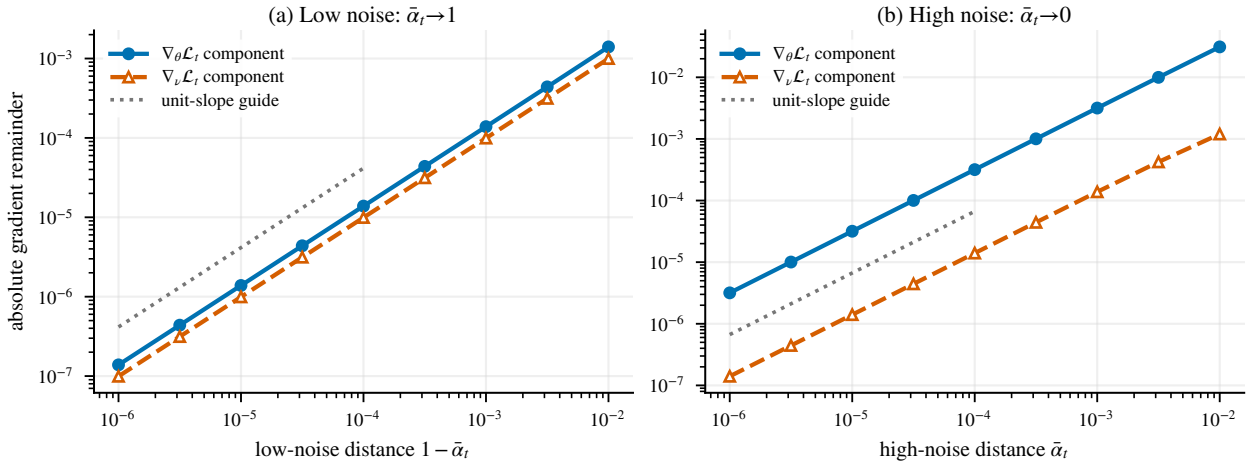


Figure 2: **First-order scaling of the endpoint gradient remainders in Propositions 4.7 and 4.8.** At one fixed model pair, the panels plot componentwise remainders for the complete low-noise expression in Proposition 4.7 and the effective high-noise coefficient expression in Proposition 4.8. The fitted log-log slopes track the guides with unit slope, consistent with the respective $\mathcal{O}(1 - \bar{\alpha}_t)$ and $\mathcal{O}(\bar{\alpha}_t)$ bounds. The panels report one selected instance, whose settings and fitted values appear in Appendix E.

5 High-Noise Gradient Dynamics and Score Matching Blindness

Building on the endpoint loss and gradient limits in Section 4, we further study the optimization dynamics and parameter-information consequences in this section. We first analyze fixed-step gradient descent on the high-noise limiting objective in Main Theorem 5.1 and then establish pointwise score matching blindness along fixed high-SNR rays in Proposition 5.2. Complete arguments and details of proofs are provided in Appendix D.

5.1 High-Noise Gradient Dynamics

As shown in Proposition 4.8, the high-noise limiting objective is given by $\mathcal{L}(\theta, \nu) := \lim_{\bar{\alpha}_t \rightarrow 0} \mathcal{L}_t(\theta, \nu)$. We consider the optimization dynamics of gradient descent with a fixed step size η : $\vartheta_{k+1} \leftarrow \vartheta_k - \eta \nabla \mathcal{L}(\vartheta_k)$, for the

iterate $\vartheta_k = (\theta_k, \nu_k)$ of parameters at time step k . The following informal Main Theorem 5.1 demonstrates the convergence rates of the optimization dynamics.

Main theorem 5.1 (Informal; Global Dynamics as $\bar{\alpha}_t \rightarrow 0$). *On the high-noise limiting loss $\mathcal{L}(\theta, \nu) = \frac{1}{2} \|\theta \tanh \nu - c^*\|^2$ with $c^* := \theta^* \tanh \nu^*$, isotropic covariates $\Sigma = I_d$, fixed-step gradient descent with an admissible initialization has nonincreasing loss and obeys the following rates.*
($c^ = \mathbf{0}$.) The loss satisfies $\mathcal{L}(\theta_K, \nu_K) = \mathcal{O}(K^{-2})$.*
($c^ \neq \mathbf{0}$, generic initializations.) Every admissible initialization outside the exceptional saddle basin has geometric loss decay after a finite transient and reaches any $\epsilon > 0$ in $\mathcal{O}(\log(1/\epsilon))$ iterations.*
($c^ \neq \mathbf{0}$, exceptional initializations.) The exceptional saddle basin consists of the admissible initializations whose trajectories converge to the saddle manifold $\{(\theta, 0) : \langle \theta, c^* \rangle = 0\}$. This basin has $(d+1)$ -dimensional Lebesgue measure zero, and every such trajectory has limiting loss $\|c^*\|^2/2$.*

Proof sketch. We establish the proof by showing that the coupled descent bounds keep the iterates in the admissible region. When $c^* = \mathbf{0}$, a gradient bound of order $\mathcal{L}^{3/2}(\vartheta_k)$ gives the K^{-2} upper bound. When $c^* \neq \mathbf{0}$, the sign recurrences give geometric decay after the signs agree. The local stable set of the saddle has lower dimension, and inverse images of null sets under the update map remain null. Hence the admissible saddle basin has zero $(d+1)$ -dimensional Lebesgue measure. See Theorem D.1 in Appendix D for the precise admissibility condition for the step size, explicit constants in convergence rates, and the detailed proofs.

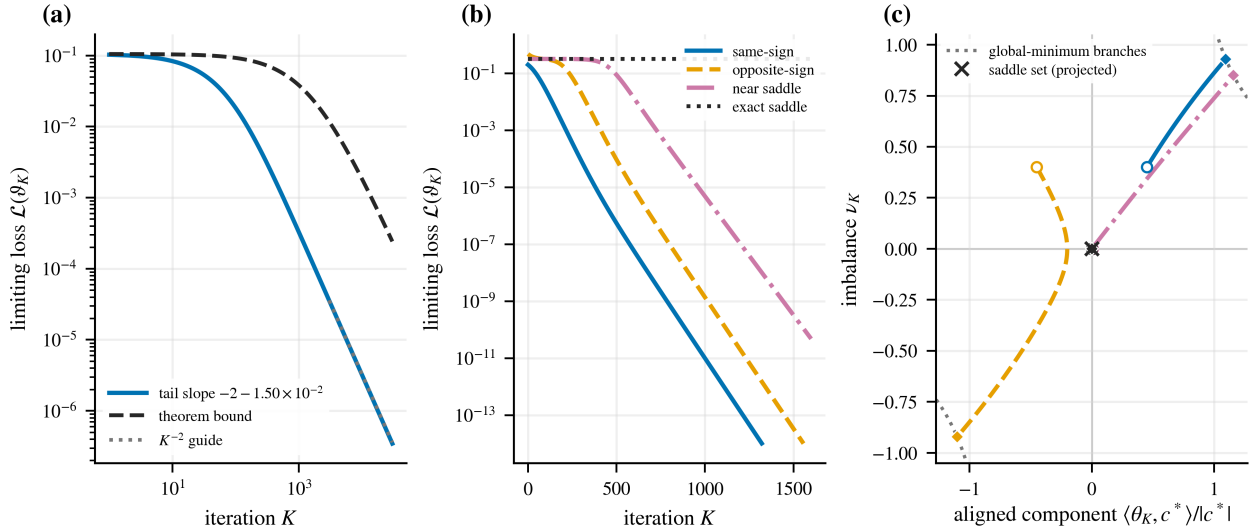


Figure 3: **Optimization Dynamics of Gradient Descent on the High-Noise Objective in Theorem 5.1.** The gradient descent update with a fixed step size is iterated from initializations satisfying the admissibility condition in Theorem D.1, and every displayed path preserves that condition and has non-increasing loss. (a) For the degenerate target, the computed loss remains below the K^{-2} upper bound in Theorem 5.1, and its fitted log-log tail slope is $-2 - 1.50 \times 10^{-2}$, tracking the anchored K^{-2} guide over the displayed tail. (b) For $c^* \neq \mathbf{0}$, the three displayed generic initializations lie outside the exceptional saddle basin and show initialization-dependent transients followed by geometric loss decay, consistent with the $\mathcal{O}(\log(1/\epsilon))$ conclusion. (c) The phase portrait in $(\langle \theta_K, c^* \rangle / \|c^*\|, \nu_K)$ coordinates uses open circles and filled diamonds to mark the initial and final iterates. The dotted curves are the two branches of the global-minimum set $\{(\theta, \nu) : \theta \tanh \nu = c^*\}$. By contrast, the saddle set $\{(\theta, 0) : \langle \theta, c^* \rangle = 0\}$ projects to the single point $(0, 0)$ in these coordinates and is marked by the black cross. The observed degenerate slope diagnoses finite-window agreement with the K^{-2} upper bound in Theorem 5.1. Appendix E provides the configurations for generating the figures.

Remark (Interpreting the Dynamics at High Noise). The objective identifies the ground truth only through the effective signal $c^* = \theta^* \tanh \nu^*$, so all ground-truth pairs with the same product induce the same target. The degenerate case includes $\theta^* = \mathbf{0}$ and balanced mixtures $\nu^* = 0$, for which the theorem provides a uniform

polynomial upper bound without a matching lower bound, so individual zero-target trajectories can decay faster than K^{-2} . The exponent is consistent with the KL gradient-inequality exponent $3/4$ of a singular loss formed by a squared product and with the associated $\mathcal{O}(K^{-2})$ bound on function values (Jozs & Ouyang, 2026). For $c^* \neq \mathbf{0}$, the almost-global conclusion applies trajectory by trajectory within the admissible class, with both the transient and contraction factor depending on the initialization.

5.2 Score Matching Blindness at High SNR

Proposition 5.2 (Pointwise Score Matching Blindness Along a Fixed Ray at High SNR). *The covariates satisfy $\mathbb{E}[\|\mathbf{x}\|^2] < \infty$. The scale of the diffusion noise level $\bar{\alpha}_t \in (0, 1]$ and finite model pair $(\theta, \nu) \in \mathbb{R}^d \times \mathbb{R}$ are fixed. Let $\theta^* \neq \mathbf{0}$ vary with $\|\theta^*\| \rightarrow \infty$ along a fixed ray, while the true mixing weights and all other model and data-generating quantities remain fixed. When $\theta \neq \mathbf{0}$, we also impose the following projection condition.*

$$\mathbb{P}(\{\langle \theta, \mathbf{x} \rangle = 0\} \cup \{\langle \theta^*, \mathbf{x} \rangle = 0\}) = 0.$$

The expectations defining $\mathcal{L}_t$ and V_t are taken under the corresponding varying ground truth. At the specified fixed model pair and fixed scale of the diffusion noise level, the following pointwise limits hold.

$$\lim_{\|\theta^*\| \rightarrow \infty} |\nabla_\nu \mathcal{L}_t(\theta, \nu)| = 0, \quad \lim_{\|\theta^*\| \rightarrow \infty} V_t(\theta, \nu) = 0.$$

Remark (Blindness at High SNR). Under the fixed-scale high-SNR regime of Proposition 5.2, Lemma D.3 gives the population EM limit $\tanh \nu^+ = N_{\theta^*}(\theta_t, \nu) \rightarrow (2/\pi) \arcsin(\rho) \tanh \nu^*$ for isotropic Gaussian covariates and $\rho := \langle \theta, \theta^* \rangle / (\|\theta\| \|\theta^*\|)$, which agrees with the infinite-SNR population update in Luo & Hashemi (2024, Corollary 3.3). Its independence from the incoming ν shows that high-SNR also affects population EM, while the dependence on ρ and ν^* retains true-imbalance information through regression alignment. The first limit in Proposition 5.2 implies blindness wrt. the mixing weight because every finite ν parametrizes an interior candidate mixing weights bijectively through $\pi(1) = (1 + \tanh \nu)/2$, $\pi(2) = (1 - \tanh \nu)/2$, its vanishing gradient means that increasing component separation eliminates the local first-order signal of the fixed-scale score matching loss for changing mixing weights at the fixed model pair, even though the true mixing weights remain fixed. The second limit in Proposition 5.2 concerns the accompanying latent-variance term: for a nonzero candidate satisfying the projection condition, its collapse records the disappearance of uncertainty about the component assignment.

6 Numerical Experiments

In the section of numerical experiments, we first assess the finite-sample behavior of the path-integrated estimator and then check the two endpoint gradient expansions, the high-noise gradient dynamics, and the pointwise high-SNR blindness result. Appendix E provides the complete configurations of numerical experiments for generating the figures, additional numerical checks, and an auxiliary finite-noise spectral diagnostic.

Under the terminal schedule of Theorem 3.2 and after applying its joint sign alignment, Figure 1 illustrates the predicted parametric scaling and joint calibration of the integrated estimator under the assumptions stated in Theorem 3.2.

Figure 2 checks the componentwise remainder orders in Propositions 4.7 and 4.8 at the fixed model pair. Panel (a) uses the complete low-noise expression, including the EM operator and latent variance, while panel (b) uses the high-noise effective-coefficient expression.

For the displayed admissible initializations, Figure 3 illustrates the polynomial upper bound, geometric decay, and exceptional saddle in Theorem 5.1.

For a one-dimensional specialization of Proposition 5.2, Figure 4 shows the imbalance gradient and posterior-variance term decreasing along the fixed-candidate, fixed-scale ray, thereby illustrating the proposition's pointwise conclusion at one fixed candidate and scale.

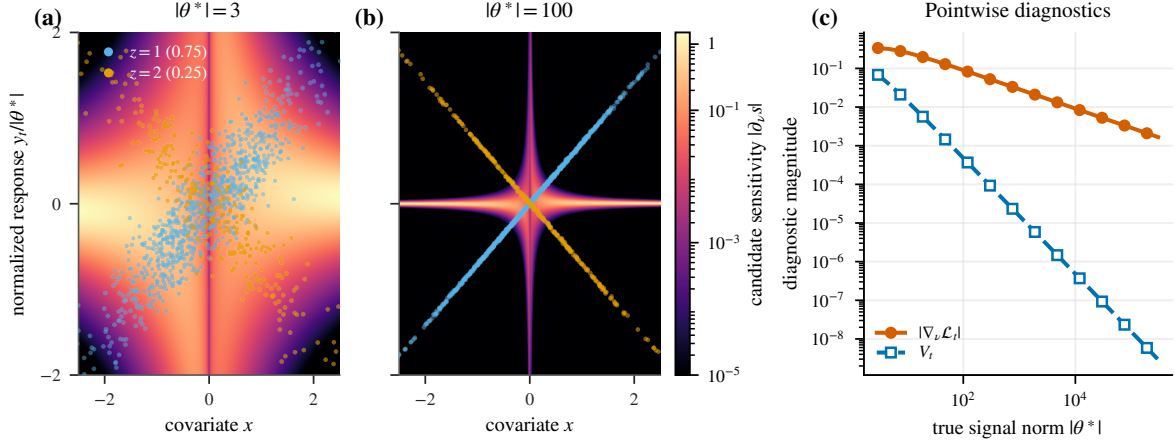


Figure 4: **Score Matching Blindness at High SNR in Proposition 5.2.** The displayed instance uses $d = 1$, $X \sim \mathcal{N}(0, 1)$, $\bar{\alpha}_t = \frac{1}{2}$, the fixed candidate $(\theta, \nu) = (0.80, -0.35)$, and $\nu^* = 0.55$, while $\theta^* > 0$ grows along the positive ray. This setting satisfies Proposition 5.2. (a)–(b) Following Wenliang & Kanagawa (2020), samples from the varying truths are overlaid on the fixed-candidate sensitivity $|\partial_{\nu} s_{\theta_t, \nu}(y_t, x)| = |\mu_t| \text{sech}^2(\mu_t y_t + \nu)$ on a log scale. The normalized responses concentrate on $y_t/||\theta^*|| \simeq \pm \sqrt{\bar{\alpha}_t} x$, and the appreciable sensitivity concentrates near the ray intersection. (c) Quadrature shows that $|nabla_v L_t|$ and the latent variance term $V_t = V(\theta_t, \nu)/\bar{\alpha}_t$ decrease along the fixed ray, providing pointwise evidence at one fixed model and scale. Appendix Figure 6 provides fixed-scale and cross-entropy controls together with an auxiliary moving-candidate diagnostic for a separate setting. At high noise, Proposition 4.8 gives $\frac{1}{2} ||\theta \tanh \nu - c^*||_{\Sigma}^2$, whose directional signal vanishes at balance. At low noise, Proposition 4.7 retains operator-Jacobian and posterior-variance corrections beyond the EM residuals.

7 Conclusions

This paper studies variance-preserving diffusion of the response in two-component mixed linear regression with unknown mixing weights and distinguishes the statistical information accumulated by integrating Score Matching along the diffusion path from the loss geometry and optimization signal available at a fixed scale of the diffusion noise level. For the first question posed in the introduction, Proposition 3.1 shows that, under the stated identifiability conditions, the terminal population KL divergence at every positive retained-signal level has the ground truth as its global minimizers. Main Theorem 3.2 proves that the path-integrated score matching estimator recovers the regression parameters and mixing weights up to their joint-sign symmetry. The same Main Theorem 3.2 establishes asymptotic normality with inverse-Fisher covariance, matching the limiting distribution of the maximum-likelihood estimator.

For the second question, Propositions 4.1 and 4.2 identify the diffused MLR Expectation–Maximization (EM) operators and express the cross-entropy gradients as EM residuals, while Main Theorem 4.3 gives equivalent cross-entropy and EM-related decompositions of the fixed-scale score matching loss. Corollaries 4.4–4.5 and Proposition 4.6 consequently show the decompositions of the fixed-scale score matching gradient as a combination of EM residuals with operator-Jacobian and latent variance terms. The endpoint and signal-strength results explain how noise changes both the optimization dynamics and the parameter information retained by score matching. At low noise, Equation (15) identifies the limiting loss, and Proposition 4.7 shows that the gradient retains the EM-related correction structure. At high noise, Proposition 4.8 shows that the limiting loss depends on the regression vector and mixing imbalance only through their effective coefficient. Under isotropic covariance, Main Theorem 5.1 proves an $\mathcal{O}(K^{-2})$ loss bound when this effective target is zero and eventual geometric decay for generic initializations when it is nonzero, with only a measure-zero exceptional basin attracted to the saddle manifold. Finally, Proposition 5.2 proves that, along a fixed high-SNR ground-truth ray and at each covered fixed candidate and diffusion scale, the imbalance gradient and latent-variance term converge pointwise to zero, eliminating the local first-order signal for changing the candidate mixing weights.

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Appendices

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The appendices are organized as follows.

- Appendix [A](#): We derive the diffused MLR density and score, establish the relative-entropy dissipation and the population and empirical likelihood bridges for path-integrated Score Matching, characterize the terminal KL zero set and its Wasserstein relation, and prove the covariance formulas for the results in Sections [2–6](#).
- Appendix [B](#): Using the likelihood bridges from Appendix [A](#), we prove consistency up to joint sign, the aligned Gaussian limit with inverse-Fisher covariance, and high-probability attainment of the empirical Score Matching minimum for the results in Section [3](#).
- Appendix [C](#): We prove the exact cross-entropy–activation and EM–latent-variance decompositions, derive their associated gradient identities, and establish the low- and high-noise endpoint results from Section [4](#).
- Appendix [D](#): We prove the convergence rates of the high-noise gradient dynamics, establish pointwise Score Matching blindness along fixed high-SNR rays, and derive the hard-assignment EM comparison for the results in Section [5](#).
- Appendix [E](#): We report the numerical configurations, implementation details, calibration calculations, and additional spectral and high-SNR blindness diagnostics for the results in Section [6](#).

A Auxiliary Identities and Bounds for Diffused Mixed Linear Regression

This appendix develops the model-specific identities and bounds used by the main proofs, proceeding from the diffused MLR representation and score formulae to the relative-entropy, terminal-identification, Wasserstein, and fourth-order covariance results.

Lemma A.1 (Equation (4) derived from MLR (2) and SDE (3)). Let y_0 be generated by MLR (2) with fixed parameters θ, π . Suppose that $\varepsilon_0 \sim \mathcal{N}(0, 1)$ is independent of $(\mathbf{x}, z)$. Let y_t be generated by SDE (3) with noise schedule β_t , whose Brownian motion is independent of $(\mathbf{x}, z, \varepsilon_0)$. Then, for every $t > 0$, $p(y_t | y_0, \mathbf{x}; \theta, \pi) = p(y_t | y_0) = \mathcal{N}(y_t; y_0\sqrt{\bar{\alpha}_t}, 1 - \bar{\alpha}_t)$, and Equation (4) holds.

$$y_t = y_0\sqrt{\bar{\alpha}_t} + \xi_t\sqrt{1 - \bar{\alpha}_t} = (-1)^{z+1}\langle\theta_t, \mathbf{x}\rangle + \varepsilon_t = (-1)^{z+1}\mu_t + \varepsilon_t,$$

where $\bar{\alpha}_t := \exp\left(-\int_0^t \beta_\tau d\tau\right)$ is the cumulative noise schedule, $\theta_t := \theta\sqrt{\bar{\alpha}_t}$ is the scaled regression parameter, and $\mu_t := \langle\theta_t, \mathbf{x}\rangle$ is the corresponding mean. The Gaussian noises $\xi_t, \varepsilon_t \sim \mathcal{N}(0, 1)$ are independent of $(\mathbf{x}, z)$.

Lemma A.2 (Negative Log-Likelihood). For the model in Section 2.1, the negative log-likelihood at the scale of the diffusion noise level t is $F(y_t, \mu_t, \nu) := -\ln p(y_t | \mathbf{x}; \theta_t, \pi)$, where $\theta_t = \theta\sqrt{\bar{\alpha}_t}$. It has the form

$$F(y_t, \mu_t, \nu) = \frac{1}{2}(y_t^2 + \mu_t^2) - \ln \frac{\cosh(\mu_t y_t + \nu)}{\cosh \nu} + \frac{1}{2} \ln(2\pi).$$

Lemma A.3 (Stein Score Function). The Stein score is $s_{\theta_t, \nu}(y_t, \mathbf{x}) := \nabla_{y_t} \ln p(y_t | \mathbf{x}; \theta_t, \pi)$. It has the form

$$s_{\theta_t, \nu}(y_t, \mathbf{x}) = -y_t + \tanh(\mu_t y_t + \nu)\mu_t.$$

Proof of Lemma A.1. Applying Itô's formula to $y_t/\sqrt{\bar{\alpha}_t}$, using $d[1/\sqrt{\bar{\alpha}_t}] = \frac{\beta_t}{2\sqrt{\bar{\alpha}_t}}dt$ for $\bar{\alpha}_t := \exp\left(-\int_0^t \beta_\tau d\tau\right)$, and substituting $dy_t = -\frac{\beta_t}{2}y_t dt + \sqrt{\beta_t}dB_t$ from SDE (3) gives

$$d\left[\frac{y_t}{\sqrt{\bar{\alpha}_t}}\right] = \frac{1}{\sqrt{\bar{\alpha}_t}}dy_t + y_t d\left[\frac{1}{\sqrt{\bar{\alpha}_t}}\right] = \sqrt{\frac{\beta_t}{\bar{\alpha}_t}}dB_t.$$

Integrating from 0 to t , setting $\theta_t := \theta\sqrt{\bar{\alpha}_t}$, and substituting $y_0 = (-1)^{z+1}\langle\theta, \mathbf{x}\rangle + \varepsilon_0$ from MLR (2) gives

$$y_t = \sqrt{\bar{\alpha}_t}y_0 + \sqrt{\bar{\alpha}_t} \int_0^t \sqrt{\frac{\beta_\tau}{\bar{\alpha}_\tau}}dB_\tau = (-1)^{z+1}\underbrace{\langle\theta_t, \mathbf{x}\rangle}_{=: \mu_t} + \underbrace{\sqrt{\bar{\alpha}_t} \left[\varepsilon_0 + \int_0^t \sqrt{\frac{\beta_\tau}{\bar{\alpha}_\tau}}dB_\tau \right]}_{=: \varepsilon_t} = (-1)^{z+1}\mu_t + \varepsilon_t.$$

Since $\sqrt{\beta_\tau/\bar{\alpha}_\tau}$ is deterministic, the stochastic integral is a zero-mean Gaussian. The initial noise satisfies $\varepsilon_0 \sim \mathcal{N}(0, 1)$, and the forward Brownian motion is independent of $(\mathbf{x}, z, \varepsilon_0)$. Hence their linear combination ε_t is also a zero-mean Gaussian, with

$$\mathbb{E}[\varepsilon_t] = \sqrt{\bar{\alpha}_t} \left(\mathbb{E}[\varepsilon_0] + \mathbb{E} \left[\int_0^t \sqrt{\frac{\beta_\tau}{\bar{\alpha}_\tau}}dB_\tau \right] \right) = \sqrt{\bar{\alpha}_t} (0 + 0) = 0.$$

Since $\varepsilon_0 \sim \mathcal{N}(0, 1)$ is independent of $\{B_\tau : 0 \leq \tau \leq t\}$, the variance of ε_t is the sum of the two variances. Itô's isometry gives $\text{Var}[\int_0^t \sqrt{\frac{\beta_\tau}{\bar{\alpha}_\tau}}dB_\tau] = \int_0^t \frac{\beta_\tau}{\bar{\alpha}_\tau}d\tau$. The identity $d\left[\frac{1}{\bar{\alpha}_\tau}\right] = \frac{\beta_\tau}{\bar{\alpha}_\tau}d\tau$ then gives $\int_0^t \sqrt{\frac{\beta_\tau}{\bar{\alpha}_\tau}}dB_\tau \sim \mathcal{N}(0, 1/\bar{\alpha}_t - 1)$ and

$$\text{Var}[\varepsilon_t] = \bar{\alpha}_t \left(\text{Var}[\varepsilon_0] + \text{Var} \left[\int_0^t \sqrt{\frac{\beta_\tau}{\bar{\alpha}_\tau}}dB_\tau \right] \right) = \bar{\alpha}_t \left(1 + \int_0^t \frac{\beta_\tau}{\bar{\alpha}_\tau}d\tau \right) = \bar{\alpha}_t \left(1 + \frac{1}{\bar{\alpha}_\tau} \Big|_0^t \right) = 1.$$

Thus, $\varepsilon_t \sim \mathcal{N}(0, 1)$ and $p(y_t | y_0, \mathbf{x}; \theta, \pi) = p(y_t | y_0) = \mathcal{N}(y_t; y_0\sqrt{\bar{\alpha}_t}, 1 - \bar{\alpha}_t)$. For $t > 0$, it follows that $\xi_t := (y_t - y_0\sqrt{\bar{\alpha}_t})/\sqrt{1 - \bar{\alpha}_t} = \sqrt{\frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t}} \int_0^t \sqrt{\frac{\beta_\tau}{\bar{\alpha}_\tau}}dB_\tau \sim \mathcal{N}(0, 1)$. The assumed independence of ε_0 , the Brownian motion, and $(\mathbf{x}, z)$ shows that ξ_t and ε_t are independent of $(\mathbf{x}, z)$. $\square$

Proofs of Lemmas A.2 and A.3. We first obtain the conditional density of y_t by marginalizing the joint density over $z \in \{1, 2\}$. The model assumptions give $\mathbb{P}(z \mid \mathbf{x}; \theta_t, \pi) = \mathbb{P}(z \mid \pi) = \pi(z)$. Hence

$$F(y_t, \mu_t, \nu) := -\ln p(y_t \mid \mathbf{x}; \theta_t, \pi) = -\ln \sum_{z \in \{1, 2\}} p(y_t, z \mid \mathbf{x}; \theta_t, \pi) = -\ln \sum_{z \in \{1, 2\}} \pi(z) \cdot p(y_t \mid z, \mathbf{x}; \theta_t, \pi).$$

Lemma A.1 gives $\varepsilon_t \sim \mathcal{N}(0, 1)$, independent of $(\mathbf{x}, z)$. Hence $p(y_t \mid z, \mathbf{x}; \theta_t, \pi) = p(y_t \mid z, \mu_t) = \mathcal{N}((-1)^{z+1} \mu_t, 1)$, where $\mu_t = \langle \theta_t, \mathbf{x} \rangle$.

$$F(y_t, \mu_t, \nu) = -\ln \sum_{z \in \{1, 2\}} \pi(z) \cdot p(y_t \mid z, \mu_t) = -\ln \sum_{z \in \{1, 2\}} \pi(z) \cdot \mathcal{N}((-1)^{z+1} \mu_t, 1).$$

The imbalance parameter $\nu := (\ln \pi(1) - \ln \pi(2))/2$ is in one-to-one correspondence with the mixing weights. In particular, $\pi(1) = \frac{e^\nu}{2 \cosh \nu}$ and $\pi(2) = \frac{e^{-\nu}}{2 \cosh \nu}$. Substitution gives

$$\begin{aligned} F(y_t, \mu_t, \nu) &= -\ln \left(\frac{e^\nu}{2 \cosh \nu} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y_t^2 - 2y_t \mu_t + \mu_t^2)} + \frac{e^{-\nu}}{2 \cosh \nu} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y_t^2 + 2y_t \mu_t + \mu_t^2)} \right) \\ &= \frac{1}{2}(y_t^2 + \mu_t^2) - \ln \left(e^{\nu + y_t \mu_t} + e^{-(\nu + y_t \mu_t)} \right) + \ln(2 \cosh \nu) + \frac{1}{2} \ln(2\pi) \\ &= \frac{1}{2}(y_t^2 + \mu_t^2) - \ln \frac{\cosh(\nu + y_t \mu_t)}{\cosh \nu} + \frac{1}{2} \ln(2\pi). \end{aligned}$$

For Lemma A.3, write ∇ for ∇_{y_t} . The Stein score is the negative gradient of F . Since μ_t , ν , and $\frac{1}{2} \ln(2\pi)$ do not depend on y_t , the chain rule gives

$$s_{\theta_t, \nu}(y_t, \mathbf{x}) := \nabla \ln p(y_t \mid \mathbf{x}; \theta_t, \pi) = -\nabla F(y_t, \mu_t, \nu) = -(y_t - \nabla \ln \cosh(\mu_t y_t + \nu)) = -y_t + \tanh(\mu_t y_t + \nu) \mu_t.$$

□

Lemma A.4 (Conditional Relative Entropy Dissipation). The derivative of the KL divergence between the true and model distributions satisfies

$$-\frac{d}{dt} \text{KL}(p(y_t \mid \mathbf{x}; \theta_t^*, \pi^*) \parallel p(y_t \mid \mathbf{x}; \theta_t, \pi)) = \frac{\beta_t}{2} \mathbb{E}_{y_t \mid \mathbf{x}; \theta_t^*, \pi^*} \|\nabla \ln p(y_t \mid \mathbf{x}; \theta_t, \pi) - \nabla \ln p(y_t \mid \mathbf{x}; \theta_t^*, \pi^*)\|^2.$$

Lemma A.5 (Log-likelihood Derivative and Score Matching Loss). Let $\mathbb{P}_n$ be the empirical measure defined before Equation (8). Then, for every $t > 0$,

$$\frac{d}{dt} \mathbb{E}_{\mathbb{P}_n} \mathbb{E}_{y_t \mid y_0} \ln p(y_t \mid \mathbf{x}; \theta_t, \pi) = \frac{\beta_t}{2} \mathbb{E}_{\mathbb{P}_n} \mathbb{E}_{y_t \mid y_0} [\|\nabla \ln p(y_t \mid \mathbf{x}; \theta_t, \pi)\|^2 - 2\langle \nabla \ln p(y_t \mid \mathbf{x}; \theta_t, \pi), \nabla \ln p(y_t \mid y_0) \rangle + 1].$$

Proof of Lemma A.4. Let $p_t := p(y_t \mid \mathbf{x}; \theta_t, \pi)$ denote the model density, and let $p_t^* := p(y_t \mid \mathbf{x}; \theta_t^*, \pi^*)$ denote the ground truth density. The negative time derivative of the KL divergence is

$$-\frac{d}{dt} \text{KL}(p_t^* \parallel p_t) = -\frac{d}{dt} \mathbb{E}_{p_t^*} \ln \frac{p_t^*}{p_t} = \frac{d}{dt} \mathbb{E}_{p_t^*} [\ln p_t] - \frac{d}{dt} \mathbb{E}_{p_t^*} [\ln p_t^*].$$

The forward Kolmogorov operator for SDE (3) is $\mathcal{A}^*[f] := -\nabla \cdot \left[f \cdot \left(-\frac{\beta_t}{2} y_t \right) \right] + \frac{1}{2} \Delta [f \cdot (\sqrt{\beta_t})^2] = \frac{\beta_t}{2} [\nabla \cdot (f y_t) + \Delta f]$. Lemma A.1 gives the transition density $p(y_t \mid y_0) = \mathcal{N}(y_t; y_0 \sqrt{\bar{\alpha}_t}, 1 - \bar{\alpha}_t)$. Its Fokker–Planck–Kolmogorov equation is

$$\partial_t p(y_t \mid y_0) = \frac{\beta_t}{2} [\nabla \cdot (p(y_t \mid y_0) y_t) + \Delta p(y_t \mid y_0)] \equiv \mathcal{A}^*[p(y_t \mid y_0)].$$

The Chapman–Kolmogorov relation uses the fixed initial law and gives

$$p_t = \mathbb{E}_{y_0 \mid \mathbf{x}; \pi} [p(y_t \mid y_0)] = \sum_{k=1}^2 \pi(k) \mathcal{N}(y_t; (-1)^{k+1} \mu_0 \sqrt{\bar{\alpha}_t}, 1).$$

Direct differentiation shows that each of these two Gaussian components satisfies the FPK equation above, and linearity of the forward operator therefore gives

$$\partial_t p_t = \mathcal{A}^*[p_t] = \frac{\beta_t}{2} [\nabla \cdot (p_t y_t) + \Delta p_t].$$

Dividing by p_t and using identities $\frac{\nabla \cdot (p_t y_t)}{p_t} = 1 + \langle \nabla \ln p_t, y_t \rangle$, $\frac{\Delta p_t}{p_t} = \Delta \ln p_t + \|\nabla \ln p_t\|^2$, we obtain the derivative of the log-density.

$$\partial_t \ln p_t = \frac{\beta_t}{2} [1 + \langle \nabla \ln p_t, y_t \rangle + \Delta \ln p_t + \|\nabla \ln p_t\|^2].$$

Applying Itô's formula to $\ln p_t$ under the forward SDE gives

$$\begin{aligned} d \ln p_t &= \partial_t \ln p_t dt + \langle \nabla \ln p_t, dy_t \rangle + \frac{1}{2} \Delta \ln p_t (dy_t)^2 \\ &= \partial_t \ln p_t dt + \left\langle \nabla \ln p_t, -\frac{\beta_t}{2} y_t dt + \sqrt{\beta_t} dB_t \right\rangle + \frac{\beta_t}{2} \Delta \ln p_t dt \\ &= \frac{\beta_t}{2} [\|\nabla \ln p_t\|^2 + 2\Delta \ln p_t + 1] dt + \sqrt{\beta_t} \langle \nabla \ln p_t, dB_t \rangle. \end{aligned}$$

Under the ground truth path measure $\mathbb{P}^*$, the state at each $\tau \in [0, t]$ has marginal density p_τ^* . Fubini's theorem moves the expectation inside the time integral, and the Brownian stochastic integral has mean zero.

$$\begin{aligned} \mathbb{E}_{p_t^*}[\ln p_t] - \mathbb{E}_{p_0^*}[\ln p_0] &= \mathbb{E}_{\mathbb{P}^*} \left[\int_0^t \frac{\beta_\tau}{2} [\|\nabla \ln p_\tau\|^2 + 2\Delta \ln p_\tau + 1] d\tau \right] \\ &= \int_0^t \frac{\beta_\tau}{2} \mathbb{E}_{p_\tau^*} [\|\nabla \ln p_\tau\|^2 + 2\Delta \ln p_\tau + 1] d\tau. \end{aligned}$$

The fundamental theorem of calculus gives

$$\frac{d}{dt} \mathbb{E}_{p_t^*}[\ln p_t] = \frac{\beta_t}{2} \mathbb{E}_{p_t^*} [\|\nabla \ln p_t\|^2 + 2\Delta \ln p_t + 1].$$

Integration by parts and the vanishing boundary terms give

$$\mathbb{E}_{p_t^*}[\Delta \ln p_t] = \mathbb{E}_{p_t^*}[\nabla \cdot (\nabla \ln p_t)] = -\mathbb{E}_{p_t^*}[\langle \nabla \ln p_t^*, \nabla \ln p_t \rangle].$$

Substitution gives

$$\frac{d}{dt} \mathbb{E}_{p_t^*}[\ln p_t] = \frac{\beta_t}{2} \mathbb{E}_{p_t^*} [\|\nabla \ln p_t\|^2 - 2\langle \nabla \ln p_t^*, \nabla \ln p_t \rangle + 1].$$

Applying the same calculation to the ground truth distribution gives

$$\frac{d}{dt} \mathbb{E}_{p_t^*}[\ln p_t^*] = \frac{\beta_t}{2} \mathbb{E}_{p_t^*} [\|\nabla \ln p_t^*\|^2 - 2\|\nabla \ln p_t^*\|^2 + 1] = \frac{\beta_t}{2} \mathbb{E}_{p_t^*} [-\|\nabla \ln p_t^*\|^2 + 1].$$

Subtracting the entropy derivative from the cross-entropy derivative gives

$$-\frac{d}{dt} \text{KL}(p_t^* \parallel p_t) = \frac{d}{dt} \mathbb{E}_{p_t^*}[\ln p_t] - \frac{d}{dt} \mathbb{E}_{p_t^*}[\ln p_t^*] = \frac{\beta_t}{2} \mathbb{E}_{p_t^*} [\|\nabla \ln p_t - \nabla \ln p_t^*\|^2].$$

Restoring the full notation $p_t \equiv p(y_t | \mathbf{x}; \theta_t, \pi)$ and $p_t^* \equiv p(y_t | \mathbf{x}; \theta_t^*, \pi^*)$ yields the lemma statement. $\square$

Proof of Lemma A.5. We apply the proof of Lemma A.4 with p_t^* replaced by $p(y_t | y_0)$. This gives

$$\frac{d}{dt} \mathbb{E}_{y_t | y_0}[\ln p_t] = \frac{\beta_t}{2} \mathbb{E}_{y_t | y_0} [\|\nabla \ln p_t\|^2 - 2\langle \nabla \ln p_t, \nabla \ln p(y_t | y_0) \rangle + 1].$$

Taking the expectation over $\mathbb{P}_n$, interchanging $\mathbb{E}_{\mathbb{P}_n}$ and $\frac{d}{dt}$, and restoring $p_t \equiv p(y_t | \mathbf{x}; \theta_t, \pi)$ proves the result. $\square$

Lemma A.6 (Decaying KL Divergence Bound). Suppose $0 < \bar{\alpha}_t < 1$. Let the two conditional diffusions start from the initial probability laws $p(y_0|\mathbf{x}; \theta^*, \pi^*)$ and $p(y_0|\mathbf{x}; \theta, \pi)$. Their time- t laws satisfy

$$\text{KL}(p(y_t|\mathbf{x}; \theta_t^*, \pi^*) \parallel p(y_t|\mathbf{x}; \theta_t, \pi)) \leq \frac{\bar{\alpha}_t}{2(1 - \bar{\alpha}_t)} W_2^2(p(y_0|\mathbf{x}; \theta^*, \pi^*), p(y_0|\mathbf{x}; \theta, \pi)).$$

Lemma A.7 (Wasserstein Distance for Conditional Distributions). Fix $\mathbf{x}$. Define the two discrete distributions $D := D(\mathbf{x}; \theta, \pi) \equiv \pi(1)\delta_{\langle \theta, \mathbf{x} \rangle} + \pi(2)\delta_{-\langle \theta, \mathbf{x} \rangle}$ and $D^* := D(\mathbf{x}; \theta^*, \pi^*) \equiv \pi^*(1)\delta_{\langle \theta^*, \mathbf{x} \rangle} + \pi^*(2)\delta_{-\langle \theta^*, \mathbf{x} \rangle}$. Then the squared 2-Wasserstein distance between $p(y_0 | \mathbf{x}; \theta^*, \pi^*)$ and $p(y_0 | \mathbf{x}; \theta, \pi)$ is upper bounded by the squared 2-Wasserstein distance between the discrete distributions.

$$W_2^2(p(y_0 | \mathbf{x}; \theta^*, \pi^*), p(y_0 | \mathbf{x}; \theta, \pi)) \leq W_2^2(D, D^*).$$

Lemma A.8 (Decaying KL Divergence w.r.t. Transition Kernel). Suppose $0 < \bar{\alpha}_t < 1$. Let $p(y_t | y_0)$ be the transition kernel at time t from a fixed initial value y_0 . Set $\mu_0 = \langle \theta, \mathbf{x} \rangle$ and $\nu = (\ln \pi(1) - \ln \pi(2))/2$. Then

$$\text{KL}(p(y_t | y_0) \parallel p(y_t | \mathbf{x}; \theta_t, \pi)) \leq \frac{\bar{\alpha}_t}{2(1 - \bar{\alpha}_t)} (y_0^2 + \mu_0^2 - 2y_0\mu_0 \tanh \nu + 1).$$

Proof of Lemma A.6. For any fixed $u \in \mathbb{R}$, Lemma A.1 gives the transition density

$$q_t(y_t | u) := \mathcal{N}(y_t; u\sqrt{\bar{\alpha}_t}, 1 - \bar{\alpha}_t).$$

Define the initial probability laws $p_0^* := p(y_0 | \mathbf{x}; \theta^*, \pi^*)$ and $p_0 := p(y_0 | \mathbf{x}; \theta, \pi)$. We use $\tilde{y}_0$ for a value under p_0^* and y_0 for a value under p_0 . Since the two transition densities have variance $1 - \bar{\alpha}_t$, the Gaussian KL formula gives

$$\text{KL}(q_t(\cdot | \tilde{y}_0) \parallel q_t(\cdot | y_0)) = \frac{\bar{\alpha}_t}{2(1 - \bar{\alpha}_t)} \|\tilde{y}_0 - y_0\|^2.$$

Let $p_t := p(y_t|\mathbf{x}; \theta_t, \pi)$ and $p_t^* := p(y_t|\mathbf{x}; \theta_t^*, \pi^*)$. Since $\theta_t = \theta\sqrt{\bar{\alpha}_t}$, we have

$$p_t = \mathbb{E}_{y_0 \sim p_0} [q_t(y_t | y_0)], \quad p_t^* = \mathbb{E}_{\tilde{y}_0 \sim p_0^*} [q_t(y_t | \tilde{y}_0)].$$

Let $\mathcal{P}(\mathbb{R})$ denote the space of probability measures on $\mathbb{R}$. For $p_0^*, p_0 \in \mathcal{P}(\mathbb{R})$, let $\Gamma(p_0^*, p_0)$ be the set of joint probability measures on $\mathbb{R} \times \mathbb{R}$ with marginals p_0^* and p_0 . Thus $\gamma \in \Gamma(p_0^*, p_0)$ if and only if every pair of Borel sets $A, B \subseteq \mathbb{R}$ satisfies $\gamma(A \times \mathbb{R}) = p_0^*(A)$, $\gamma(\mathbb{R} \times B) = p_0(B)$. Theorem 4.1 of Villani (2009) shows that a lower semicontinuous quadratic cost bounded from below attains its infimum. Thus an optimal coupling $\gamma^* \in \Gamma(p_0^*, p_0)$ satisfies

$$\mathbb{E}_{(\tilde{y}_0, y_0) \sim \gamma^*} [\|\tilde{y}_0 - y_0\|^2] = \inf_{\gamma \in \Gamma(p_0^*, p_0)} \mathbb{E}_{(\tilde{y}_0, y_0) \sim \gamma} [\|\tilde{y}_0 - y_0\|^2] =: W_2^2(p_0^*, p_0).$$

The optimal coupling γ^* expresses both marginals p_t^* and p_t as expectations on the same probability space.

$$p_t^* = \mathbb{E}_{(\tilde{y}_0, y_0) \sim \gamma^*} [q_t(y_t | \tilde{y}_0)], \quad p_t = \mathbb{E}_{(\tilde{y}_0, y_0) \sim \gamma^*} [q_t(y_t | y_0)].$$

The integral log-sum inequality is the continuous-density form of joint convexity of relative entropy (Cover & Thomas, 2006, Theorem 2.7.2). It moves the expectation outside the divergence and gives the following upper bound.

$$\text{KL}(p_t^* \parallel p_t) \leq \mathbb{E}_{(\tilde{y}_0, y_0) \sim \gamma^*} \left[\text{KL}(q_t(\cdot | \tilde{y}_0) \parallel q_t(\cdot | y_0)) \right].$$

Substituting the conditional KL divergence derived above gives the stated upper bound.

$$\text{KL}(p_t^* \parallel p_t) \leq \mathbb{E}_{(\tilde{y}_0, y_0) \sim \gamma^*} \left[\frac{\bar{\alpha}_t}{2(1 - \bar{\alpha}_t)} \|\tilde{y}_0 - y_0\|^2 \right] = \frac{\bar{\alpha}_t}{2(1 - \bar{\alpha}_t)} W_2^2(p_0^*, p_0).$$

Restoring the full notation for the four probability laws yields the lemma statement. $\square$

Proof of Lemma A.7. Fix $\mathbf{x}$ and both parameter pairs. Write $p_0 := p(y_0 | \mathbf{x}; \theta, \pi)$ and $p_0^* := p(y_0 | \mathbf{x}; \theta^*, \pi^*)$, and let D, D^* be the discrete laws in the lemma statement. Set $\bar{y}_0 = (-1)^{z+1} \langle \theta, \mathbf{x} \rangle$ in Equation (4). Lemma A.1 gives $y_0 = \bar{y}_0 + \varepsilon_0$, with $\bar{y}_0 \perp \varepsilon_0$ and $\varepsilon_0 \sim \eta := \mathcal{N}(0, 1)$, and hence

$$\bar{y}_0 \sim D, \quad y_0 = \bar{y}_0 + \varepsilon_0 \sim p_0 = D * \eta.$$

For the ground truth law, let $y_0^* = \bar{y}_0^* + \varepsilon_0^*$, where $\bar{y}_0^* \sim D^*$, $\varepsilon_0^* \sim \eta$, and the two terms are independent. Thus

$$y_0^* = \bar{y}_0^* + \varepsilon_0^* \sim p_0^* = D^* * \eta.$$

Since η is an even regularizing kernel, Lemma 5.2 of Santambrogio (2015) gives

$$W_2^2(p_0^*, p_0) = W_2^2(D^* * \eta, D * \eta) \leq W_2^2(D^*, D) = W_2^2(D, D^*).$$

This establishes the desired upper bound and completes the proof. $\square$

Proof of Lemma A.8. Let u denote a dummy initial value. We write $p_t := p(y_t | \mathbf{x}; \theta_t, \pi) = \mathbb{E}_{u | \mathbf{x}; \theta, \pi} [p(y_t | u)]$ and $p(y_t | y_0) = \mathbb{E}_{u \sim \delta_{y_0}} [p(y_t | u)]$. Lemma A.6 applies with the two initial laws $p(u | \mathbf{x}; \theta, \pi)$ and δ_{y_0} . It gives

$$\text{KL}(p(y_t | y_0) \parallel p(y_t | \mathbf{x}; \theta_t, \pi)) \leq \frac{\bar{\alpha}_t}{2(1 - \bar{\alpha}_t)} W_2^2(\delta_{y_0}, p(u | \mathbf{x}; \theta, \pi)).$$

The squared 2-Wasserstein distance between δ_{y_0} and the two-Gaussian mixture is

$$\begin{aligned} W_2^2(\delta_{y_0}, p(u | \mathbf{x}; \theta, \pi)) &= \mathbb{E}_{u | \mathbf{x}; \theta, \pi} [(y_0 - u)^2] \\ &= \pi(1) [(y_0 - \mu_0)^2 + 1] + \pi(2) [(y_0 + \mu_0)^2 + 1] = y_0^2 + \mu_0^2 - 2y_0\mu_0 \tanh \nu + 1. \end{aligned}$$

The first equality holds because a point mass has one coupling with the second marginal. The second equality uses $u = (-1)^{z+1} \mu_0 + \varepsilon_0$ and $\mathbb{E}[\varepsilon_0^2] = 1$. $\square$

Lemma A.9 (Two Losses of Stein Score Function). Set $\theta_t = \theta \sqrt{\bar{\alpha}_t}$ and $\nu = (\ln \pi(1) - \ln \pi(2))/2$. Define the candidate score by

$$s_{\theta_t, \nu}(y_t, \mathbf{x}) := \nabla_{y_t} \ln p(y_t | \mathbf{x}; \theta_t, \pi).$$

Set $\theta_t^* = \theta^* \sqrt{\bar{\alpha}_t}$ and $\nu^* = (\ln \pi^*(1) - \ln \pi^*(2))/2$. Define the ground truth score by

$$s_{\theta_t^*, \nu^*}(y_t, \mathbf{x}) := \nabla_{y_t} \ln p(y_t | \mathbf{x}; \theta_t^*, \pi^*).$$

Then

$$\begin{aligned} &\mathbb{E}_{y_t | \mathbf{x}; \theta_t^*, \pi^*} \|s_{\theta_t, \nu}(y_t, \mathbf{x}) - s_{\theta_t^*, \nu^*}(y_t, \mathbf{x})\|^2 \\ &= \mathbb{E}_{y_0 | \mathbf{x}; \theta^*, \pi^*} \mathbb{E}_{y_t | y_0} [\|s_{\theta_t, \nu}(y_t, \mathbf{x})\|^2] \\ &\quad - 2\mathbb{E}_{y_0 | \mathbf{x}; \theta^*, \pi^*} \mathbb{E}_{y_t | y_0} [\langle s_{\theta_t, \nu}(y_t, \mathbf{x}), \nabla \ln p(y_t | y_0) \rangle] + C(\mu_t^*, \nu^*). \end{aligned}$$

Here $C(\mu_t^*, \nu^*) := 1 - [\mu_t^*]^2 \mathbb{E}_{y_t | \mathbf{x}; \theta_t^*, \pi^*} [\text{sech}^2(\mu_t^* y_t + \nu^*)] \geq 0$ depends on $\mu_t^* = \langle \theta_t^*, \mathbf{x} \rangle$ and ν^* alone.

Proof. Differentiating the Gaussian-mixture marginal and applying Bayes' rule gives the denoising score identity, with $\nabla := \nabla_{y_t}$,

$$s_{\theta_t^*, \nu^*}(y_t, \mathbf{x}) \equiv \nabla \ln p(y_t | \mathbf{x}; \theta_t^*, \pi^*) = \mathbb{E}[\nabla \ln p(y_t | y_0) | y_t, \mathbf{x}; \theta_t^*, \pi^*].$$

The tower property therefore rewrites the cross-term as

$$\mathbb{E}_{y_t | \mathbf{x}; \theta_t^*, \pi^*} [\langle s_{\theta_t, \nu}, s_{\theta_t^*, \nu^*} \rangle] = \mathbb{E}_{y_0 | \mathbf{x}; \theta^*, \pi^*} \mathbb{E}_{y_t | y_0} [\langle s_{\theta_t, \nu}, \nabla \ln p(y_t | y_0) \rangle].$$

Expanding the squared score difference and applying this cross-term identity gives

$$\begin{aligned} C &:= \mathbb{E}_{y_t | \mathbf{x}; \theta_t^*, \pi^*} [\|s_{\theta_t, \nu} - s_{\theta_t^*, \nu^*}\|^2] - \mathbb{E}_{y_0 | \mathbf{x}; \theta^*, \pi^*} \mathbb{E}_{y_t | y_0} [\|s_{\theta_t, \nu}\|^2] \\ &\quad + 2\mathbb{E}_{y_0 | \mathbf{x}; \theta^*, \pi^*} \mathbb{E}_{y_t | y_0} [\langle s_{\theta_t, \nu}, \nabla \ln p(y_t | y_0) \rangle] \\ &= \mathbb{E}_{y_t | \mathbf{x}; \theta_t^*, \pi^*} [\|s_{\theta_t^*, \nu^*}(y_t, \mathbf{x})\|^2], \end{aligned}$$

where suppressed score arguments in the first two lines are $(y_t, \mathbf{x})$. Integration by parts, the vanishing boundary term, and Lemma A.3 give

$$\mathbb{E}_{y_t|\mathbf{x};\theta_t^*,\pi^*}[\|s_{\theta_t^*,\nu^*}(y_t, \mathbf{x})\|^2] = -\mathbb{E}_{y_t|\mathbf{x};\theta_t^*,\pi^*}[\nabla s_{\theta_t^*,\nu^*}(y_t, \mathbf{x})] = 1 - [\mu_t^*]^2 \mathbb{E}_{y_t|\mathbf{x};\theta_t^*,\pi^*}[\text{sech}^2(\mu_t^* y_t + \nu^*)].$$

Lemma A.2 shows that the conditional density is determined by y_t , μ_t^* , and ν^* . After integrating over y_t , C therefore depends on μ_t^* and ν^* alone. $\square$

Lemma A.10 (Asymptotic Behavior of the Scaled Score Matching Loss). Let y_0 be generated by MLR (2) under the ground truth parameters θ^*, π^* . Let y_t follow SDE (3) at time $t \geq 0$, with cumulative noise schedule $\bar{\alpha}_t \in (0, 1]$. Let $\mu_0 := \langle \theta, \mathbf{x} \rangle$ and $\mu_0^* := \langle \theta^*, \mathbf{x} \rangle$ denote the estimated and ground truth means at $t = 0$, such that the means at time t are $\mu_t = \mu_0 \sqrt{\bar{\alpha}_t}$ and $\mu_t^* = \mu_0^* \sqrt{\bar{\alpha}_t}$. Let $\nu := (\ln \pi(1) - \ln \pi(2))/2$ and $\nu^* := (\ln \pi^*(1) - \ln \pi^*(2))/2$ denote the estimated and ground truth imbalance parameters of the mixing weights. Then

$$\begin{aligned} \lim_{\bar{\alpha}_t \rightarrow 0} \frac{1}{\bar{\alpha}_t} \mathbb{E}_{y_t|\mathbf{x};\theta_t^*,\pi^*}[\|s_{\theta_t,\nu}(y_t, \mathbf{x}) - s_{\theta_t^*,\nu^*}(y_t, \mathbf{x})\|^2] &= (\mu_0 \tanh \nu - \mu_0^* \tanh \nu^*)^2, \\ \lim_{\bar{\alpha}_t \rightarrow 1} \frac{1}{\bar{\alpha}_t} \mathbb{E}_{y_t|\mathbf{x};\theta_t^*,\pi^*}[\|s_{\theta_t,\nu}(y_t, \mathbf{x}) - s_{\theta_t^*,\nu^*}(y_t, \mathbf{x})\|^2] &= \mathbb{E}_{y_0|\mathbf{x};\theta^*,\pi^*} [(\mu_0 \tanh(\mu_0 y_0 + \nu) - \mu_0^* \tanh(\mu_0^* y_0 + \nu^*))^2]. \end{aligned}$$

Proof of Lemma A.10. Using the Stein score function $s_{\theta_t,\nu}(y_t, \mathbf{x}) = -y_t + \mu_t \tanh(\mu_t y_t + \nu)$ established in Lemma A.3, the difference between the estimated and ground truth score functions cancels the linear $-y_t$ terms. We place the marginals at all scales of the diffusion noise level on a common probability space. Let $\bar{y}_0^* := (-1)^{z+1} \mu_0^*$, and let $\varepsilon \sim \mathcal{N}(0, 1)$ be independent of z . At each scale, $\sqrt{\bar{\alpha}_t} \bar{y}_0^* + \varepsilon$ has the same law as $y_t | \mathbf{x}; \theta_t^*, \pi^*$. Substituting this representation and $\mu_t = \mu_0 \sqrt{\bar{\alpha}_t}$ gives

$$\frac{1}{\bar{\alpha}_t} \|s_{\theta_t,\nu}(y_t, \mathbf{x}) - s_{\theta_t^*,\nu^*}(y_t, \mathbf{x})\|^2 = \underbrace{(\mu_0 \tanh(\bar{\alpha}_t \mu_0 \bar{y}_0^* + \sqrt{\bar{\alpha}_t} \mu_0 \varepsilon + \nu) - \mu_0^* \tanh(\bar{\alpha}_t \mu_0^* \bar{y}_0^* + \sqrt{\bar{\alpha}_t} \mu_0^* \varepsilon + \nu^*))^2}_{=: \ell_{\mu_0, \mu_0^*, \nu, \nu^*}(\bar{\alpha}_t, \bar{y}_0^*, \varepsilon)}.$$

The triangle inequality and $|\tanh(u)| \leq 1$ give the following bound for each fixed $\mathbf{x}$.

$$|\ell_{\mu_0, \mu_0^*, \nu, \nu^*}(\bar{\alpha}_t, \bar{y}_0^*, \varepsilon)| \leq (|\mu_0| + |\mu_0^*|)^2 < \infty \implies \mathbb{E}_{y_t|\mathbf{x};\theta_t^*,\pi^*} \left[\frac{1}{\bar{\alpha}_t} \|s_{\theta_t,\nu}(y_t, \mathbf{x}) - s_{\theta_t^*,\nu^*}(y_t, \mathbf{x})\|^2 \right] < \infty.$$

This integrable bound permits the limit to pass through the expectation by the dominated convergence theorem (Durrett, 2019, Theorem 1.5.8). The expectation over the common probability space is the iterated expectation over $\bar{y}_0^*$ and ε . For each $c \in \{0, 1\}$, the dominated convergence theorem gives

$$\lim_{\bar{\alpha}_t \rightarrow c} \mathbb{E}_{y_t|\mathbf{x};\theta_t^*,\pi^*} \left[\frac{1}{\bar{\alpha}_t} \|s_{\theta_t,\nu}(y_t, \mathbf{x}) - s_{\theta_t^*,\nu^*}(y_t, \mathbf{x})\|^2 \right] = \mathbb{E}_{\bar{y}_0^*|\mu_0^*,\nu^*} \mathbb{E}_\varepsilon \left[\lim_{\bar{\alpha}_t \rightarrow c} \ell_{\mu_0, \mu_0^*, \nu, \nu^*}(\bar{\alpha}_t, \bar{y}_0^*, \varepsilon) \right].$$

The pointwise limit for $c = 0$ is

$$\lim_{\bar{\alpha}_t \rightarrow 0} \ell_{\mu_0, \mu_0^*, \nu, \nu^*}(\bar{\alpha}_t, \bar{y}_0^*, \varepsilon) = (\mu_0 \tanh \nu - \mu_0^* \tanh \nu^*)^2.$$

The pointwise limit for $c = 1$ is

$$\lim_{\bar{\alpha}_t \rightarrow 1} \ell_{\mu_0, \mu_0^*, \nu, \nu^*}(\bar{\alpha}_t, \bar{y}_0^*, \varepsilon) = (\mu_0 \tanh(\mu_0(\bar{y}_0^* + \varepsilon) + \nu) - \mu_0^* \tanh(\mu_0^*(\bar{y}_0^* + \varepsilon) + \nu^*))^2.$$

Since $\bar{y}_0^* + \varepsilon$ has the ground truth law of $y_0 | \mathbf{x}; \theta^*, \pi^*$, the last display gives the second equality. $\square$

Lemma A.11 (Wasserstein-2 Distance between Two Discrete Distributions). Fix $\mathbf{x}$. Define the discrete distributions $D \equiv D(\mathbf{x}; \theta, \pi) := \pi(1)\delta_{\langle \theta, \mathbf{x} \rangle} + \pi(2)\delta_{-\langle \theta, \mathbf{x} \rangle}$ and $D^* \equiv D(\mathbf{x}; \theta^*, \pi^*) := \pi^*(1)\delta_{\langle \theta^*, \mathbf{x} \rangle} + \pi^*(2)\delta_{-\langle \theta^*, \mathbf{x} \rangle}$. Then for $\mu_0 \equiv \langle \theta, \mathbf{x} \rangle$, $\mu_0^* \equiv \langle \theta^*, \mathbf{x} \rangle$ and $\nu \equiv (\ln \pi(1) - \ln \pi(2))/2$, $\nu^* \equiv (\ln \pi^*(1) - \ln \pi^*(2))/2$, we have

$$W_2^2(D, D^*) = (|\mu_0| - |\mu_0^*|)^2 + 2|\mu_0||\mu_0^*| |\text{sgn}(\mu_0) \tanh \nu - \text{sgn}(\mu_0^*) \tanh \nu^*|.$$

Proof of Lemma A.11. We adopt $\text{sgn}(0) = 0$ and set

$$a := |\mu_0|, \quad a^* := |\mu_0^*|, \quad q := \text{sgn}(\mu_0) \tanh \nu, \quad q^* := \text{sgn}(\mu_0^*) \tanh \nu^*.$$

If $aa^* = 0$, the claimed identity follows. Otherwise, D is the law of aS , where $S \in \{-1, 1\}$, $\mathbb{P}(S = 1) = (1 + q)/2$, and $\tanh \nu = \pi(1) - \pi(2)$. Likewise, D^* is the law of a^*S^* with $\mathbb{P}(S^* = 1) = (1 + q^*)/2$. For any coupling of S and S^* ,

$$\mathbb{E}[(aS - a^*S^*)^2] = a^2 + (a^*)^2 - 2aa^*\mathbb{E}[SS^*].$$

The maximal coupling has $\mathbb{P}(S \neq S^*) = |q - q^*|/2$, and hence $\mathbb{E}[SS^*] = 1 - |q - q^*|$. It therefore minimizes the quadratic cost, giving

$$W_2^2(D, D^*) = (a - a^*)^2 + 2aa^*|q - q^*|,$$

which is the stated formula. $\square$

Lemma A.12 (Expectations of Gaussians, Lemma E.2 of [Luo & Hashemi \(2025\)](#)). We use the following identities from Lemma E.2 of [Luo & Hashemi \(2025\)](#). Let g and g' be standard Gaussian variables with $\mathbb{E}[gg'] = \sin \varphi$, where $\varphi \in [-\pi/2, \pi/2]$. Then

$$\mathbb{E}[\text{sgn}(gg')] = \frac{2}{\pi} \varphi, \quad \mathbb{E}[|gg'|] = \frac{2}{\pi} [\varphi \sin \varphi + \cos \varphi], \quad \mathbb{E}[g^2 \text{sgn}(gg')] = \frac{2}{\pi} [\varphi + \sin \varphi \cos \varphi].$$

Lemma A.13 (Expectation of Wasserstein-2 Distance between Two Discrete Distributions). Suppose that $\mathbf{x}$ is independent of (z, ε_0) and $\mathbb{E}[\mathbf{x}\mathbf{x}^\top] = I_d$. Then D, D^* from Lemma A.11 satisfy

$$\begin{aligned} \mathbb{E}_{\mathbf{x}} W_2^2(D, D^*) &= \|\theta\|^2 + \|\theta^*\|^2 - 2\langle \theta, \theta^* \rangle \text{sgn}(\nu\nu^*) \tanh(|\nu| \wedge |\nu^*|) - 2\mathbb{E}_{\mathbf{x}}[|\mu_0\mu_0^*|](1 - \tanh(|\nu| \vee |\nu^*|)) \\ &\leq \|\theta\|^2 + \|\theta^*\|^2 - 2\langle \theta, \theta^* \rangle \text{sgn}(\nu\nu^*) \tanh(|\nu| \wedge |\nu^*|) - 2|\langle \theta, \theta^* \rangle|(1 - \tanh(|\nu| \vee |\nu^*|)). \end{aligned}$$

In particular, suppose that $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, I_d)$ and $\theta, \theta^* \neq \mathbf{0}$. Define $\rho := \frac{\langle \theta, \theta^* \rangle}{\|\theta\|\|\theta^*\|}$ and $\varphi := \arcsin |\rho|$, so $\sin \varphi = |\rho|$. Then

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{N}(\mathbf{0}, I_d)} W_2^2(D, D^*) \\ &= \|\theta\|^2 + \|\theta^*\|^2 - 2\|\theta\|\|\theta^*\| \left[\text{sgn}(\rho\nu\nu^*) \tanh(|\nu| \wedge |\nu^*|) \sin \varphi + (1 - \tanh(|\nu| \vee |\nu^*|)) \frac{2}{\pi} [\varphi \sin \varphi + \cos \varphi] \right]. \end{aligned}$$

Proof of Lemma A.13. Write $A := \mu_0 = \langle \theta, \mathbf{x} \rangle$, $B := \mu_0^* = \langle \theta^*, \mathbf{x} \rangle$, and

$$m := \tanh(|\nu| \wedge |\nu^*|), \quad M := \tanh(|\nu| \vee |\nu^*|).$$

Since $|\tanh \nu| = \tanh |\nu|$, the identity $|a - b| = \max\{|a|, |b|\} - \text{sgn}(ab) \min\{|a|, |b|\}$ gives

$$|AB| |\text{sgn}(A) \tanh \nu - \text{sgn}(B) \tanh \nu^*| = |AB| M - AB \text{sgn}(\nu\nu^*) m.$$

Lemma A.11 therefore yields the pointwise identity

$$W_2^2(D, D^*) = A^2 + B^2 - 2AB \text{sgn}(\nu\nu^*) m - 2|AB|(1 - M).$$

Taking expectations and using $\mathbb{E}[\mathbf{x}\mathbf{x}^\top] = I_d$ proves the equality. Moreover, $\mathbb{E}|AB| \geq |\mathbb{E}[AB]| = |\langle \theta, \theta^* \rangle|$. Since $1 - M \geq 0$, this proves the stated inequality.

For the Gaussian specialization, the normalized projections $G := A/\|\theta\|$ and $G^* := B/\|\theta^*\|$ are standard Gaussian with correlation ρ . Lemma A.12, applied with the signed angle $\arcsin \rho$, gives

$$\mathbb{E}|AB| = \frac{2\|\theta\|\|\theta^*\|}{\pi} (\varphi \sin \varphi + \cos \varphi), \quad \mathbb{E}[AB] = \text{sgn}(\rho) \|\theta\|\|\theta^*\| \sin \varphi.$$

Taking expectations in the pointwise identity and substituting these moments proves the Gaussian formula. If either parameter vector is zero, the general identity instead gives $\mathbb{E}W_2^2(D, D^*) = \|\theta\|^2 + \|\theta^*\|^2$. $\square$

Proposition A.14 (Proposition 3.1). *Suppose that $0 < \bar{\alpha}_T \leq 1$, $\theta^* \neq \mathbf{0}$, $\nu^* \in \mathbb{R}$, and that $\mathbf{x}$ has a density that is positive almost everywhere on $\mathbb{R}^d$ and finite second moments. Define the terminal population KL divergence*

$$\text{KL}_T(\theta, \nu) := \text{KL}(p(y_T, \mathbf{x} \mid \theta_T^*, \pi^*) \parallel p(y_T, \mathbf{x} \mid \theta_T, \pi)).$$

Then

$$\arg \min_{(\theta, \nu) \in \mathbb{R}^d \times \mathbb{R}} \text{KL}_T(\theta, \nu) = \{(\theta^*, \nu^*), (-\theta^*, -\nu^*)\}.$$

Proof of Proposition 3.1. Because the marginal distribution of $\mathbf{x}$ does not depend on the model parameters, the chain rule for KL divergence yields

$$\text{KL}_T(\theta, \nu) = \mathbb{E}_{\mathbf{x}} \text{KL}(p(y_T \mid \mathbf{x}; \theta_T^*, \pi^*) \parallel p(y_T \mid \mathbf{x}; \theta_T, \pi)).$$

Applying Pinsker's inequality (Cover & Thomas, 2006) for each $\mathbf{x}$ and then taking the expectation over $\mathbf{x}$ gives

$$\text{KL}_T(\theta, \nu) \geq 2\mathbb{E}_{\mathbf{x}} \text{TV}^2(p(\cdot \mid \mathbf{x}; \theta_T^*, \pi^*), p(\cdot \mid \mathbf{x}; \theta_T, \pi)).$$

We next identify the equality case. Suppose that $\text{KL}_T(\theta, \nu) = 0$. Since the conditional KL divergence is non-negative, it follows that

$$p(\cdot \mid \mathbf{x}; \theta_T, \pi) = p(\cdot \mid \mathbf{x}; \theta_T^*, \pi^*) \quad \text{as distributions for almost every } \mathbf{x}.$$

By Lemma A.1, the first two raw moments of the terminal conditional distribution are

$$\mathbb{E}_{y_T \mid \mathbf{x}; \theta_T, \pi}[y_T] = \mu_T \tanh \nu, \quad \mathbb{E}_{y_T \mid \mathbf{x}; \theta_T, \pi}[y_T^2] = 1 + \mu_T^2.$$

Equality of the terminal conditional distributions and $\bar{\alpha}_T > 0$ therefore imply

$$\mu_0 \tanh \nu = \mu_0^* \tanh \nu^*, \quad \mu_0^2 = (\mu_0^*)^2 \quad \text{for almost every } \mathbf{x}.$$

The second identity can be rewritten as

$$\mathbf{x}^\top (\theta \theta^\top - \theta^* (\theta^*)^\top) \mathbf{x} = 0 \quad \text{almost surely.}$$

Since the density of $\mathbf{x}$ is positive almost everywhere on $\mathbb{R}^d$, the quadratic polynomial on the left-hand side is the zero polynomial. Hence

$$\theta \theta^\top = \theta^* (\theta^*)^\top.$$

Because $\theta^* \neq \mathbf{0}$, equality of these rank-one matrices implies $\theta = s\theta^*$ for some $s \in \{+1, -1\}$. Substitution into the first-moment identity gives $s \tanh \nu = \tanh \nu^*$. Oddness and strict monotonicity of the hyperbolic tangent give $\nu = s\nu^*$. The joint sign change $(\theta, \nu) \mapsto (-\theta, -\nu)$ exchanges the two mixture components and leaves the terminal conditional distribution unchanged. Thus

$$\text{KL}_T(\theta, \nu) = 0 \iff (\theta, \nu) \in \{(\theta^*, \nu^*), (-\theta^*, -\nu^*)\}.$$

Lemma A.11 and the same positive-density argument show that $\mathbb{E}_{\mathbf{x}} W_2^2(D, D^*) = 0$ has the same two solutions. Combining these equivalences yields the claimed characterization of the global minimizers. $\square$

Lemma A.15 (Integrated Relative de Bruijn Identity for Conditional MLR). Let the population Score Matching loss be $\mathcal{L}_t(\theta, \nu) := \frac{1}{2\bar{\alpha}_t} \mathbb{E}_{y_t, \mathbf{x}; \theta_t^*, \pi^*} [\|s_{\theta_t, \nu}(y_t, \mathbf{x}) - s_{\theta_t^*, \nu^*}(y_t, \mathbf{x})\|^2]$ as defined in Equation (6). Suppose that $\mathbf{x}$ is independent of (z, ε_0) and that $\bar{\alpha}_t \in (0, 1]$. Then every $T \geq 0$ satisfies

$$\text{KL}(p(y_0, \mathbf{x} \mid \theta^*, \pi^*) \parallel p(y_0, \mathbf{x} \mid \theta, \pi)) = \text{KL}(p(y_T, \mathbf{x} \mid \theta_T^*, \pi^*) \parallel p(y_T, \mathbf{x} \mid \theta_T, \pi)) + \int_{\bar{\alpha}_T}^{\bar{\alpha}_0} \mathcal{L}_t(\theta, \nu) d\bar{\alpha}_t.$$

Lemma A.16 (Log-likelihood and Finite Sample Score Matching Loss). Let $\mathbb{P}_n$ be the empirical measure defined before Equation (8), and let $\mathcal{J}_t^n$ be the empirical auxiliary loss in Equation (8). Then every $T > 0$ with $0 < \bar{\alpha}_T < 1$ satisfies

$$-\mathbb{E}_{\mathbb{P}_n} \ln p(y_0 \mid \mathbf{x}; \theta, \pi) = \mathbb{E}_{\mathbb{P}_n} \text{KL}(p(y_T \mid y_0) \parallel p(y_T \mid \mathbf{x}; \theta_T, \pi)) + \int_{\bar{\alpha}_T}^{\bar{\alpha}_0} \mathcal{J}_t^n(\theta, \nu) d\bar{\alpha}_t + \frac{1}{2} \left(\ln \frac{1 - \bar{\alpha}_T}{\bar{\alpha}_T} + \ln(2\pi) + 1 \right).$$

Proof. The independence assumption gives $p(\mathbf{x} \mid z; \theta, \pi) = p(\mathbf{x} \mid z; \theta_t, \pi) = p(\mathbf{x})$. Therefore $p(y_0, \mathbf{x} \mid \theta, \pi) = p(\mathbf{x})p(y_0 \mid \mathbf{x}; \theta, \pi)$ and $p(y_t, \mathbf{x} \mid \theta_t, \pi) = p(\mathbf{x})p(y_t \mid \mathbf{x}; \theta_t, \pi)$.

Under the ground truth, $p(y_0, \mathbf{x} \mid \theta^*, \pi^*) = p(\mathbf{x})p(y_0 \mid \mathbf{x}; \theta^*, \pi^*)$ and $p(y_t, \mathbf{x} \mid \theta_t^*, \pi^*) = p(\mathbf{x})p(y_t \mid \mathbf{x}; \theta_t^*, \pi^*)$. Hence $\mathbb{E}_{y_0, \mathbf{x} \mid \theta^*, \pi^*}[\cdot] = \mathbb{E}_{\mathbf{x}}\mathbb{E}_{y_0 \mid \mathbf{x}; \theta^*, \pi^*}[\cdot]$ and $\mathbb{E}_{y_t, \mathbf{x} \mid \theta_t^*, \pi^*}[\cdot] = \mathbb{E}_{\mathbf{x}}\mathbb{E}_{y_t \mid \mathbf{x}; \theta_t^*, \pi^*}[\cdot]$.

Interchanging $\frac{d}{dt}$ and $\mathbb{E}_{\mathbf{x}}$, applying Lemma A.4, and using $\frac{d\bar{\alpha}_t}{dt} = -\beta_t\bar{\alpha}_t$ gives

$$\begin{aligned} & -\frac{d}{dt} \text{KL}(p(y_t, \mathbf{x} \mid \theta_t^*, \pi^*) \parallel p(y_t, \mathbf{x} \mid \theta_t, \pi)) = -\mathbb{E}_{\mathbf{x}} \frac{d}{dt} \text{KL}(p(y_t \mid \mathbf{x}; \theta_t^*, \pi^*) \parallel p(y_t \mid \mathbf{x}; \theta_t, \pi)) \\ & = \frac{\beta_t}{2} \mathbb{E}_{\mathbf{x}} \mathbb{E}_{y_t \mid \mathbf{x}; \theta_t^*, \pi^*} [\|s_{\theta_t, \nu}(y_t, \mathbf{x}) - s_{\theta_t^*, \nu^*}(y_t, \mathbf{x})\|^2] = \beta_t \bar{\alpha}_t \cdot \mathcal{L}_t(\theta, \nu) = -\mathcal{L}_t(\theta, \nu) \frac{d\bar{\alpha}_t}{dt}. \end{aligned}$$

Integrating from 0 to T , with $\theta_0 = \theta$ and $\theta_0^* = \theta^*$, gives

$$\text{KL}(p(y_0, \mathbf{x} \mid \theta^*, \pi^*) \parallel p(y_0, \mathbf{x} \mid \theta, \pi)) - \text{KL}(p(y_T, \mathbf{x} \mid \theta_T^*, \pi^*) \parallel p(y_T, \mathbf{x} \mid \theta_T, \pi)) = \int_{\bar{\alpha}_T}^{\bar{\alpha}_0} \mathcal{L}_t(\theta, \nu) d\bar{\alpha}_t.$$

Rearranging the terms proves the result. $\square$

Proof of Lemma A.16. Fix $\epsilon \in (0, T)$. We apply Lemma A.5 to $\mathcal{J}_t^n$, use $\frac{d\bar{\alpha}_t}{dt} = -\beta_t\bar{\alpha}_t$, and integrate from ϵ to T . Under the common Gaussian coupling, $y_\epsilon \rightarrow y_0$ in L^2 and $\theta_\epsilon \rightarrow \theta$ as $\epsilon \downarrow 0$. The log-density is continuous and has a quadratic envelope for each fixed empirical sample, so dominated convergence applies. We therefore let $\epsilon \downarrow 0$, interpret the integral at its upper endpoint as an improper integral, and obtain

$$-\mathbb{E}_{\mathbb{P}_n} \ln p(y_0 \mid \mathbf{x}; \theta, \pi) + \mathbb{E}_{\mathbb{P}_n} \mathbb{E}_{y_T \mid y_0} \ln p(y_T \mid \mathbf{x}; \theta_T, \pi) = \int_{\bar{\alpha}_T}^{\bar{\alpha}_0} \mathcal{J}_t^n(\theta, \nu) d\bar{\alpha}_t + \frac{1}{2} \ln \frac{1}{\bar{\alpha}_T}.$$

The Gaussian transition satisfies

$$\begin{aligned} p(y_T \mid y_0) &= \mathcal{N}(y_T; y_0 \sqrt{\bar{\alpha}_T}, 1 - \bar{\alpha}_T), \\ -\ln p(y_T \mid y_0) &= \frac{1}{2} \ln(2\pi(1 - \bar{\alpha}_T)) + \frac{\|y_T - y_0 \sqrt{\bar{\alpha}_T}\|^2}{2(1 - \bar{\alpha}_T)}, \\ \mathbb{E}_{y_T \mid y_0} [\|y_T - y_0 \sqrt{\bar{\alpha}_T}\|^2] &= 1 - \bar{\alpha}_T. \end{aligned}$$

These identities give

$$\begin{aligned} -\mathbb{E}_{y_T \mid y_0} \ln p(y_T \mid \mathbf{x}; \theta_T, \pi) &= \text{KL}(p(y_T \mid y_0) \parallel p(y_T \mid \mathbf{x}; \theta_T, \pi)) - \mathbb{E}_{y_T \mid y_0} [\ln p(y_T \mid y_0)] \\ &= \text{KL}(p(y_T \mid y_0) \parallel p(y_T \mid \mathbf{x}; \theta_T, \pi)) + \frac{1}{2} (\ln(1 - \bar{\alpha}_T) + \ln(2\pi) + 1). \end{aligned}$$

Taking $\mathbb{E}_{\mathbb{P}_n}$, substituting into the first identity, and rearranging the terms proves the result. $\square$

Lemma A.17 (Derivatives of Negative Log-Likelihood). Let $F(y_t, \mu_t, \nu) := -\ln p(y_t \mid \mathbf{x}; \theta_t, \pi)$ be the negative log-likelihood from Lemma A.2. Then

$$y_t \partial_{y_t} F(y_t, \mu_t, \nu) - y_t^2 = \mu_t \partial_{\mu_t} F(y_t, \mu_t, \nu) - \mu_t^2 = (\partial_\nu F(y_t, \mu_t, \nu) - \tanh \nu) \mu_t y_t = -\tanh(\mu_t y_t + \nu) \mu_t y_t.$$

Moreover, $\partial_{y_t y_t}^2 F(y_t, \mu_t, \nu) = 1 - \mu_t^2 \text{sech}^2(\mu_t y_t + \nu)$, and

$$\|\partial_{y_t} F(y_t, \mu_t, \nu)\|^2 - \partial_{y_t y_t}^2 F(y_t, \mu_t, \nu) = 2y_t \partial_{y_t} F(y_t, \mu_t, \nu) + (\mu_t^2 - y_t^2 - 1).$$

Proof. Differentiating the expression for F from Lemma A.2 with respect to y_t , μ_t , and ν gives

$$\partial_{y_t} F(y_t, \mu_t, \nu) = y_t - \tanh(\mu_t y_t + \nu) \mu_t, \quad \partial_{\mu_t} F(y_t, \mu_t, \nu) = \mu_t - \tanh(\mu_t y_t + \nu) y_t,$$

$$\partial_\nu F(y_t, \mu_t, \nu) = \tanh \nu - \tanh(\mu_t y_t + \nu).$$

We complete the proof of the first identity by rearranging terms in the above expressions and multiplying by y_t, μ_t and $\mu_t y_t$ to obtain $-\tanh(\mu_t y_t + \nu) \mu_t y_t$. For the second identity, use $\tanh^2(\cdot) = 1 - \text{sech}^2(\cdot)$, $d \tanh(\cdot)/d(\cdot) = \text{sech}^2(\cdot)$, and $\partial_{y_t} F(y_t, \mu_t, \nu) = y_t - \tanh(\mu_t y_t + \nu) \mu_t$. Then

$$\begin{aligned} \partial_{y_t y_t}^2 F(y_t, \mu_t, \nu) &= 1 - \mu_t^2 \text{sech}^2(\mu_t y_t + \nu) = 1 - \mu_t^2 + \mu_t^2 \tanh^2(\mu_t y_t + \nu) = 1 - \mu_t^2 + \|y_t - \partial_{y_t} F(y_t, \mu_t, \nu)\|^2 \\ &= \|\partial_{y_t} F(y_t, \mu_t, \nu)\|^2 - 2y_t \partial_{y_t} F(y_t, \mu_t, \nu) + (y_t^2 + 1 - \mu_t^2). \end{aligned}$$

Therefore, the last identity follows by rearranging terms in the above expressions. $\square$

Lemma A.18 (Spiked Covariance under General Gaussian Covariance). Let $\mathbf{x} \sim \mathcal{N}(0, \Sigma)$, where $\Sigma \succ 0$, and define $\mathcal{C}^* := \mathbb{E}[\mathbf{x}\mathbf{x}^\top (\mathbf{x}^\top \theta^*)^2]$. With $c := (\theta^*)^\top \Sigma \theta^*$ and $a := \Sigma^{1/2} \theta^*$,

$$\mathcal{C}^* = c\Sigma + 2\Sigma\theta^*(\Sigma\theta^*)^\top, \quad \Sigma^{-1/2}\mathcal{C}^*\Sigma^{-1/2} = cI_d + 2aa^\top.$$

If $\theta^* \neq \mathbf{0}$, the whitened matrix has principal eigenspace $\text{span}\{a\}$, with eigenvalue $3c$, and eigenvalue c on $a^\perp$, with multiplicity $d - 1$. The equivalent generalized eigenproblem $\mathcal{C}^*v = \lambda\Sigma v$ has leading eigenspace $\text{span}\{\theta^*\}$, lower eigenspace $\{v : v^\top \Sigma \theta^* = 0\}$, and generalized spectral gap $2c$. When $\Sigma = I_d$, these are the ordinary eigenspaces of $\mathcal{C}^*$.

Proof. Isserlis' identity gives $\mathbb{E}[X_i X_j X_k X_l] = \Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}$ (Isserlis, 1918). Contracting this identity with $\theta_k^* \theta_l^*$ yields $\mathcal{C}^* = c\Sigma + 2\Sigma\theta^*(\Sigma\theta^*)^\top$, and whitening yields the displayed rank-one form. Since $\|a\|^2 = c$, that matrix maps a to $3ca$ and every $z \perp a$ to cz . Under $w = \Sigma^{1/2}v$, $\mathcal{C}^*v = \lambda\Sigma v$ is equivalent to $\Sigma^{-1/2}\mathcal{C}^*\Sigma^{-1/2}w = \lambda w$, which gives the generalized eigenspaces and gap. $\square$

Lemma A.19 (Spiked Covariance for Elliptical Distributions). Let $\mathbf{x} \in \mathbb{R}^d$ follow a zero-mean elliptical distribution with covariance $\Sigma \succ 0$ and finite fourth moments. Define the excess kurtosis parameter

$$\kappa := \frac{\mathbb{E}[\|\Sigma^{-1/2}\mathbf{x}\|^4]}{d(d+2)} - 1 > -1,$$

where $\Sigma^{-1/2}$ is the symmetric inverse square root of Σ . Then the fourth-order moments satisfy

$$\mathbb{E}[X_i X_j X_k X_l] = (1 + \kappa)(\Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}).$$

The fourth cumulant tensor

$$\Delta_{ijkl} := \mathbb{E}[X_i X_j X_k X_l] - (\Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk})$$

satisfies $\Delta_{ijkl} = \kappa(\Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk})$.

For $\mathcal{C}^* := \mathbb{E}[\mathbf{x}\mathbf{x}^\top (\mathbf{x}^\top \theta^*)^2]$, set $c := (\theta^*)^\top \Sigma \theta^*$ and $a := \Sigma^{1/2} \theta^*$. Then

$$\mathcal{C}^* = (1 + \kappa)[c\Sigma + 2\Sigma\theta^*(\Sigma\theta^*)^\top], \quad \Sigma^{-1/2}\mathcal{C}^*\Sigma^{-1/2} = (1 + \kappa)(cI_d + 2aa^\top).$$

If $\theta^* \neq \mathbf{0}$, the whitened eigenvalues are $3(1 + \kappa)c$ on $\text{span}\{a\}$ and $(1 + \kappa)c$ on $a^\perp$. The equivalent generalized problem $\mathcal{C}^*v = \lambda\Sigma v$ has leading eigenspace $\text{span}\{\theta^*\}$, lower eigenspace $\{v : v^\top \Sigma \theta^* = 0\}$, and generalized spectral gap $2(1 + \kappa)c$. For $\Sigma = I_d$, these are ordinary eigenspaces.

Proof. Write $\mathbf{x} = R\Sigma^{1/2}\mathbf{u}$, where $R \geq 0$ is independent of the uniform direction $\mathbf{u} \in \mathbb{S}^{d-1}$ (Fang et al., 1990, Theorem 2.5(iii), p. 30). Rotational invariance implies that the fourth-moment tensor of $\mathbf{u}$ has the form $c_u(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ for a scalar c_u . Taking the double trace and using $\|\mathbf{u}\| = 1$ gives $c_u = 1/[d(d+2)]$, and therefore

$$\mathbb{E}[u_i u_j u_k u_l] = \frac{\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}}{d(d+2)}.$$

Together with $R = \|\Sigma^{-1/2}\mathbf{x}\|$, this identity gives the stated fourth-moment formula. Contracting it with $\theta_k^* \theta_l^*$ gives the formula for $\mathcal{C}^*$. The remaining claims follow from the whitened calculation in Lemma A.18, multiplied by $1 + \kappa$. $\square$

B Proofs of Maximum-Likelihood Asymptotics for Path-Integrated Score Matching

This appendix uses the likelihood bridges from Appendix A to restate and prove Main Theorem 3.2, including consistency modulo joint sign, the aligned Gaussian limit with inverse-Fisher covariance, and the attainment argument required by the empirical estimator.

Main theorem B.1 (Main Theorem 3.2). *The ground-truth parameters $(\theta^*, \nu^*) \in \mathbb{R}^d \times \mathbb{R}$ satisfy $\theta^* \neq \mathbf{0}$. The deterministic terminal-horizon sequence $\{T_n\}_{n \geq 1}$ satisfies $\ln \frac{1-\bar{\alpha}_{T_n}}{\bar{\alpha}_{T_n}} - \ln n \rightarrow \infty$, and the covariate $\mathbf{x}$ has a density that is positive almost everywhere and satisfies $\mathbb{E}\|\mathbf{x}\|^2 < \infty$. The estimator $\hat{\vartheta}_{n,T_n}^{\text{SM}} \equiv (\hat{\theta}_{n,T_n}^{\text{SM}}, \hat{\nu}_{n,T_n}^{\text{SM}})$ is consistent up to a joint sign change. We choose $s_n \in \arg \min_{s \in \{+1, -1\}} \left\| s \left(\hat{\theta}_{n,T_n}^{\text{SM}}, \hat{\nu}_{n,T_n}^{\text{SM}} \right) - (\theta^*, \nu^*) \right\|$. Then, as $n \rightarrow \infty$,*

$$s_n \left(\hat{\theta}_{n,T_n}^{\text{SM}}, \hat{\nu}_{n,T_n}^{\text{SM}} \right) \xrightarrow{\mathbb{P}} (\theta^*, \nu^*).$$

If $\mathbb{E}\|\mathbf{x}\|^4 < \infty$, then the aligned Score Matching estimator is asymptotically normal. As $n \rightarrow \infty$,

$$\sqrt{n} \left[s_n \left(\hat{\theta}_{n,T_n}^{\text{SM}}, \hat{\nu}_{n,T_n}^{\text{SM}} \right) - (\theta^*, \nu^*) \right] \xrightarrow{d} \mathcal{N}(\mathbf{0}, I(\theta^*, \nu^*)^{-1}),$$

where I denotes the Fisher information matrix.

Proof of Theorem 3.2. This proof adapts the DDPM-to-MLE argument of Chewi et al. (2025a) to conditional MLR with joint sign symmetry. The model-specific part identifies the two label-equivalent limits. The local M-estimator argument is then applied after sign alignment.

Proof-specific notation. In addition to the notation fixed in Section 2.4, define

- $\mathcal{E}_n(\vartheta) := \hat{\mathcal{R}}_n^{\text{MLE}}(\vartheta) - \hat{\mathcal{R}}_n^{\text{MLE}}(\vartheta^*)$ and $\mathcal{E}(\vartheta) := \mathbb{E}[\mathcal{E}_n(\vartheta)]$, where the expectation is over the i.i.d. sample. These quantities are the empirical contrast and the population excess risk.
- $c_n := \bar{\alpha}_{T_n} / [2(1 - \bar{\alpha}_{T_n})]$ and $\Sigma_n := \mathbb{E}_{\mathbb{P}_n}[\mathbf{x}\mathbf{x}^\top]$.
- $\tilde{\vartheta}_{n,T_n}^{\text{SM}} := s_n \hat{\vartheta}_{n,T_n}^{\text{SM}}$. This estimator is sign-aligned in both coordinates, with s_n as defined in the theorem.
- $m_{\vartheta}(y_0, \mathbf{x}) := \ln p(y_0 \mid \mathbf{x}; \vartheta)$. We use $\dot{m}(y_0, \mathbf{x})$ for a measurable local Lipschitz envelope of m_{ϑ} near ϑ^* .

We set $T = T_n$ in Lemma A.16 and define $\mathbf{c}(\bar{\alpha}_{T_n}) := \frac{1}{2} \left(\ln \frac{1-\bar{\alpha}_{T_n}}{\bar{\alpha}_{T_n}} + \ln(2\pi) + 1 \right)$. The lemma gives the empirical bridge

$$\hat{\mathcal{R}}_{n,T_n}^{\text{SM}}(\vartheta) = \hat{\mathcal{R}}_n^{\text{MLE}}(\vartheta) - \Delta_n(\vartheta) - \mathbf{c}(\bar{\alpha}_{T_n}).$$

Whenever the empirical minimum is attained, exact optimality gives $\hat{\mathcal{R}}_{n,T_n}^{\text{SM}}(\hat{\vartheta}_{n,T_n}^{\text{SM}}) \leq \hat{\mathcal{R}}_{n,T_n}^{\text{SM}}(\vartheta^*)$. Substitution isolates the excess risk $\mathcal{E}_n(\hat{\vartheta}_{n,T_n}^{\text{SM}}) \equiv \hat{\mathcal{R}}_n^{\text{MLE}}(\hat{\vartheta}_{n,T_n}^{\text{SM}}) - \hat{\mathcal{R}}_n^{\text{MLE}}(\vartheta^*)$. Since $\Delta_n(\vartheta^*) \geq 0$, we obtain

$$\mathcal{E}_n(\hat{\vartheta}_{n,T_n}^{\text{SM}}) \leq \Delta_n(\hat{\vartheta}_{n,T_n}^{\text{SM}}) - \Delta_n(\vartheta^*) \leq \Delta_n(\hat{\vartheta}_{n,T_n}^{\text{SM}}).$$

Step 1. Boundedness of $\hat{\vartheta}_{n,T_n}^{\text{SM}}$ and control of the excess risk.

Lemma A.8 bounds the decaying KL divergence Δ_n . We first work on samples with a nonempty score matching argmin. Set $c_n \equiv \frac{\bar{\alpha}_{T_n}}{2(1-\bar{\alpha}_{T_n})}$. The Cauchy–Schwarz inequality for the empirical covariance matrix

Σ_n , together with $|\tanh(\cdot)| \leq 1$, gives

$$\begin{aligned}\mathcal{E}_n(\hat{\vartheta}_{n,T_n}^{\text{SM}}) &\leq \Delta_n(\hat{\vartheta}_{n,T_n}^{\text{SM}}) \\ &\leq c_n \mathbb{E}_{\mathbb{P}_n} \left[y_0^2 + (\hat{\theta}_{n,T_n}^{\text{SM}})^\top \mathbf{x} \mathbf{x}^\top \hat{\theta}_{n,T_n}^{\text{SM}} - 2y_0((\hat{\theta}_{n,T_n}^{\text{SM}})^\top \mathbf{x}) \tanh \hat{\nu}_{n,T_n}^{\text{SM}} + 1 \right] \\ &\leq c_n \left(\|\hat{\theta}_{n,T_n}^{\text{SM}}\|_{\Sigma_n}^2 + 2\|\hat{\theta}_{n,T_n}^{\text{SM}}\|_{\Sigma_n} \sqrt{\mathbb{E}_{\mathbb{P}_n}[y_0^2]} + \mathbb{E}_{\mathbb{P}_n}[y_0^2] + 1 \right).\end{aligned}$$

Lemma A.2 gives the negative log-likelihood. We apply $|x| \geq \ln \frac{\cosh(x+x')}{\cosh x'} \geq -|x|$ and the Cauchy–Schwarz inequality for Σ_n to obtain the lower bound

$$\begin{aligned}\mathcal{E}_n(\hat{\vartheta}_{n,T_n}^{\text{SM}}) &= \mathbb{E}_{\mathbb{P}_n} \left[\frac{1}{2}(\hat{\theta}_{n,T_n}^{\text{SM}})^\top \mathbf{x} \mathbf{x}^\top \hat{\theta}_{n,T_n}^{\text{SM}} - \frac{1}{2}(\theta^*)^\top \mathbf{x} \mathbf{x}^\top \theta^* \right. \\ &\quad \left. - \ln \frac{\cosh((\hat{\theta}_{n,T_n}^{\text{SM}})^\top \mathbf{x} y_0 + \hat{\nu}_{n,T_n}^{\text{SM}})}{\cosh \hat{\nu}_{n,T_n}^{\text{SM}}} + \ln \frac{\cosh((\theta^*)^\top \mathbf{x} y_0 + \nu^*)}{\cosh \nu^*} \right] \\ &\geq \frac{1}{2} \|\hat{\theta}_{n,T_n}^{\text{SM}}\|_{\Sigma_n}^2 - \|\hat{\theta}_{n,T_n}^{\text{SM}}\|_{\Sigma_n} \sqrt{\mathbb{E}_{\mathbb{P}_n}[y_0^2]} \\ &\quad - \|\theta^*\|_{\Sigma_n} \sqrt{\mathbb{E}_{\mathbb{P}_n}[y_0^2]} - \frac{1}{2} \|\theta^*\|_{\Sigma_n}^2.\end{aligned}$$

Combining the quadratic lower bound with the c_n -scaled upper bound gives the following constraint on the empirical norm of the score matching estimator.

$$\underbrace{\left(\frac{1}{2} - c_n \right)}_{:=A_n} \|\hat{\theta}_{n,T_n}^{\text{SM}}\|_{\Sigma_n}^2 - \underbrace{(1 + 2c_n) \sqrt{\mathbb{E}_{\mathbb{P}_n}[y_0^2]}}_{:=B_n} \|\hat{\theta}_{n,T_n}^{\text{SM}}\|_{\Sigma_n} \leq \underbrace{c_n(\mathbb{E}_{\mathbb{P}_n}[y_0^2] + 1) + \|\theta^*\|_{\Sigma_n} \sqrt{\mathbb{E}_{\mathbb{P}_n}[y_0^2]} + \frac{1}{2} \|\theta^*\|_{\Sigma_n}^2}_{C_n}.$$

Under the time-scaling condition $\ln \frac{1-\bar{\alpha}_{T_n}}{\bar{\alpha}_{T_n}} - \ln n \rightarrow \infty$, it follows that $c_n \equiv \frac{\bar{\alpha}_{T_n}}{2(1-\bar{\alpha}_{T_n})} = o(1/n)$. Since $A_n = \frac{1}{2} - c_n \rightarrow \frac{1}{2}$, we have $A_n > 0$ for all large n . Since $\mathbf{x}$ has finite population second moments $\mathbb{E}[\mathbf{x} \mathbf{x}^\top]$, Markov's inequality applied to the finite population expectations gives $\|\theta^*\|_{\Sigma_n}^2 = \mathcal{O}_P(1)$ and $\mathbb{E}_{\mathbb{P}_n}[y_0^2] = \mathcal{O}_P(1)$. First, consider the empirical weighted norm $\|\theta^*\|_{\Sigma_n}^2 = (\theta^*)^\top \Sigma_n \theta^*$ with $\Sigma_n \equiv \mathbb{E}_{\mathbb{P}_n}[\mathbf{x} \mathbf{x}^\top]$. Its expectation under the data-generating distribution is finite. Markov's inequality gives, for every $M > 0$,

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(\|\theta^*\|_{\Sigma_n}^2 > M) \leq \lim_{M \rightarrow \infty} \frac{\mathbb{E}[(\theta^*)^\top \mathbb{E}_{\mathbb{P}_n}[\mathbf{x} \mathbf{x}^\top] \theta^*]}{M} = \lim_{M \rightarrow \infty} \frac{(\theta^*)^\top \mathbb{E}[\mathbf{x} \mathbf{x}^\top] \theta^*}{M} = 0.$$

Second, consider the empirical second moment $\mathbb{E}_{\mathbb{P}_n}[y_0^2]$. Under the ground-truth distribution, the response satisfies $y_0 = (-1)^{z+1} \langle \theta^*, \mathbf{x} \rangle + \varepsilon_0$, where $(\mathbf{x}, z) \perp\!\!\!\perp \varepsilon_0$ and $\varepsilon_0 \sim \mathcal{N}(0, 1)$. Taking the expectation gives $\mathbb{E}[\mathbb{E}_{\mathbb{P}_n}[y_0^2]] = \mathbb{E}[y_0^2] = (\theta^*)^\top \mathbb{E}[\mathbf{x} \mathbf{x}^\top] \theta^* + 1$. The finiteness of $\mathbb{E}[\mathbf{x} \mathbf{x}^\top]$ and Markov's inequality give

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(\mathbb{E}_{\mathbb{P}_n}[y_0^2] > M) \leq \lim_{M \rightarrow \infty} \frac{\mathbb{E}[\mathbb{E}_{\mathbb{P}_n}[y_0^2]]}{M} = \lim_{M \rightarrow \infty} \frac{(\theta^*)^\top \mathbb{E}[\mathbf{x} \mathbf{x}^\top] \theta^* + 1}{M} = 0.$$

The two inequalities above guarantee $\|\theta^*\|_{\Sigma_n}^2 = \mathcal{O}_P(1)$, $\mathbb{E}_{\mathbb{P}_n}[y_0^2] = \mathcal{O}_P(1)$. It follows that

$$B_n := (1 + 2c_n) \sqrt{\mathbb{E}_{\mathbb{P}_n}[y_0^2]} = \mathcal{O}(1) \sqrt{\mathcal{O}_P(1)} = \mathcal{O}_P(1),$$

$$C_n := c_n(\mathbb{E}_{\mathbb{P}_n}[y_0^2] + 1) + \|\theta^*\|_{\Sigma_n} \sqrt{\mathbb{E}_{\mathbb{P}_n}[y_0^2]} + \frac{1}{2} \|\theta^*\|_{\Sigma_n}^2 = o(n^{-1}) \mathcal{O}_P(1) + \mathcal{O}_P(1) = \mathcal{O}_P(1).$$

This leaves a quadratic inequality of the form $A_n \|\hat{\theta}_{n,T_n}^{\text{SM}}\|_{\Sigma_n}^2 - B_n \|\hat{\theta}_{n,T_n}^{\text{SM}}\|_{\Sigma_n} \leq C_n$, where $A_n \rightarrow \frac{1}{2}$ and $B_n, C_n = \mathcal{O}_P(1)$. Hence $A_n \geq \frac{1}{4}$ for all large n . Solving the quadratic inequality shows that the empirical norm is bounded in probability.

$$\|\hat{\theta}_{n,T_n}^{\text{SM}}\|_{\Sigma_n} \leq \frac{B_n + \sqrt{B_n^2 + 4A_n C_n}}{2A_n} \leq \frac{B_n}{A_n} + \sqrt{\frac{C_n}{A_n}} \leq 4\mathcal{O}_P(1) + \sqrt{4\mathcal{O}_P(1)} = \mathcal{O}_P(1).$$

On samples with an empty argmin, the convention in Section 2.4 gives $\hat{\vartheta}_{n,T_n}^{\text{SM}} = (\mathbf{0}, 0)$. Hence the empirical norm bound holds unconditionally. The transition-kernel KL bound also holds for the fallback value. In particular, $\Delta_n((\mathbf{0}, 0)) \leq c_n(\mathbb{E}_{\mathbb{P}_n}[y_0^2] + 1) = o_P(n^{-1})$. Substitution into the KL discrepancy bound, together with $c_n = o(1/n)$, gives

$$\Delta_n(\hat{\vartheta}_{n,T_n}^{\text{SM}}) \leq c_n \mathcal{O}_P(1) = o_P(n^{-1}).$$

On samples with a nonempty argmin, we also have $\mathcal{E}_n(\hat{\vartheta}_{n,T_n}^{\text{SM}}) \leq \Delta_n(\hat{\vartheta}_{n,T_n}^{\text{SM}})$. Moreover, since $\mathbf{x}$ has a density that is positive almost everywhere on $\mathbb{R}^d$, $\Sigma := \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$ is positive definite, and its smallest eigenvalue satisfies $\lambda_{\min}(\Sigma) > 0$. Weyl's Perturbation Theorem (Theorem 4.3.1 on page 239 of [Horn & Johnson \(2012\)](#)) and the Weak Law of Large Numbers (Theorem 2.2.3 of [Durrett \(2019\)](#)) give $|\lambda_{\min}(\Sigma_n) - \lambda_{\min}(\Sigma)| \leq \|\Sigma_n - \Sigma\|_2 = o_P(1)$. Therefore, $|\lambda_{\min}(\Sigma_n) - \lambda_{\min}(\Sigma)|/\lambda_{\min}(\Sigma) = o_P(1)$. For all large n , $\lambda_{\min}(\Sigma_n) \geq \lambda_{\min}(\Sigma) - |\lambda_{\min}(\Sigma_n) - \lambda_{\min}(\Sigma)| = (1 - o_P(1))\lambda_{\min}(\Sigma) > 0$ with high probability and

$$\|\hat{\theta}_{n,T_n}^{\text{SM}}\| \leq \sqrt{\frac{(\hat{\theta}_{n,T_n}^{\text{SM}})^\top \Sigma_n (\hat{\theta}_{n,T_n}^{\text{SM}})}{\lambda_{\min}(\Sigma_n)}} \leq \frac{\|\hat{\theta}_{n,T_n}^{\text{SM}}\|_{\Sigma_n}}{\sqrt{(1 - o_P(1))\lambda_{\min}(\Sigma)}} = \mathcal{O}_P(1).$$

Step 2. Convergence to the ground truth and boundedness of $\hat{\vartheta}_{n,T_n}^{\text{SM}}$.

Taking the expectation of the empirical average under the i.i.d. ground-truth sample gives the population excess risk

$$\mathcal{E}(\vartheta) = \mathbb{E}_{\mathbf{x}}[\text{KL}(p(\cdot | \mathbf{x}; \vartheta^*) \parallel p(\cdot | \mathbf{x}; \vartheta))] \geq 0.$$

The equality condition for KL divergence shows that

$$\mathcal{E}(\vartheta) = 0 \Leftrightarrow p(\cdot | \mathbf{x}; \vartheta^*) = p(\cdot | \mathbf{x}; \vartheta) \quad \text{for almost every } \mathbf{x}.$$

Their conditional moment generating function is

$$\mathbb{E}_{y_0|\mathbf{x};\vartheta}[\exp(ty_0)] = [\cosh(t\theta^\top \mathbf{x}) + \tanh \nu \cdot \sinh(t\theta^\top \mathbf{x})] \exp(t^2/2).$$

In particular, equality of the conditional distributions implies equality of their first two raw moments.

$$(\theta^\top \mathbf{x}) \tanh \nu = ((\theta^*)^\top \mathbf{x}) \tanh \nu^*, \quad (\theta^\top \mathbf{x})^2 = ((\theta^*)^\top \mathbf{x})^2 \quad \text{almost surely.}$$

The second identity is equivalent to

$$\mathbf{x}^\top (\theta\theta^\top - \theta^*(\theta^*)^\top) \mathbf{x} = 0 \quad \text{almost surely.}$$

Since $\mathbf{x}$ has a density that is positive almost everywhere on $\mathbb{R}^d$, this quadratic polynomial is zero on $\mathbb{R}^d$, and hence $\theta\theta^\top = \theta^*(\theta^*)^\top$. Because $\theta^* \neq \mathbf{0}$, it follows that $\theta = s\theta^*$ for some $s \in \{+1, -1\}$. Substitution into the first-moment identity gives $s \tanh \nu = \tanh \nu^*$, so the oddness and strict monotonicity of $\tanh$ imply $\nu = s\nu^*$. The reverse implication follows because the joint sign flip exchanges the two mixture components and leaves the conditional distribution unchanged. This agrees with Proposition 3.1 at $\bar{\alpha}_0 = 1$ and establishes identifiability for every $\vartheta \in \mathbb{R}^d \times \mathbb{R}$.

$$\mathcal{E}(\vartheta) = 0 \iff \vartheta \equiv (\theta, \nu) = s(\theta^*, \nu^*) = s\vartheta^*, \quad s \in \{+1, -1\}.$$

Step 1 gives $\|\hat{\theta}_{n,T_n}^{\text{SM}}\| = \mathcal{O}_P(1)$. Thus, for every $\delta > 0$, we can choose a finite radius $R_\delta > \|\theta^*\|$ and an index n_δ such that

$$\inf_{n \geq n_\delta} \mathbb{P}(\hat{\theta}_{n,T_n}^{\text{SM}} \in K_{R_\delta}) \geq 1 - \delta, \quad K_{R_\delta} := \{\theta \in \mathbb{R}^d \mid \|\theta\| \leq R_\delta\}.$$

We work on the compact parameter set $K_\delta := K_{R_\delta} \times \bar{\mathbb{R}}$, where $\bar{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$. At its two boundary points, the conditional model has the limits $p(\cdot | \mathbf{x}; \theta, \pm\infty) = \mathcal{N}(\pm\langle \theta, \mathbf{x} \rangle, 1)$. Fatou's lemma (Theorem 1.5.5 of [Durrett \(2019\)](#)) yields

$$\liminf_{|\nu| \rightarrow \infty} \mathcal{E}(\theta, \nu) \geq \mathbb{E}_{\mathbf{x}} \left[\min_{s \in \{+1, -1\}} \text{KL}(p(\cdot | \mathbf{x}; \vartheta^*) \parallel \mathcal{N}(s\langle \theta, \mathbf{x} \rangle, 1)) \right] > 0.$$

Because the ground-truth parameter is finite ($|\nu^*| < \infty$ and $\theta^* \neq \mathbf{0}$), the true conditional distribution is a strict two-component mixture for almost every $\mathbf{x}$. The identifiability of Gaussian mixtures implies that the KL divergence to either single Gaussian is positive for almost every $\mathbf{x}$. Define

$$g(\theta) := \mathbb{E}_{\mathbf{x}} \left[\min_{s \in \{+1, -1\}} \text{KL}(p(\cdot | \mathbf{x}; \vartheta^*) \parallel \mathcal{N}(s\langle \theta, \mathbf{x} \rangle, 1)) \right].$$

The integrand is continuous in θ . Convexity of KL divergence in its first argument bounds it, for $\theta \in K_{R_\delta}$, by $\frac{1}{2}(\|\theta^*\| + R_\delta)^2 \|\mathbf{x}\|^2$. The finite second moment of $\mathbf{x}$ and dominated convergence therefore make g continuous on K_{R_δ} . We also have $g(\theta) > 0$ at every $\theta \in K_{R_\delta}$. Compactness now gives $\kappa_\delta := \min_{\theta \in K_{R_\delta}} g(\theta) > 0$.

$$\inf_{\theta \in K_{R_\delta}} \liminf_{|\nu| \rightarrow \infty} \mathcal{E}(\theta, \nu) \geq \kappa_\delta > 0, \quad \min_{\vartheta \in K_\delta} \mathcal{E}(\vartheta) = \mathcal{E}(\vartheta^*) = \mathcal{E}(-\vartheta^*) = 0.$$

Hence $\arg \min_{\vartheta \in K_\delta} \mathcal{E}(\vartheta) = \{\vartheta^*, -\vartheta^*\}$. The ambient parameter space $\mathbb{R}^d \times \overline{\mathbb{R}}$ contains K_δ . The global maximizers of the expected conditional log-likelihood on the ambient space are $\mathcal{S}^* := \{\vartheta^*, -\vartheta^*\}$.

$$\mathcal{S}^* = \left\{ \vartheta' \in \mathbb{R}^d \times \overline{\mathbb{R}} \mid \mathbb{E}_{y_0, \mathbf{x} | \vartheta^*} [m_{\vartheta'}] = \sup_{\vartheta \in \mathbb{R}^d \times \overline{\mathbb{R}}} \mathbb{E}_{y_0, \mathbf{x} | \vartheta^*} [m_\vartheta] \right\} = \left\{ \vartheta' \in \mathbb{R}^d \times \overline{\mathbb{R}} \mid \mathcal{E}(\vartheta') = 0 \right\}.$$

Let $U \subset K_\delta$ be a neighborhood and set $R_U := \sup_{(\theta, \nu) \in U} \|\theta\| < \infty$. Lemma A.2 and the inequality $\ln[\cosh(x + x')/\cosh x'] \leq |x|$, followed by passage to the boundary limits when needed, give

$$\begin{aligned} \mathbb{E} \left[\sup_{\vartheta \in U} \{-m_\vartheta(y_0, \mathbf{x})\} \right] &\leq \frac{1}{2} \mathbb{E}[y_0^2] + \frac{1}{2} R_U^2 \mathbb{E}[\|\mathbf{x}\|^2] + R_U \mathbb{E}[|y_0| \|\mathbf{x}\|] + \frac{1}{2} \ln(2\pi) \\ &\leq \frac{1}{2} \mathbb{E}[y_0^2] + \frac{1}{2} R_U^2 \mathbb{E}[\|\mathbf{x}\|^2] + R_U \sqrt{\mathbb{E}[y_0^2] \mathbb{E}[\|\mathbf{x}\|^2]} + \frac{1}{2} \ln(2\pi) < \infty. \end{aligned}$$

This integrable envelope gives the local domination condition in Wald's consistency theorem for M-estimators with nonunique maxima (Theorem 5.14 on page 48 of [van der Vaart \(1998\)](#)). The conditional log-likelihood is continuous in each finite parameter and has continuous limits at $\nu = \pm\infty$, so it is upper semicontinuous on K_δ .

The empirical bridge and score matching optimality give the required likelihood near-optimality. Let $\mathcal{A}_n$ denote the event that the empirical score matching argmin is nonempty, and define $D_n := \sup_{\vartheta} \mathbb{E}_{\mathbb{P}_n} m_\vartheta - \mathbb{E}_{\mathbb{P}_n} m_{\hat{\vartheta}_{n, T_n}^{\text{SM}}}$ on the entire sample space using the estimator convention in Section 2.4. Equivalently, $D_n = \hat{\mathcal{R}}_n^{\text{MLE}}(\hat{\vartheta}_{n, T_n}^{\text{SM}}) - \inf_{\vartheta} \hat{\mathcal{R}}_n^{\text{MLE}}(\vartheta)$. On $\mathcal{A}_n$, the score matching optimality gives $\hat{\mathcal{R}}_n^{\text{SM}}(\hat{\vartheta}) - \hat{\mathcal{R}}_n^{\text{SM}}(\vartheta) \leq 0$, the bridge constant cancels, and $\Delta_n(\vartheta) \geq 0$; hence, for every ϑ ,

$$\hat{\mathcal{R}}_n^{\text{MLE}}(\hat{\vartheta}) - \hat{\mathcal{R}}_n^{\text{MLE}}(\vartheta) = \hat{\mathcal{R}}_n^{\text{SM}}(\hat{\vartheta}) - \hat{\mathcal{R}}_n^{\text{SM}}(\vartheta) + \Delta_n(\hat{\vartheta}) - \Delta_n(\vartheta) \leq \Delta_n(\hat{\vartheta}).$$

Taking the supremum over ϑ gives $0 \leq D_n \leq \Delta_n(\hat{\vartheta})$ on $\mathcal{A}_n$. Step 1 established $\Delta_n(\hat{\vartheta}) = o_P(n^{-1})$ on the entire sample space, while $\mathbb{P}(\mathcal{A}_n^c) \rightarrow 0$ is ensured by the application of Lemma B.2. Therefore, for every $\epsilon > 0$, $\mathbb{P}(nD_n > \epsilon) \leq \mathbb{P}(\mathcal{A}_n^c) + \mathbb{P}\{n\Delta_n(\hat{\vartheta}) > \epsilon\} \rightarrow 0$, which proves $D_n = o_P(n^{-1})$. We apply Wald's theorem to the compact set $K_\delta \equiv K_{R_\delta} \times \overline{\mathbb{R}}$. The equivalence $\hat{\vartheta}_{n, T_n}^{\text{SM}} \in K_\delta \Leftrightarrow \hat{\theta}_{n, T_n}^{\text{SM}} \in K_{R_\delta}$ gives

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left\{ \min_{s \in \{+1, -1\}} \|s \hat{\vartheta}_{n, T_n}^{\text{SM}} - \vartheta^*\| \geq \epsilon \right\} \cap \left\{ \hat{\theta}_{n, T_n}^{\text{SM}} \in K_{R_\delta} \right\} \right) = 0 \quad \text{for every } \epsilon > 0.$$

The union bound $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) \leq \mathbb{P}(A \cap B) + 1 - \mathbb{P}(B)$, together with the definitions of R_δ and K_{R_δ} , gives

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\min_{s \in \{+1, -1\}} \|s \hat{\vartheta}_{n, T_n}^{\text{SM}} - \vartheta^*\| \geq \epsilon \right) \leq 1 - \liminf_{n \rightarrow \infty} \mathbb{P} \left(\hat{\theta}_{n, T_n}^{\text{SM}} \in K_{R_\delta} \right) \leq \delta \quad \text{for every } \epsilon > 0.$$

Letting δ tend to zero shows that $\lim_{n \rightarrow \infty} \mathbb{P} \left(\min_{s \in \{+1, -1\}} \|s \hat{\vartheta}_{n, T_n}^{\text{SM}} - \vartheta^*\| \geq \epsilon \right) = 0$, and hence

$$\min_{s \in \{+1, -1\}} \|s \hat{\vartheta}_{n, T_n}^{\text{SM}} - \vartheta^*\| \xrightarrow{\mathbb{P}} 0.$$

This also implies $\min_{s \in \{+1, -1\}} \|s \hat{\theta}_{n, T_n}^{\text{SM}} - \theta^*\| = o_P(1)$. The preceding convergence and the triangle inequality also give $\|\hat{\vartheta}_{n, T_n}^{\text{SM}}\| = \mathcal{O}_P(1)$ and $|\hat{\nu}_{n, T_n}^{\text{SM}}| = \mathcal{O}_P(1)$.

$$\|\hat{\vartheta}_{n, T_n}^{\text{SM}}\| \leq \|\vartheta^*\| + \min_{s \in \{+1, -1\}} \|s \hat{\vartheta}_{n, T_n}^{\text{SM}} - \vartheta^*\| = \mathcal{O}(1) + o_P(1) = \mathcal{O}_P(1),$$

$$|\hat{\nu}_{n, T_n}^{\text{SM}}| = \sqrt{\|\hat{\vartheta}_{n, T_n}^{\text{SM}}\|^2 - \|\hat{\theta}_{n, T_n}^{\text{SM}}\|^2} \leq \|\hat{\vartheta}_{n, T_n}^{\text{SM}}\| = \mathcal{O}_P(1).$$

Step 3. Asymptotic normality.

We apply Theorem 5.23 of [van der Vaart \(1998, p. 53\)](#). The following paragraphs verify its assumptions for the aligned estimator.

The aligned estimator is consistent. Step 2 gives

$$\tilde{\vartheta}_{n, T_n}^{\text{SM}} := s_n \hat{\vartheta}_{n, T_n}^{\text{SM}} \xrightarrow{\mathbb{P}} \vartheta^*.$$

The aligned estimator is near optimal. For $m_{\vartheta}(y_0, \mathbf{x}) := \ln p(y_0 \mid \mathbf{x}; \vartheta)$, Lemma [A.2](#) gives the symmetry $\hat{\mathcal{R}}_n^{\text{MLE}}(\vartheta) = \hat{\mathcal{R}}_n^{\text{MLE}}(-\vartheta)$. Therefore, the near-optimality bound from Step 2 also holds after sign alignment.

$$\mathbb{E}_{\mathbb{P}_n} m_{\tilde{\vartheta}_{n, T_n}^{\text{SM}}} \geq \sup_{\vartheta} \mathbb{E}_{\mathbb{P}_n} m_{\vartheta} - o_P(1/n).$$

The population criterion has a nonsingular quadratic expansion. The conditional log-likelihood $m_{\vartheta}(y_0, \mathbf{x}) = \ln p(y_0 \mid \mathbf{x}; \vartheta)$ is constructed from $\ln \cosh(\cdot)$ and linear maps (Lemma [A.2](#)). Hence the map $\vartheta \mapsto m_{\vartheta}(y_0, \mathbf{x})$ is smooth in $\vartheta \equiv (\theta, \nu)$ for all $y_0, \mathbf{x}$. In particular, the derivative ∇m_{ϑ^*} with respect to ϑ at $\vartheta = \vartheta^*$ exists. Writing $u = \theta^\top \mathbf{x}$, the Hessian blocks of the conditional log-likelihood are

$$\begin{aligned} \nabla_{\theta\theta}^2 m_{\vartheta} &= \mathbf{x}\mathbf{x}^\top [y_0^2 \operatorname{sech}^2(y_0 u + \nu) - 1], \\ \nabla_{\theta\nu}^2 m_{\vartheta} &= \mathbf{x} y_0 \operatorname{sech}^2(y_0 u + \nu), \\ \nabla_{\nu\nu}^2 m_{\vartheta} &= \operatorname{sech}^2(y_0 u + \nu) - \operatorname{sech}^2 \nu. \end{aligned}$$

Their norms have a common integrable bound over ϑ given by a linear combination of $\|\mathbf{x}\|^2(y_0^2 + 1)$, $\|\mathbf{x}\| |y_0|$, and 1. Under the fourth-moment condition and the representation $y_0 = (-1)^{z+1}(\theta^*)^\top \mathbf{x} + \varepsilon_0$, this envelope is integrable. Because the Hessian entries are continuous in ϑ , the Dominated Convergence Theorem ([Durrett, 2019, Theorem 1.5.8](#)) justifies differentiating under the expectation twice and shows that the population criterion has a second-order Taylor expansion at ϑ^* , governed by

$$V_{\vartheta^*} := \nabla^2 \mathbb{E}_{y_0, \mathbf{x} \mid \vartheta^*} [m_{\vartheta}] \Big|_{\vartheta=\vartheta^*} = \mathbb{E}_{y_0, \mathbf{x} \mid \vartheta^*} [\nabla^2 m_{\vartheta^*}].$$

The information matrix equality gives

$$I(\vartheta^*) := \mathbb{E}_{y_0, \mathbf{x} \mid \vartheta^*} [\nabla m_{\vartheta^*} \nabla m_{\vartheta^*}^\top] = -\mathbb{E}_{y_0, \mathbf{x} \mid \vartheta^*} [\nabla^2 m_{\vartheta^*}] = -\mathbb{E}_{y_0, \mathbf{x} \mid \vartheta^*} [\nabla_{\vartheta}^2 \ln p(y_0 \mid \mathbf{x}; \vartheta)] = -V_{\vartheta^*}.$$

We verify nonsingularity through the score coordinates. For $(a, b) \in \mathbb{R}^d \times \mathbb{R}$, set $m = (\theta^*)^\top \mathbf{x}$ and $q = a^\top \mathbf{x}$. If $(a, b)^\top I(\vartheta^*)(a, b) = 0$, then the corresponding linear combination of the score coordinates vanishes almost surely.

$$(q y_0 + b) \tanh(m y_0 + \nu^*) - q m - b \tanh \nu^* = 0.$$

For almost every $\mathbf{x}$, the conditional density $y_0 \mapsto p(y_0 \mid \mathbf{x}; \vartheta^*)$ is positive on $\mathbb{R}$. The displayed expression is continuous in y_0 , so it vanishes for every y_0 . For every such $\mathbf{x}$ with $m \neq 0$, division by y_0 followed by $y_0 \rightarrow +\infty$ gives $q = 0$, after which the nonconstancy of $y_0 \mapsto \tanh(m y_0 + \nu^*)$ gives $b = 0$. Because $\theta^* \neq \mathbf{0}$ and the density of $\mathbf{x}$ is positive almost everywhere, $m \neq 0$ almost surely, and $a^\top \mathbf{x} = 0$ almost surely forces $a = 0$. Thus the Fisher information matrix $I(\vartheta^*)$ is positive definite and nonsingular.

A square-integrable Lipschitz envelope exists. Let U be a fixed convex neighborhood of ϑ^* with radius R , and define $\dot{m}(y_0, \mathbf{x}) := \sup_{\vartheta \in U} \|\nabla m_{\vartheta}(y_0, \mathbf{x})\|$. The Mean Value Theorem (Theorem 9.19 on page 218

of Rudin (1976)) gives $|m_{\vartheta_1}(y_0, \mathbf{x}) - m_{\vartheta_2}(y_0, \mathbf{x})| \leq \dot{m}(y_0, \mathbf{x}) \|\vartheta_1 - \vartheta_2\|$ for every $\vartheta_1, \vartheta_2 \in U$. Lemma A.2 gives $\nabla_{\theta} m_{\vartheta} = \nabla_{\theta} \ln p(y_0 | \mathbf{x}; \vartheta) = \mathbf{x}(y_0 \tanh(y_0 \theta^\top \mathbf{x} + \nu) - \theta^\top \mathbf{x})$ and $\nabla_{\nu} m_{\vartheta} = \nabla_{\nu} \ln p(y_0 | \mathbf{x}; \vartheta) = \tanh(y_0 \theta^\top \mathbf{x} + \nu) - \tanh \nu$. We apply the triangle inequality and $|\tanh(\cdot)| \leq 1$. The bounds $\sup_{\vartheta \in U} \|\theta\| \leq R + \|\theta^*\|$ and $|y_0| \leq \|\theta^*\| \|\mathbf{x}\| + |\varepsilon_0|$ yield

$$\dot{m}(y_0, \mathbf{x}) \leq \|\mathbf{x}\|(\|\theta^*\| \|\mathbf{x}\| + |\varepsilon_0| + (R + \|\theta^*\|) \|\mathbf{x}\|) + 2 \leq \left(R + 2\|\theta^*\| + \frac{1}{2}\right) \|\mathbf{x}\|^2 + \frac{1}{2} \varepsilon_0^2 + 2.$$

Because the fourth moments are finite ($\mathbb{E}[\|\mathbf{x}\|_2^4] < \infty$) and $\mathbb{E}[\varepsilon_0^4] = 3$, the Lipschitz envelope satisfies the square-integrability requirement ($\mathbb{E}_{y_0, \mathbf{x} | \vartheta^*}[\dot{m}^2(y_0, \mathbf{x})] < \infty$), noting that $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$.

The aligned estimator $\tilde{\vartheta}_{n, T_n}^{\text{SM}} := s_n \hat{\vartheta}_{n, T_n}^{\text{SM}}$ is consistent. The event argument in Step 2 shows that its empirical likelihood suboptimality is $o_P(n^{-1})$. The verified conditions allow Theorem 5.23 of van der Vaart (1998) to be applied. It gives

$$\sqrt{n} \left(\tilde{\vartheta}_{n, T_n}^{\text{SM}} - \vartheta^* \right) \xrightarrow{d} \mathcal{N} \left(\mathbf{0}, (-V_{\vartheta^*})^{-1} \mathbb{E}_{y_0, \mathbf{x} | \vartheta^*} [\nabla m_{\vartheta^*} \nabla m_{\vartheta^*}^\top] (-V_{\vartheta^*})^{-1} \right) = \mathcal{N}(\mathbf{0}, I(\theta^*, \nu^*)^{-1}).$$

This completes the proof, noting that $I(\theta^*, \nu^*) \equiv I(\vartheta^*) = \mathbb{E}_{y_0, \mathbf{x} | \vartheta^*} [\nabla m_{\vartheta^*} \nabla m_{\vartheta^*}^\top] = -V_{\vartheta^*}$. $\square$

Lemma B.2 (High-probability attainment of the empirical minimum). The dimension d is fixed, and the observations are i.i.d. from the MLR model with $\theta^* \neq \mathbf{0}$ and $\nu^* \in \mathbb{R}$. The covariate $\mathbf{x}$ has a density that is positive almost everywhere on $\mathbb{R}^d$ and satisfies $\mathbb{E}\|\mathbf{x}\|^2 < \infty$. The terminal horizon satisfies $\bar{\alpha}_{T_n}/(1 - \bar{\alpha}_{T_n}) = o(n^{-1})$. Then the probability that the empirical Score Matching objective attains its minimum over $\mathbb{R}^{d+1}$ tends to one. A sample-measurable exact minimizer can be selected whenever this minimum exists.

Proof. Set $a_n := \bar{\alpha}_{T_n}$, $\sigma_n := \sqrt{a_n(1 - a_n)}$, $\Sigma_n := \mathbb{E}_{\mathbb{P}_n}[\mathbf{x}\mathbf{x}^\top]$, and $m_n := \mathbb{E}_{\mathbb{P}_n}[\mathbf{x}y_0]$. For $u_\theta := \theta^\top \mathbf{x}$, define $r_\theta := y_0 u_\theta$ and $q_{\theta, \xi} := a_n r_\theta + \sigma_n u_\theta \xi$, where $\xi \sim \mathcal{N}(0, 1)$. The terminal-horizon condition gives $a_n \rightarrow 0$. The empirical bridge in Lemma A.16 and the Gaussian mixture formula in Lemma A.2 show that, up to a parameter-independent random term, the empirical score matching criterion is

$$\mathbf{G}_n(\theta, \nu) = \frac{1 - a_n}{2} \theta^\top \Sigma_n \theta - \mathbb{E}_{\mathbb{P}_n} \log \cosh(r_\theta + \nu) + \mathbb{E}_{\mathbb{P}_n} \mathbb{E}_\xi \log \cosh(q_{\theta, \xi} + \nu).$$

Set $\rho := \tanh \nu$ and $\psi_\rho(z) := \log(\cosh z + \rho \sinh z)$. The identity $\log \cosh(\nu + z) - \log \cosh \nu = \psi_\rho(z)$ gives the continuous compactification

$$\bar{\mathbf{G}}_n(\theta, \rho) = \frac{1 - a_n}{2} \theta^\top \Sigma_n \theta - \mathbb{E}_{\mathbb{P}_n} \psi_\rho(r_\theta) + \mathbb{E}_{\mathbb{P}_n} \mathbb{E}_\xi \psi_\rho(q_{\theta, \xi}), \quad \rho \in [-1, 1].$$

Indeed, $\cosh z + \rho \sinh z > 0$ on $[-1, 1]$, and $\psi_{\pm 1}(z) = \pm z$. For $|\rho| < 1$, the derivative is $\psi'_\rho(z) = \tanh(z + \arctanh \rho)$, while the endpoint derivatives are ± 1 . Thus ψ_ρ is 1-Lipschitz. Define $L_n := \{\mathbb{E}_{\mathbb{P}_n}[(1 - a_n)|y_0| + \sigma_n \mathbb{E}[\xi]^2]\}^{1/2}$. Cauchy-Schwarz gives

$$\bar{\mathbf{G}}_n(\theta, \rho) \geq \frac{1 - a_n}{2} \|\theta\|_{\Sigma_n}^2 - L_n \|\theta\|_{\Sigma_n}.$$

Thus the compactified criterion is coercive in fitted-value space $\text{Im}(X)$, where the design matrix $X \in \mathbb{R}^{n \times d}$ has rows $(\mathbf{x}^{(i)})^\top$. It has a compact nonempty argmin there, and X^+ gives its minimum-norm lift to $\mathbb{R}^d$. For $n \geq d$, the positive-density assumption makes Σ_n positive definite with probability one, so the criterion is coercive in θ itself.

At $\rho = \pm 1$, direct substitution gives

$$\bar{\mathbf{G}}_n(\theta, \pm 1) = (1 - a_n) \left(\frac{1}{2} \theta^\top \Sigma_n \theta \mp \theta^\top m_n \right), \quad b_n = -\frac{1 - a_n}{2} m_n^\top \Sigma_n^+ m_n.$$

Here b_n is the common boundary minimum because $m_n \in \text{Range}(\Sigma_n)$. For $n \geq d$, we have $\Sigma_n^+ = \Sigma_n^{-1}$ almost surely. The weak law gives

$$\Sigma_n \xrightarrow{\mathbb{P}} \Sigma, \quad m_n \xrightarrow{\mathbb{P}} \Sigma \theta^* \tanh \nu^*, \quad b_n \xrightarrow{\mathbb{P}} b_\infty := -\frac{1}{2} \tanh^2(\nu^*) (\theta^*)^\top \Sigma \theta^*.$$

Put $u^* := (\theta^*)^\top \mathbf{x}$, $r^* := y_0 u^*$, and $H_n := \mathbb{E}_{\mathbb{P}_n} \mathbb{E}_\xi \log \cosh(\nu^* + a_n r^* + \sigma_n u^* \xi)$. The Lipschitz bound yields

$$|H_n - \log \cosh \nu^*| \leq a_n \mathbb{E}_{\mathbb{P}_n} |r^*| + \sigma_n \mathbb{E} |\xi| \mathbb{E}_{\mathbb{P}_n} |u^*| = o_P(1).$$

The empirical moments on the right are $\mathcal{O}_P(1)$. The weak law applied to the other terms therefore gives

$$\mathbf{G}_n(\theta^*, \nu^*) \xrightarrow{\mathbb{P}} g_* := \mathbb{E} \left[\frac{1}{2} (u^*)^2 - \log \frac{\cosh(r^* + \nu^*)}{\cosh \nu^*} \right].$$

The boundary gap has the distributional representation

$$b_\infty - g_* = \mathbb{E}_{\mathbf{x}} \text{KL}(p(\cdot | \mathbf{x}; \theta^*, \nu^*) \| \mathcal{N}(\tanh \nu^* u^*, 1)) > 0.$$

The inequality is strict because the true conditional variance is $1 + \text{sech}^2(\nu^*) (u^*)^2 > 1$ for almost every $\mathbf{x}$, whereas the boundary Gaussian has variance one. Consequently,

$$\mathbb{P}\{\mathbf{G}_n(\theta^*, \nu^*) < b_n\} \longrightarrow 1.$$

On this event, every compactified minimizer has $|\rho| < 1$ and maps through $\nu = \text{arctanh } \rho$ to an exact finite minimizer. Hence

$$\mathbb{P} \left\{ \arg \min_{\vartheta \in \mathbb{R}^{d+1}} \widehat{\mathcal{R}}_{n, T_n}^{\text{SM}}(\vartheta) \neq \emptyset \right\} \longrightarrow 1.$$

For measurability, write $u = X\theta \in \mathbb{R}^n$. The identity $u \in \text{Im}(X)$ if and only if $XX^+u = u$ gives a measurable graph for the fitted-value space. We extend the compactified criterion by $+\infty$ off this space. The measurable minimum theorem then gives a measurable compact argmin correspondence on $\mathbb{R}^n \times [-1, 1]$. We minimize $|\rho|$ over this correspondence, take a measurable selector $(\tilde{u}_n, \tilde{\rho}_n)$, and set $\tilde{\theta}_n := X^+ \tilde{u}_n$. This lift is measurable because $X \mapsto X^+$ is Borel measurable. The selector satisfies

$$\left\{ \arg \min_{\vartheta \in \mathbb{R}^{d+1}} \widehat{\mathcal{R}}_{n, T_n}^{\text{SM}}(\vartheta) \neq \emptyset \right\} = \{|\tilde{\rho}_n| < 1\}.$$

The event on the left is therefore measurable, and $\text{arctanh } \tilde{\rho}_n$ gives the claimed exact selection on that event. $\square$

Remark (Fixed-Sample Nonattainment). Finite-parameter attainment need not hold on the whole sample space for a fixed n . For example, take $n = d = 1$, $0 < a_1 < 1$, $x \neq 0$, and $y_0 \neq 0$, and write $u = \theta x$. Jensen's inequality and the 1-Lipschitz property of $\log \cosh$ give

$$\mathbf{G}_1(\theta, \nu) \geq \frac{1 - a_1}{2} (|u| - |y_0|)^2 - \frac{1 - a_1}{2} y_0^2.$$

The inequality is strict for every finite ν when $u \neq 0$, while the criterion is zero at $u = 0$. The boundary choices $(u, \rho) = (y_0, 1)$ and $(-y_0, -1)$ attain the negative lower bound. Thus the finite-parameter argmin is empty for this sample, although Lemma B.2 shows that this obstruction has vanishing probability in the stated asymptotic regime.

C Proofs of Cross-Entropy and EM Decompositions for Fixed-Scale Score Matching

This appendix restates and proves the exact fixed-scale decompositions, gradient identities, and noise-limit results from Section 4: Propositions 4.1, 4.2, 4.6, 4.7, and 4.8, Main Theorem 4.3, and Corollaries 4.4 and 4.5.

Proposition C.1 (Proposition 4.1). *The EM update rules for the regression parameters θ_t and the imbalance parameter ν are*

$$M(\theta_t, \nu) = \Sigma^{-1} \mathbb{E}_*[\mathbf{x} y_t \tanh(\mu_t y_t + \nu)], \quad N(\theta_t, \nu) = \mathbb{E}_*[\tanh(\mu_t y_t + \nu)].$$

They also satisfy

$$\langle \Sigma M(\theta_t, \nu), \theta_t \rangle = \mathbb{E}_*[\mu_t y_t \tanh(\mu_t y_t + \nu)] = \mathbb{E}_*[y_t^2 - y_t \partial_{y_t} F(y_t, \mu_t, \nu)].$$

Their derivatives satisfy

$$\Sigma \nabla_\nu M(\theta_t, \nu) = \nabla_{\theta_t} N(\theta_t, \nu) = \mathbb{E}_*[\mathbf{x} y_t \operatorname{sech}^2(\mu_t y_t + \nu)].$$

Proof for Proposition 4.1. The calculation adapts Appendix B of Luo & Hashemi (2024) and Proposition 3 of Luo & Hashemi (2025) to the diffused parameter θ_t . Those results use the clean-scale parameter. Here we replace it with θ_t from Equation (4).

The marginal law of $\mathbf{x}$ does not depend on (θ_t, ν) , so $-\ln p(y_t, \mathbf{x} \mid \theta_t, \pi) = F(y_t, \mu_t, \nu) - \ln p(\mathbf{x})$. Since $\mu_t = \langle \theta_t, \mathbf{x} \rangle$, differentiation gives

$$M(\theta_t, \nu) = \theta_t - \mathbb{E}[\mathbf{x} \mathbf{x}^\top]^{-1} \nabla_{\theta_t} \mathbb{E}_{\mathbf{x}} \mathbb{E}_{y_t | \mathbf{x}; \theta_t^*, \pi^*} [-\ln p(y_t, \mathbf{x} \mid \theta_t, \pi)] = \theta_t - \Sigma^{-1} \mathbb{E}_{\mathbf{x}} \mathbb{E}_{y_t | \mathbf{x}; \theta_t^*, \pi^*} [\mathbf{x} \partial_{\mu_t} F(y_t, \mu_t, \nu)],$$

$$N(\theta_t, \nu) = \tanh \nu - \nabla_\nu \mathbb{E}_{\mathbf{x}} \mathbb{E}_{y_t | \mathbf{x}; \theta_t^*, \pi^*} [-\ln p(y_t, \mathbf{x} \mid \theta_t, \pi)] = \tanh \nu - \mathbb{E}_{\mathbf{x}} \mathbb{E}_{y_t | \mathbf{x}; \theta_t^*, \pi^*} [\partial_\nu F(y_t, \mu_t, \nu)].$$

Lemma A.17 gives $\partial_{\mu_t} F = \mu_t - y_t \tanh(\mu_t y_t + \nu)$ and $\partial_\nu F = \tanh \nu - \tanh(\mu_t y_t + \nu)$. Substitution yields

$$M(\theta_t, \nu) = \Sigma^{-1} \mathbb{E}_*[\mathbf{x} y_t \tanh(\mu_t y_t + \nu)], \quad N(\theta_t, \nu) = \mathbb{E}_*[\tanh(\mu_t y_t + \nu)].$$

The same derivative formulas imply $\mu_t \partial_{\mu_t} F - \mu_t^2 = y_t \partial_{y_t} F - y_t^2 = -\mu_t y_t \tanh(\mu_t y_t + \nu)$. Therefore

$$\langle \Sigma M(\theta_t, \nu), \theta_t \rangle = \mathbb{E}_*[\mu_t y_t \tanh(\mu_t y_t + \nu)] = \mathbb{E}_*[y_t^2 - y_t \partial_{y_t} F(y_t, \mu_t, \nu)].$$

The identity $\tanh'(u) = \operatorname{sech}^2(u)$ and the relation $\nabla_{\theta_t} \mu_t = \mathbf{x}$ give the derivative formula below. The dominated convergence theorem justifies moving the derivatives through $\mathbb{E}_*$ (Durrett, 2019, Theorem 1.5.8).

$$\Sigma \nabla_\nu M(\theta_t, \nu) = \mathbb{E}_*[\mathbf{x} y_t \nabla_\nu \tanh(\mu_t y_t + \nu)] = \mathbb{E}_*[\mathbf{x} y_t \operatorname{sech}^2(\mu_t y_t + \nu)] = \mathbb{E}_*[\nabla_{\theta_t} \tanh(\mu_t y_t + \nu)] = \nabla_{\theta_t} N(\theta_t, \nu).$$

□

Proposition C.2 (Proposition 4.2). *The cross-entropy $\mathcal{H}(\theta_t, \nu)$ defined in Section 2.3 satisfies*

$$\nabla_{\theta_t} \mathcal{H}(\theta_t, \nu) = \Sigma (\theta_t - M(\theta_t, \nu)), \quad \nabla_\nu \mathcal{H}(\theta_t, \nu) = \tanh \nu - N(\theta_t, \nu).$$

Proof for Proposition 4.2. The calculation in the preceding proof gives

$$M(\theta_t, \nu) = \theta_t - \Sigma^{-1} \mathbb{E}_*[\mathbf{x} \partial_{\mu_t} F(y_t, \mu_t, \nu)], \quad N(\theta_t, \nu) = \tanh \nu - \mathbb{E}_*[\partial_\nu F(y_t, \mu_t, \nu)].$$

The same integrable bounds permit differentiation through the expectation (Durrett, 2019, Theorem 1.5.8). Since $\nabla_{\theta_t} \mu_t = \mathbf{x}$, rearrangement gives

$$\Sigma (\theta_t - M(\theta_t, \nu)) = \mathbb{E}_*[\mathbf{x} \partial_{\mu_t} F(y_t, \mu_t, \nu)] = \mathbb{E}_*[\nabla_{\theta_t} F(y_t, \mu_t, \nu)] = \nabla_{\theta_t} \mathbb{E}_*[F(y_t, \mu_t, \nu)] = \nabla_{\theta_t} \mathcal{H}(\theta_t, \nu),$$

$$\tanh \nu - N(\theta_t, \nu) = \mathbb{E}_*[\partial_\nu F(y_t, \mu_t, \nu)] = \nabla_\nu \mathbb{E}_*[F(y_t, \mu_t, \nu)] = \nabla_\nu \mathcal{H}(\theta_t, \nu).$$

□

Main theorem C.3 (Main Theorem 4.3). *We define the expected squared activation $\mathcal{A}(\theta_t, \nu) := \frac{1}{2}\mathbb{E}_*[\mu_t^2 \tanh^2(\mu_t y_t + \nu)]$ and its ground truth counterpart $\mathcal{A}(\theta_t^*, \nu^*) := \frac{1}{2}\mathbb{E}_*[(\mu_t^*)^2 \tanh^2(\mu_t^* y_t + \nu^*)]$. The Score Matching loss satisfies*

$$\bar{\alpha}_t \mathcal{L}_t(\theta, \nu) = \langle \nabla_{\theta_t} \mathcal{H}(\theta_t, \nu), \theta_t \rangle - \mathcal{A}(\theta_t, \nu) + \mathcal{A}(\theta_t^*, \nu^*).$$

We also define the latent variance $V(\theta_t, \nu) \equiv \frac{1}{2}\mathbb{E}_*[\mu_t^2 \text{sech}^2(\mu_t y_t + \nu)]$. The equivalent representation is

$$\bar{\alpha}_t \mathcal{L}_t(\theta, \nu) = \|\theta_t\|_\Sigma^2/2 - \langle \Sigma M(\theta_t, \nu), \theta_t \rangle + \|\theta_t^*\|_\Sigma^2/2 + V(\theta_t, \nu) - V(\theta_t^*, \nu^*).$$

Here $V(\theta_t^*, \nu^*) \equiv \frac{1}{2}\mathbb{E}_*[(\mu_t^*)^2 \text{sech}^2(\mu_t^* y_t + \nu^*)]$ does not depend on the candidate parameters.

Proof for Main Theorem 4.3. The candidate score is $s_{\theta_t, \nu}(y_t, \mathbf{x}) = -y_t + \mu_t \tanh(\mu_t y_t + \nu)$. Together with its ground-truth counterpart, it gives

$$\mathcal{L}_t(\theta, \nu) = \frac{1}{2\bar{\alpha}_t} \mathbb{E}_{y_t, \mathbf{x} | \theta_t^*, \pi^*} \|s_{\theta_t, \nu}(y_t, \mathbf{x}) - s_{\theta_t^*, \nu^*}(y_t, \mathbf{x})\|^2.$$

Lemmas A.2 and A.3 give $s_{\theta_t, \nu} = -\partial_{y_t} F(y_t, \mu_t, \nu)$ and $p(y_t | \mathbf{x}; \theta_t^*, \pi^*) = \exp[-F(y_t, \mu_t^*, \nu^*)]$. Hence

$$\mathcal{L}_t(\theta, \nu) = \frac{1}{2\bar{\alpha}_t} \mathbb{E}_{\mathbf{x}} \int_{\mathbb{R}} \|\partial_{y_t} F(y_t, \mu_t, \nu) + \partial_{y_t} F(y_t, \mu_t^*, \nu^*)\|^2 \exp(-F(y_t, \mu_t^*, \nu^*)) dy_t.$$

Integration by parts in y_t is valid because the boundary terms vanish. Applying it once to the mixed term and once to the ground-truth score term gives

$$2\bar{\alpha}_t \mathcal{L}_t(\theta, \nu) = \mathbb{E}_* [\|\partial_{y_t} F(y_t, \mu_t, \nu)\|^2 - 2\partial_{y_t y_t}^2 F(y_t, \mu_t, \nu) + \partial_{y_t y_t}^2 F(y_t, \mu_t^*, \nu^*)].$$

Lemma A.17 gives $\|\partial_{y_t} F\|^2 - \partial_{y_t y_t}^2 F = 2[y_t \partial_{y_t} F - y_t^2] + (\mu_t^2 + y_t^2 - 1)$ and $\partial_{y_t y_t}^2 F = 1 - \mu_t^2 \text{sech}^2(\mu_t y_t + \nu)$. Equation (4) also gives $\mathbb{E}[y_t^2 | \mathbf{x}] = (\mu_t^*)^2 + 1$. These identities yield

$$2\bar{\alpha}_t \mathcal{L}_t(\theta, \nu) = 2\mathbb{E}_* [y_t \partial_{y_t} F(y_t, \mu_t, \nu) - y_t^2] + \mathbb{E}_{\mathbf{x}} [\mu_t^2 + (\mu_t^*)^2] + \mathbb{E}_* [\mu_t^2 \text{sech}^2(\mu_t y_t + \nu)] - \mathbb{E}_* [(\mu_t^*)^2 \text{sech}^2(\mu_t^* y_t + \nu^*)].$$

Proposition 4.1 identifies the first expectation as $-\langle \Sigma M(\theta_t, \nu), \theta_t \rangle$. The covariance identity gives $\mathbb{E}[\mu_t^2 + (\mu_t^*)^2] = \|\theta_t\|_\Sigma^2 + \|\theta_t^*\|_\Sigma^2$. Thus

$$2\bar{\alpha}_t \mathcal{L}_t(\theta, \nu) = \|\theta_t\|_\Sigma^2 - 2\langle \Sigma M(\theta_t, \nu), \theta_t \rangle + \|\theta_t^*\|_\Sigma^2 + \mathbb{E}_* [\mu_t^2 \text{sech}^2(\mu_t y_t + \nu)] - \mathbb{E}_* [(\mu_t^*)^2 \text{sech}^2(\mu_t^* y_t + \nu^*)].$$

This is the latent-variance representation. For the activation form, we use Proposition 4.2, the identity $\text{sech}^2(u) = 1 - \tanh^2(u)$, and the two covariance identities. We obtain

$$2\bar{\alpha}_t \mathcal{L}_t(\theta, \nu) = 2\langle \nabla_{\theta_t} \mathcal{H}(\theta_t, \nu), \theta_t \rangle - \mathbb{E}_* [\mu_t^2 \tanh^2(\mu_t y_t + \nu)] + \mathbb{E}_* [(\mu_t^*)^2 \tanh^2(\mu_t^* y_t + \nu^*)].$$

Division by two and the definitions of $\mathcal{A}(\theta_t, \nu)$ and $\mathcal{A}(\theta_t^*, \nu^*)$ complete the proof. $\square$

Corollary C.4 (Corollary 4.4). *The gradients of $\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)$ with respect to θ_t and ν are*

$$\begin{aligned} \nabla_{\theta_t} [\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)] &= \nabla_{\theta_t} \mathcal{H}(\theta_t, \nu) + \nabla_{\theta_t}^2 \mathcal{H}(\theta_t, \nu) \theta_t - \nabla_{\theta_t} \mathcal{A}(\theta_t, \nu), \\ \nabla_{\nu} [\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)] &= \langle \nabla_{\theta_t \nu}^2 \mathcal{H}(\theta_t, \nu), \theta_t \rangle - \nabla_{\nu} \mathcal{A}(\theta_t, \nu). \end{aligned}$$

Proof for Corollary 4.4. Differentiate the first identity in Main Theorem 4.3 with respect to θ_t and ν . The product rule gives $\nabla_{\theta_t} \langle \nabla_{\theta_t} \mathcal{H}(\theta_t, \nu), \theta_t \rangle = \nabla_{\theta_t} \mathcal{H}(\theta_t, \nu) + \nabla_{\theta_t}^2 \mathcal{H}(\theta_t, \nu) \theta_t$ and $\nabla_{\nu} \langle \nabla_{\theta_t} \mathcal{H}(\theta_t, \nu), \theta_t \rangle = \langle \nabla_{\theta_t \nu}^2 \mathcal{H}(\theta_t, \nu), \theta_t \rangle$. These formulas prove the two identities. $\square$

Corollary C.5 (Corollary 4.5). *The gradients of $\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)$ with respect to θ_t and ν are*

$$\begin{aligned} \nabla_{\theta_t} [\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)] &= \Sigma(\theta_t - M(\theta_t, \nu)) - (\nabla_{\theta_t} M(\theta_t, \nu))^\top \Sigma \theta_t + \nabla_{\theta_t} V(\theta_t, \nu), \\ \nabla_{\nu} [\bar{\alpha}_t \mathcal{L}_t(\theta, \nu)] &= -\langle \nabla_{\theta_t} N(\theta_t, \nu), \theta_t \rangle + \nabla_{\nu} V(\theta_t, \nu). \end{aligned}$$

Proof for Corollary 4.5. Differentiate the second identity in Main Theorem 4.3 with respect to θ_t and ν . The product rule gives $\nabla_{\theta_t} \|\theta_t\|_{\Sigma}^2 = 2\Sigma\theta_t$ and $\nabla_{\theta_t} \langle \Sigma M(\theta_t, \nu), \theta_t \rangle = \Sigma M(\theta_t, \nu) + (\nabla_{\theta_t} M(\theta_t, \nu))^{\top} \Sigma \theta_t$. Proposition 4.1 gives $\nabla_{\nu} \langle \Sigma M(\theta_t, \nu), \theta_t \rangle = \langle \Sigma \nabla_{\nu} M(\theta_t, \nu), \theta_t \rangle = \langle \nabla_{\theta_t} N(\theta_t, \nu), \theta_t \rangle$. These formulas prove the two identities. $\square$

We define the normalized EM operators at time step t by

$$\bar{M}_t(\theta, \nu) := M(\theta_t, \nu) / \sqrt{\bar{\alpha}_t}, \quad \bar{N}_t(\theta, \nu) := (N(\theta_t, \nu) - (1 - \bar{\alpha}_t) \tanh \nu) / \bar{\alpha}_t.$$

They satisfy $\theta - \bar{M}_t(\theta, \nu) = (\theta_t - M(\theta_t, \nu)) / \sqrt{\bar{\alpha}_t}$, $\tanh \nu - \bar{N}_t(\theta, \nu) = (\tanh \nu - N(\theta_t, \nu)) / \bar{\alpha}_t$, and $\Sigma \nabla_{\nu} \bar{M}_t(\theta, \nu) = \nabla_{\theta} \bar{N}_t(\theta, \nu)$. We also set $V_t(\theta, \nu) := V(\theta_t, \nu) / \bar{\alpha}_t$.

Proposition C.6 (Proposition 4.6). *The gradients of the Score Matching loss in Equation (6) are*

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_t(\theta, \nu) &= \Sigma(\theta - \bar{M}_t(\theta, \nu)) - (\nabla_{\theta} \bar{M}_t(\theta, \nu))^{\top} \Sigma \theta + \nabla_{\theta} V_t(\theta, \nu), \\ \nabla_{\nu} \mathcal{L}_t(\theta, \nu) &= -\langle \nabla_{\theta} \bar{N}_t(\theta, \nu), \theta \rangle + \nabla_{\nu} V_t(\theta, \nu). \end{aligned}$$

Remark. Couple the diffused responses by $y_t = \sqrt{\bar{\alpha}_t} y_0 + \sqrt{1 - \bar{\alpha}_t} \xi_t$. Then $y_t \rightarrow y_0$ almost surely as $\bar{\alpha}_t \rightarrow 1$. The functions $\tanh$ and sech^2 are bounded. The operator integrands and their first parameter derivatives therefore have a common integrable envelope when $\mathbb{E}[\|\mathbf{x}\|^4] < \infty$. The Gaussian noise has moments of every order. Dominated convergence now gives $\bar{M}_t(\theta, \nu) \rightarrow M(\theta, \nu)$, $\bar{N}_t(\theta, \nu) \rightarrow N(\theta, \nu)$, and $V_t(\theta, \nu) \rightarrow V(\theta, \nu)$, together with the parameter derivative limits used in Proposition 4.6. Substitution into that proposition gives

$$\begin{aligned} \lim_{\bar{\alpha}_t \rightarrow 1} \nabla_{\theta} \mathcal{L}_t(\theta, \nu) &= \Sigma(\theta - M(\theta, \nu)) - (\nabla_{\theta} M(\theta, \nu))^{\top} \Sigma \theta + \nabla_{\theta} V(\theta, \nu), \\ \lim_{\bar{\alpha}_t \rightarrow 1} \nabla_{\nu} \mathcal{L}_t(\theta, \nu) &= -\langle \nabla_{\theta} N(\theta, \nu), \theta \rangle + \nabla_{\nu} V(\theta, \nu). \end{aligned}$$

Proposition 4.7 strengthens these limits by proving an $\mathcal{O}(1 - \bar{\alpha}_t)$ remainder.

Proof for Proposition 4.6. Recall the normalized EM operators $\bar{M}_t(\theta, \nu) := M(\theta_t, \nu) / \sqrt{\bar{\alpha}_t}$, $\bar{N}_t(\theta, \nu) := (N(\theta_t, \nu) - (1 - \bar{\alpha}_t) \tanh \nu) / \bar{\alpha}_t$, and $V_t(\theta, \nu) := V(\theta_t, \nu) / \bar{\alpha}_t$. These definitions guarantee $\theta - \bar{M}_t(\theta, \nu) = (\theta_t - M(\theta_t, \nu)) / \sqrt{\bar{\alpha}_t}$, $\tanh \nu - \bar{N}_t(\theta, \nu) = (\tanh \nu - N(\theta_t, \nu)) / \bar{\alpha}_t$, and $\Sigma \nabla_{\nu} \bar{M}_t(\theta, \nu) = \nabla_{\theta} \bar{N}_t(\theta, \nu)$. Dividing the identities of Corollary 4.5 by $\bar{\alpha}_t$ and applying the chain rule with $\theta_t = \sqrt{\bar{\alpha}_t} \theta$ yields the stated gradients of $\mathcal{L}_t(\theta, \nu)$. $\square$

Proposition C.7 (Proposition 4.7). *If $\mathbb{E}[\|\mathbf{x}\|^4] < \infty$, then the gradients of the Score Matching loss satisfy the following expansions as $\bar{\alpha}_t \rightarrow 1$.*

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_t(\theta, \nu) &= \Sigma(\theta - M(\theta, \nu)) - (\nabla_{\theta} M(\theta, \nu))^{\top} \Sigma \theta + \nabla_{\theta} V(\theta, \nu) + \mathcal{O}(1 - \bar{\alpha}_t), \\ \nabla_{\nu} \mathcal{L}_t(\theta, \nu) &= -\langle \nabla_{\theta} N(\theta, \nu), \theta \rangle + \nabla_{\nu} V(\theta, \nu) + \mathcal{O}(1 - \bar{\alpha}_t). \end{aligned}$$

Proof of Proposition 4.7. We write $\mu_0 = \langle \theta, \mathbf{x} \rangle$, $\mu_0^* = \langle \theta^*, \mathbf{x} \rangle$, and $a = \bar{\alpha}_t$. Since the loss depends only on the marginal law at each scale, Equation (4) permits the common coupling

$$y_t = (-1)^{z+1} \sqrt{a} \mu_0^* + \varepsilon, \quad \mu_t = \sqrt{a} \mu_0, \quad \mu_t^* = \sqrt{a} \mu_0^*,$$

where $\varepsilon \sim \mathcal{N}(0, 1)$ is independent of $(\mathbf{x}, z)$. Equations (5) and (6) show that cancellation of the linear terms in the two scores gives the normalized score difference

$$G_t := \mu_0 \tanh(\mu_t y_t + \nu) - \mu_0^* \tanh(\mu_t^* y_t + \nu^*), \quad \mathcal{L}_t(\theta, \nu) = \frac{1}{2} \mathbb{E}_{\mathbf{x}, z, \varepsilon} [G_t^2].$$

The bounds below justify the identities

$$\nabla_{\theta} \mathcal{L}_t = \mathbb{E}[G_t \nabla_{\theta} G_t], \quad \nabla_{\nu} \mathcal{L}_t = \mathbb{E}[G_t \partial_{\nu} G_t].$$

We now control the dependence on a near one. For $a \in [1/2, 1]$, differentiation of the two activation arguments gives

$$\partial_a(\mu_t y_t) = \frac{\mu_t y_t}{2a} + \frac{(-1)^{z+1}}{2} \mu_0 \mu_0^*, \quad \partial_a(\mu_t^* y_t) = \frac{\mu_t^* y_t}{2a} + \frac{(-1)^{z+1}}{2} (\mu_0^*)^2.$$

For each fixed $c \in \mathbb{R}$, the function $\tanh(u + c)$ and the quantities $u^k \tanh^{(j)}(u + c)$, with $j \in \{1, 2\}$ and $k \in \{0, 1, 2\}$, are bounded over $u \in \mathbb{R}$. The product and chain rules therefore give a constant C , independent of a , such that

$$|G_t| + \|\nabla_\theta G_t\| + |\partial_\nu G_t| \leq C\|\mathbf{x}\|, \quad |\partial_a G_t| + \|\partial_a \nabla_\theta G_t\| + |\partial_a \partial_\nu G_t| \leq C(\|\mathbf{x}\| + \|\mathbf{x}\|^3).$$

The a -derivatives of the two loss-gradient integrands $G_t \nabla_\theta G_t$ and $G_t \partial_\nu G_t$ are thus bounded by $C(\|\mathbf{x}\|^2 + \|\mathbf{x}\|^4)$. The fourth-moment assumption makes this envelope integrable. Differentiation under the expectation is valid (Folland, 1999, Theorem 2.27(b)), and

$$\sup_{a \in [1/2, 1]} \{ \|\partial_a \nabla_\theta \mathcal{L}_t(\theta, \nu)\| + |\partial_a \nabla_\nu \mathcal{L}_t(\theta, \nu)| \} < \infty.$$

Integration from a to one shows that both loss gradients equal their endpoint values plus $\mathcal{O}(1 - a)$. At $a = 1$, the normalized operators reduce to M , N , and V . Proposition 4.6 identifies the endpoint gradients with the two leading terms in the statement. Substituting $a = \bar{\alpha}_t$ completes the proof. $\square$

Proposition C.8 (Proposition 4.8). *If $\mathbb{E}[\|\mathbf{x}\|^2] < \infty$, then $\lim_{\bar{\alpha}_t \rightarrow 0} \mathcal{L}_t(\theta, \nu) = \frac{1}{2} \|\theta \tanh \nu - \theta^* \tanh \nu^*\|_\Sigma^2$. The condition $\mathbb{E}[\|\mathbf{x}\|^4] < \infty$ suffices for the $\mathcal{O}(\bar{\alpha}_t)$ gradient expansions*

$$\nabla_\theta \mathcal{L}_t(\theta, \nu) = \nabla_\theta \lim_{\bar{\alpha}_t \rightarrow 0} \mathcal{L}_t(\theta, \nu) + \mathcal{O}(\bar{\alpha}_t), \quad \nabla_\nu \mathcal{L}_t(\theta, \nu) = \nabla_\nu \lim_{\bar{\alpha}_t \rightarrow 0} \mathcal{L}_t(\theta, \nu) + \mathcal{O}(\bar{\alpha}_t)$$

as $\bar{\alpha}_t \rightarrow 0$.

Proof of Proposition 4.8. With $\mu_0 = \langle \theta, \mathbf{x} \rangle$ and $\mu_0^* = \langle \theta^*, \mathbf{x} \rangle$, set

$$f(w) := \frac{1}{2} [\mu_0 \tanh(\nu + \mu_0 w) - \mu_0^* \tanh(\nu^* + \mu_0^* w)]^2.$$

Under the ground-truth model, Equation (4) and the normalization in Equation (6) give

$$\mathcal{L}_t(\theta, \nu) = \mathbb{E}_{\mathbf{x}, z, \varepsilon_t} [f(\sqrt{\bar{\alpha}_t} \varepsilon_t + \bar{\alpha}_t (-1)^{z+1} \mu_0^*)].$$

Since $|f(w)| \leq \frac{1}{2}(|\mu_0| + |\mu_0^*|)^2$, dominated convergence under $\mathbb{E}[\|\mathbf{x}\|^2] < \infty$ yields

$$\lim_{\bar{\alpha}_t \rightarrow 0} \mathcal{L}_t(\theta, \nu) = \mathbb{E}_{\mathbf{x}}[f(0)] = \frac{1}{2} \|\theta \tanh \nu - \theta^* \tanh \nu^*\|_\Sigma^2.$$

Conditional on $(\mathbf{x}, z)$, Gaussian integration by parts gives, for $a > 0$,

$$\frac{d}{da} \mathbb{E}_{\varepsilon_t} [f(\sqrt{a} \varepsilon_t + a(-1)^{z+1} \mu_0^*)] = \mathbb{E}_{\varepsilon_t} \left[(-1)^{z+1} \mu_0^* f'(\sqrt{a} \varepsilon_t + a(-1)^{z+1} \mu_0^*) + \frac{1}{2} f''(\sqrt{a} \varepsilon_t + a(-1)^{z+1} \mu_0^*) \right].$$

For $j \in \{1, 2, 3\}$, $k \in \{0, 1, 2\}$, and η in any bounded set, the functions $\tanh$ and $v^k \tanh^{(j)}(\eta + v)$ have a common finite bound. Product and chain rules therefore give a constant $C < \infty$, common to a bounded neighborhood of (θ, ν) , such that the derivatives in μ_0 and ν of the integrand on the right satisfy

$$\|\mathbf{x}\| \left| \partial_{\mu_0} \left\{ (-1)^{z+1} \mu_0^* f'(w) + \frac{1}{2} f''(w) \right\} \right| + \left| \partial_\nu \left\{ (-1)^{z+1} \mu_0^* f'(w) + \frac{1}{2} f''(w) \right\} \right| \leq C(1 + \|\mathbf{x}\|^4)$$

for every $w \in \mathbb{R}$. The fourth-moment condition gives continuity of the parameter gradients at $a = 0$ and permits differentiation under the expectations and integration of the preceding Gaussian identity from zero to $\bar{\alpha}_t$. Since $\nabla_\theta = \mathbf{x} \partial_{\mu_0}$, this gives

$$\nabla_\theta \mathcal{L}_t(\theta, \nu) = \nabla_\theta \lim_{\bar{\alpha}_t \rightarrow 0} \mathcal{L}_t(\theta, \nu) + \mathcal{O}(\bar{\alpha}_t), \quad \nabla_\nu \mathcal{L}_t(\theta, \nu) = \nabla_\nu \lim_{\bar{\alpha}_t \rightarrow 0} \mathcal{L}_t(\theta, \nu) + \mathcal{O}(\bar{\alpha}_t),$$

which proves the claim. $\square$

D Proofs for High-Noise Gradient Dynamics and Score Matching Blindness

In this appendix we give the full quantitative statement (Theorem D.1) and proof of Main Theorem 5.1, prove the complementary high-SNR blindness result in Proposition 5.2, and establish the companion hard-assignment limit that supports the comparison with population EM.

Main theorem D.1 (Formal statement of Main Theorem 5.1). *Assume $\Sigma = \mathbb{E}[\mathbf{x}\mathbf{x}^\top] = I_d$, set $c^* := \theta^* \tanh \nu^*$, and define the high-noise loss*

$$\mathcal{L}(\theta, \nu) := \lim_{\alpha_t \rightarrow 0} \mathcal{L}_t(\theta, \nu) = \frac{1}{2} \|\theta \tanh \nu - c^*\|^2.$$

For $\vartheta = (\theta, \nu)$, set $J(\vartheta) := \frac{1}{2}(\|\theta\|^2 - \sinh^2 \nu)$ and $x_+ := \max\{x, 0\}$. Given $\eta > 0$, let $\Phi(\vartheta) := \vartheta - \eta \nabla \mathcal{L}(\vartheta)$ and

$$\mathcal{D}_\eta := \{\vartheta \in \mathbb{R}^{d+1} : 32\eta \max\{\mathcal{L}(\vartheta), [J(\vartheta) + \mathcal{L}(\vartheta)]_+, \|c^*\|^2, 1\} \leq 1\}.$$

The set $\mathcal{D}_\eta$ is forward invariant under Φ , and the loss is nonincreasing along its trajectories. If $\vartheta_0 \in \mathcal{D}_\eta$ and $\vartheta_{k+1} = \Phi(\vartheta_k)$, then, at every index $k \in \mathbb{Z}_{\geq 0}$, $\vartheta_k \in \mathcal{D}_\eta$ and $\mathcal{L}(\vartheta_{k+1}) \leq \mathcal{L}(\vartheta_k)$. Write $\mathcal{L}_0 := \mathcal{L}(\vartheta_0)$ and $B_0 := 4 \max\{\sqrt{\mathcal{L}_0}, \sqrt{[J(\vartheta_0) + \mathcal{L}_0]_+}, \|c^*\|, 1\}$.

(a) **Degenerate target.** If $c^* = \mathbf{0}$ and $\mathcal{L}_0 > 0$, then

$$\mathcal{L}(\vartheta_K) \leq \left(\mathcal{L}_0^{-1/2} + \frac{\eta K}{\sqrt{6} B_0} \right)^{-2}, \quad K \in \mathbb{Z}_{\geq 0}.$$

Thus $\mathcal{L}(\vartheta_K) = \mathcal{O}(K^{-2})$. If $\mathcal{L}_0 = 0$, all subsequent losses vanish.

(b) **Null saddle basin.** Suppose $c^* \neq \mathbf{0}$, and define

$$\mathcal{S} := \{(\theta, 0) : \langle \theta, c^* \rangle = 0\}, \quad \mathcal{W}^s(\mathcal{S}) := \left\{ \vartheta \in \mathbb{R}^{d+1} : \exists \vartheta_s \in \mathcal{S}, \lim_{k \rightarrow \infty} \Phi^k(\vartheta) = \vartheta_s \right\}.$$

The exceptional set $\mathcal{D}_\eta \cap \mathcal{W}^s(\mathcal{S})$ has zero $(d+1)$ -dimensional Lebesgue measure. Initializations in this set satisfy $\mathcal{L}(\vartheta_k) \rightarrow \|c^*\|^2/2$.

(c) **Generic geometric convergence.** Suppose $c^* \neq \mathbf{0}$ and $\vartheta_0 \in \mathcal{D}_\eta \setminus \mathcal{W}^s(\mathcal{S})$. Then there exist $\tau \in \mathbb{Z}_{\geq 0}$ and $\underline{u} > 0$, both depending on ϑ_0 , such that

$$\mathcal{L}(\vartheta_{\tau+j}) \leq e^{-\eta \underline{u}^2 j} \mathcal{L}(\vartheta_\tau), \quad j \in \mathbb{Z}_{\geq 0}.$$

With $K_\epsilon := \inf\{k \in \mathbb{Z}_{\geq 0} : \mathcal{L}(\vartheta_k) \leq \epsilon\}$, this bound gives $K_\epsilon = \mathcal{O}_{\vartheta_0, \eta, c^*}(\log(1/\epsilon))$ as $\epsilon \downarrow 0$.

Proof of Theorem 5.1. Fix $\vartheta_0 = (\theta_0, \nu_0) \in \mathcal{D}_\eta$, and write $J_0 := J(\vartheta_0)$, $\mathcal{L}_0 := \mathcal{L}(\vartheta_0)$, $B := B_0$, and $C := \|c^*\|$. The admissibility condition gives

$$2\eta B^2 = 32\eta \max\{\mathcal{L}_0, [J_0 + \mathcal{L}_0]_+, C^2, 1\} \leq 1, \quad \eta \leq \frac{1}{2B^2} \leq \frac{1}{32}.$$

Define the confinement set and its ambient convex cylinder by

$$\Omega := \{\vartheta : \mathcal{L}(\vartheta) \leq \mathcal{L}_0, J(\vartheta) + \mathcal{L}(\vartheta) \leq J_0 + \mathcal{L}_0\}, \quad \overline{\Omega} := \{(\theta, \nu) : \|\theta\| \leq B, \nu \in \mathbb{R}\}.$$

We first prove $\Omega \subset \overline{\Omega}$. Neither set is claimed to be compact because ν need not be bounded.

The definition of B gives

$$B \geq 4, \quad C \leq \frac{B}{4}, \quad C^2 \leq \frac{B^2}{16}, \quad \mathcal{L}_0 \leq \frac{B^2}{16}, \quad [J_0 + \mathcal{L}_0]_+ \leq \frac{B^2}{16}.$$

For $(\theta, \nu) \in \Omega$, the triangle inequality and $(a + b)^2 \leq 2a^2 + 2b^2$ give

$$\|\theta\| \tanh |\nu| \leq \|\theta \tanh \nu - c^*\| + C \leq \sqrt{2\mathcal{L}_0} + C \leq \sqrt{2C^2 + 4\mathcal{L}_0}.$$

If $\nu = 0$, then $\|\theta\|^2 + C^2 = 2[J(\theta, 0) + \mathcal{L}(\theta, 0)]$, so membership in Ω gives

$$\|\theta\|^2 + C^2 \leq 2(J_0 + \mathcal{L}_0) \implies J_0 + \mathcal{L}_0 \geq 0, \quad \|\theta\|^2 \leq 2(J_0 + \mathcal{L}_0) \leq \frac{1}{8}B^2.$$

If $\nu \neq 0$, then $\tanh^{-2} \nu = 1 + \sinh^{-2} \nu$, so the loss bound above gives the first inequality below, while the energy bound defining Ω gives the second inequality,

$$\|\theta\|^2 \leq (2C^2 + 4\mathcal{L}_0)(1 + \sinh^{-2} \nu), \quad \|\theta\|^2 = \sinh^2 \nu + 2J(\vartheta) \leq \sinh^2 \nu + 2(J_0 + \mathcal{L}_0).$$

Moreover,

$$2C^2 + 4\mathcal{L}_0 \leq \frac{3}{8}B^2, \quad 2(J_0 + \mathcal{L}_0) \leq 2[J_0 + \mathcal{L}_0]_+ \leq \frac{1}{8}B^2.$$

Here we used $J_0 + \mathcal{L}_0 \leq [J_0 + \mathcal{L}_0]_+$. Taking the minimum of the two preceding bounds, using $\min\{a+b, c+d\} \leq \max\{a, c\} + \min\{b, d\}$, and then using $\min\{a/x, x\} \leq \sqrt{a}$ for $a \geq 0$ and $x > 0$ therefore yields

$$\|\theta\|^2 \leq \max\{2C^2 + 4\mathcal{L}_0, 2(J_0 + \mathcal{L}_0)\} + \min\left\{\frac{2C^2 + 4\mathcal{L}_0}{\sinh^2 \nu}, \sinh^2 \nu\right\} \leq \frac{3}{8}B^2 + \sqrt{\frac{3}{8}}B < \frac{3}{4}B^2.$$

The last strict inequality follows from $B \geq 4$. Thus $\sup_{\Omega} \|\theta\| \leq \sqrt{3}B/2 < B$, proving $\Omega \subset \bar{\Omega}$. Put $u := \tanh \nu \in (-1, 1)$, so $\partial_{\nu} u = 1 - u^2$. The first derivatives are

$$\nabla_{\theta} \mathcal{L} = u(u\theta - c^*), \quad \nabla_{\nu} \mathcal{L} = (1 - u^2)\langle \theta, u\theta - c^* \rangle = (1 - u^2)(u\|\theta\|^2 - \langle \theta, c^* \rangle).$$

Differentiating once more gives

$$\nabla_{\theta\theta}^2 \mathcal{L} = u^2 I_d, \quad \nabla_{\theta\nu}^2 \mathcal{L} = (1 - u^2)\{(u\theta - c^*) + u\theta\} = (1 - u^2)(2u\theta - c^*),$$

and

$$\nabla_{\nu\nu}^2 \mathcal{L} = -2u(1 - u^2)(u\|\theta\|^2 - \langle \theta, c^* \rangle) + (1 - u^2)^2\|\theta\|^2 = (1 - u^2)\{2u\langle \theta, c^* \rangle + (1 - 3u^2)\|\theta\|^2\}.$$

Thus

$$\nabla^2 \mathcal{L} = \begin{pmatrix} u^2 I_d & (1 - u^2)(2u\theta - c^*) \\ (1 - u^2)(2u\theta - c^*)^\top & (1 - u^2)\{2u\langle \theta, c^* \rangle + (1 - 3u^2)\|\theta\|^2\} \end{pmatrix}.$$

By symmetry it suffices to consider $u \in [0, 1]$. Since $\frac{d}{du}[2u(1 - u^2)] = 2(1 - 3u^2)$, the first expression is maximized at $u = 1/\sqrt{3}$. In addition, $(1 - u^2)(1 - 3u^2) = 3(u^2 - 2/3)^2 - 1/3 \in [-1/3, 1]$. Hence

$$\max_{|u| \leq 1} |2u(1 - u^2)| = \frac{4}{3\sqrt{3}}, \quad \max_{|u| \leq 1} |(1 - u^2)(1 - 3u^2)| = 1.$$

On $\bar{\Omega}$, $\|\theta\| \leq B$, $C \leq B/4$, and $B \geq 4$, and therefore

$$\|\nabla_{\theta\theta}^2 \mathcal{L}\|_2 = u^2 \leq 1 \leq \frac{B^2}{16},$$

$$\|\nabla_{\theta\nu}^2 \mathcal{L}\|_2 \leq \frac{4}{3\sqrt{3}}\|\theta\| + C \leq \frac{4}{3\sqrt{3}}B + \frac{B}{4} \leq \left(\frac{1}{3\sqrt{3}} + \frac{1}{16}\right)B^2,$$

and

$$|\nabla_{\nu\nu}^2 \mathcal{L}| \leq \frac{4}{3\sqrt{3}}\|\theta\|C + \|\theta\|^2 \leq \frac{4}{3\sqrt{3}}B\frac{B}{4} + B^2 = \left(1 + \frac{1}{3\sqrt{3}}\right)B^2.$$

The last step in the mixed-block bound uses $B \leq B^2/4$, which also gives $B/4 \leq B^2/16$. Decomposing the Hessian into its four block-supported summands and applying the triangle inequality gives $\|\nabla^2 \mathcal{L}\|_2 \leq$

$\|\nabla_{\theta\theta}^2 \mathcal{L}\|_2 + 2\|\nabla_{\theta\nu}^2 \mathcal{L}\|_2 + \|\nabla_{\nu\nu}^2 \mathcal{L}\|$. Since $1/\sqrt{3} < 3/4$, the resulting coefficient satisfies $19/16 + 1/\sqrt{3} < 31/16 < 2$. These bounds give

$$\sup_{\vartheta \in \bar{\Omega}} \|\nabla^2 \mathcal{L}(\vartheta)\|_2 \leq \left\{ \frac{1}{16} + 2 \left(\frac{1}{3\sqrt{3}} + \frac{1}{16} \right) + \left(1 + \frac{1}{3\sqrt{3}} \right) \right\} B^2 = \left(\frac{19}{16} + \frac{1}{\sqrt{3}} \right) B^2 < 2B^2.$$

Step 1. Descent, confinement, and admissibility. Since $\vartheta_0 \in \Omega$, the induction starts at $k = 0$. Suppose as the induction hypothesis that $\vartheta_k \in \Omega$, and set $u_k := \tanh \nu_k$ and $q_k := u_k \theta_k - c^*$. The first component of the gradient is $\nabla_{\theta} \mathcal{L}(\vartheta_k) = u_k q_k$. Since $\eta \leq 1/32$, the residual bound is

$$\|q_k\| \leq |u_k| \|\theta_k\| + C \leq \frac{\sqrt{3}}{2} B + \frac{B}{4} = \left(\frac{\sqrt{3}}{2} + \frac{1}{4} \right) B < \frac{5}{4} B,$$

and the resulting one-step movement satisfies

$$\|\theta_{k+1}\| = \|\theta_k - \eta u_k q_k\| \leq \|\theta_k\| + \eta |u_k| \|q_k\| \leq \left(\frac{\sqrt{3}}{2} + \frac{1}{32} \frac{5}{4} \right) B = \left(\frac{\sqrt{3}}{2} + \frac{5}{128} \right) B < B.$$

Here the last strict inequality follows from $\sqrt{3}/2 < 7/8$ and $7/8 + 5/128 = 117/128 < 1$. Since $\bar{\Omega}$ imposes no restriction on ν , this bound is sufficient to place ϑ_{k+1} in $\bar{\Omega}$. Thus $\vartheta_k, \vartheta_{k+1} \in \bar{\Omega}$, and convexity of $\bar{\Omega}$ places the segment joining them in $\bar{\Omega}$. Taylor's inequality on this segment and the preceding uniform Hessian bound give

$$\mathcal{L}(\vartheta_{k+1}) - \mathcal{L}(\vartheta_k) \leq \langle \nabla \mathcal{L}(\vartheta_k), -\eta \nabla \mathcal{L}(\vartheta_k) \rangle + B^2 \|\eta \nabla \mathcal{L}(\vartheta_k)\|^2 = -\eta(1 - \eta B^2) \|\nabla \mathcal{L}(\vartheta_k)\|^2 \leq -\frac{\eta}{2} \|\nabla \mathcal{L}(\vartheta_k)\|^2.$$

Here the Taylor remainder coefficient is $B^2 = \frac{1}{2}(2B^2)$, and the final inequality uses $\eta B^2 \leq 1/2$.

The component updates are

$$\theta_{k+1} = \theta_k - \eta u_k q_k, \quad \nu_{k+1} = \nu_k - \eta(1 - u_k^2) \langle \theta_k, q_k \rangle.$$

This identity gives

$$\|\theta_{k+1}\|^2 - \|\theta_k\|^2 = -2\eta u_k \langle \theta_k, q_k \rangle + \eta^2 u_k^2 \|q_k\|^2 = \eta^2 u_k^2 \|q_k\|^2 + \frac{2u_k}{1 - u_k^2} (\nu_{k+1} - \nu_k).$$

Convexity of $\nu \mapsto \sinh^2 \nu$ gives the exact supporting-line comparison

$$\sinh^2 \nu_{k+1} - \sinh^2 \nu_k \geq \sinh(2\nu_k) (\nu_{k+1} - \nu_k) = \frac{2u_k}{1 - u_k^2} (\nu_{k+1} - \nu_k).$$

Combining these two displays and then using the descent inequality above yields

$$J(\vartheta_{k+1}) - J(\vartheta_k) \leq \frac{\eta^2}{2} u_k^2 \|q_k\|^2 = \frac{\eta^2}{2} \|\nabla_{\theta} \mathcal{L}(\vartheta_k)\|^2 \leq \frac{\eta^2}{2} \|\nabla \mathcal{L}(\vartheta_k)\|^2 \leq \eta [\mathcal{L}(\vartheta_k) - \mathcal{L}(\vartheta_{k+1})].$$

Because $\eta < 1$, combining the descent and energy-drift inequalities gives

$$[J(\vartheta_{k+1}) + \mathcal{L}(\vartheta_{k+1})] - [J(\vartheta_k) + \mathcal{L}(\vartheta_k)] \leq -(1 - \eta) [\mathcal{L}(\vartheta_k) - \mathcal{L}(\vartheta_{k+1})] \leq 0.$$

Hence $\vartheta_{k+1} \in \Omega$, and induction proves confinement. In particular, monotonicity of the two trajectory-dependent quantities gives the admissibility chain

$$\max\{\mathcal{L}(\vartheta_{k+1}), [J(\vartheta_{k+1}) + \mathcal{L}(\vartheta_{k+1})]_+, C^2, 1\} \leq \max\{\mathcal{L}_0, [J_0 + \mathcal{L}_0]_+, C^2, 1\} \leq \frac{1}{32\eta}.$$

Thus admissibility is preserved. In summary,

$$\vartheta_k \in \Omega \subset \bar{\Omega}, \quad \|\theta_k\| \leq \frac{\sqrt{3}}{2} B, \quad \vartheta_k \in \mathcal{D}_{\eta}, \quad k \in \mathbb{Z}_{\geq 0}.$$

Step 2. Degenerate target. Suppose $c^* = \mathbf{0}$. If $\mathcal{L}_0 = 0$, the descent inequality from Step 1 shows that all subsequent losses vanish. We may therefore assume $\mathcal{L}_0 > 0$ and work until a possible finite iterate with zero loss, after which the same conclusion applies.

Since $c^* = \mathbf{0}$, direct calculation gives

$$\nabla_\theta \mathcal{L} = u^2 \theta, \quad \nabla_\nu \mathcal{L} = u(1 - u^2) \|\theta\|^2, \quad \|\nabla \mathcal{L}\|^2 = u^4 \|\theta\|^2 + u^2(1 - u^2)^2 \|\theta\|^4 = 2\mathcal{L}\{u^2 + \|\theta\|^2(1 - u^2)^2\}.$$

For $\vartheta \in \Omega$, if $u^2 > 1/2$, then

$$u^2 + \|\theta\|^2(1 - u^2)^2 \geq u^2 > \frac{1}{2} \geq \frac{\|\theta\| \|u\|}{\sqrt{3}B}.$$

If $u^2 \leq 1/2$, then $(|u| - \|\theta\|/2)^2 \geq 0$ gives

$$u^2 + \|\theta\|^2(1 - u^2)^2 \geq u^2 + \frac{1}{4} \|\theta\|^2 \geq \|\theta\| \|u\| \geq \frac{\|\theta\| \|u\|}{\sqrt{3}B}.$$

Here we used the confinement bound $\|\theta\| \leq \sqrt{3}B/2$ and $B \geq 4$. We obtain

$$\|\nabla \mathcal{L}(\vartheta)\|^2 = 2\mathcal{L}(\vartheta)\{u^2 + \|\theta\|^2(1 - u^2)^2\} \geq \frac{2\mathcal{L}(\vartheta)\sqrt{2\mathcal{L}(\vartheta)}}{\sqrt{3}B} = 2\sqrt{\frac{2}{3}} \frac{\mathcal{L}(\vartheta)^{3/2}}{B}.$$

As long as $\mathcal{L}(\vartheta_{k+1}) > 0$, the supporting-line inequality for $x \mapsto x^{-1/2}$, together with the descent inequality and the preceding gradient lower bound, gives

$$\mathcal{L}(\vartheta_{k+1})^{-1/2} - \mathcal{L}(\vartheta_k)^{-1/2} \geq \frac{\mathcal{L}(\vartheta_k) - \mathcal{L}(\vartheta_{k+1})}{2\mathcal{L}(\vartheta_k)^{3/2}} \geq \frac{\eta \|\nabla \mathcal{L}(\vartheta_k)\|^2}{4\mathcal{L}(\vartheta_k)^{3/2}} \geq \frac{\eta}{\sqrt{6}B}.$$

If $\mathcal{L}(\vartheta_K) > 0$, telescoping and inversion prove

$$\mathcal{L}(\vartheta_K) \leq \left(\mathcal{L}_0^{-1/2} + \frac{\eta K}{\sqrt{6}B_0} \right)^{-2}.$$

If $\mathcal{L}(\vartheta_K) = 0$, the same bound is immediate, and the descent inequality makes all later losses vanish. Thus the displayed sublinear bound holds for every $K \in \mathbb{Z}_{\geq 0}$.

For Steps 3 through 6, assume $c^* \neq \mathbf{0}$, so $C > 0$.

Step 3. Saddle points and the boundary of the sign regions. If $\vartheta = (\theta, 0) \in \mathcal{S}$, then $u = 0$, $\nabla_\theta \mathcal{L}(\vartheta) = u(u\theta - c^*) = 0$, and $\nabla_\nu \mathcal{L}(\vartheta) = (1 - u^2)\langle \theta, u\theta - c^* \rangle = -\langle \theta, c^* \rangle = 0$. Hence $\Phi(\vartheta) = \vartheta$, so an initialization in $\mathcal{S}$ remains fixed.

Fix any boundary index $r \in \mathbb{Z}_{\geq 0}$ with $\langle \theta_r, c^* \rangle \nu_r = 0$ and $\vartheta_r \notin \mathcal{S}$. The boundary point enters the matching-sign region after one update. If $\langle \theta_r, c^* \rangle = 0$ and $\nu_r \neq 0$, then $u_r \nu_r > 0$, $0 < u_r(1 - u_r^2)/\nu_r \leq 1$, and $\eta \|\theta_r\|^2 \leq 3\eta B^2/4 \leq 3/8$. Therefore,

$$\langle \theta_{r+1}, c^* \rangle \nu_r = \eta C^2 u_r \nu_r > 0, \quad \frac{\nu_{r+1}}{\nu_r} = 1 - \eta \|\theta_r\|^2 \frac{u_r(1 - u_r^2)}{\nu_r} \geq 1 - \eta \|\theta_r\|^2 \geq \frac{5}{8} > 0, \quad \langle \theta_{r+1}, c^* \rangle \nu_{r+1} > 0.$$

If $\langle \theta_r, c^* \rangle \neq 0$ and $\nu_r = 0$, then $u_r = 0$ and $q_r = -c^*$, so the same conclusion follows from the update

$$\theta_{r+1} = \theta_r - \eta u_r q_r = \theta_r, \quad \nu_{r+1} = \nu_r - \eta(1 - u_r^2)\langle \theta_r, q_r \rangle = \eta \langle \theta_r, c^* \rangle, \quad \langle \theta_{r+1}, c^* \rangle \nu_{r+1} = \eta \langle \theta_r, c^* \rangle^2 > 0.$$

Thus, in either boundary case,

$$\langle \theta_{r+1}, c^* \rangle \nu_{r+1} > 0.$$

Step 4. Matching signs. Because $\mathcal{L}(\theta, \nu) = \mathcal{L}(-\theta, -\nu)$ and $\Phi(-\theta, -\nu) = -\Phi(\theta, \nu)$, it suffices to analyze a time r at which $\langle \theta_r, c^* \rangle > 0$ and $\nu_r > 0$. In this and the next step, we use the normalized projection $w_k := \langle \theta_k, c^* \rangle / C^2$. The component updates from Step 1 give

$$w_{k+1} = (1 - \eta u_k^2) w_k + \eta u_k, \quad \nu_{k+1} - \nu_k = \eta(1 - u_k^2)(C^2 w_k - u_k \|\theta_k\|^2).$$

Define $w_{\min,r} := \min\{w_r, 1\} > 0$ and $u_{\min,r} := \min\{u_r, C^2 w_{\min,r}/(2B^2)\} > 0$. The joint induction starts at $k = r$. Suppose that $w_k \geq w_{\min,r}$ and $u_k \geq u_{\min,r}$ for some $k \geq r$. Since $1 - \eta u_k^2 > 0$, the first matching-sign recurrence gives

$$w_{k+1} = (1 - \eta u_k^2)w_k + \eta u_k \geq [(1 - \eta u_k^2) + \eta u_k] \min\{w_k, 1\} = [1 + \eta u_k(1 - u_k)] \min\{w_k, 1\} \geq w_{\min,r}.$$

Moreover, $0 < C^2 w_{\min,r}/B^2 \leq C^2/B^2 \leq 1/16$. If $u_k \leq C^2 w_{\min,r}/B^2$, the second matching-sign recurrence and the confinement bound $\|\theta_k\|^2 \leq 3B^2/4$ give

$$C^2 w_k - u_k \|\theta_k\|^2 \geq C^2 w_{\min,r} - \frac{3}{4} C^2 w_{\min,r} = \frac{1}{4} C^2 w_{\min,r} > 0 \implies \nu_{k+1} \geq \nu_k \implies u_{k+1} \geq u_k \geq u_{\min,r}.$$

Otherwise, $u_k > C^2 w_{\min,r}/B^2$. Dropping the nonnegative term involving $C^2 w_k$ in that recurrence and using $2\eta B^2 \leq 1$ give

$$\nu_{k+1} \geq \operatorname{arctanh}(u_k) - \eta(1 - u_k^2)u_k \|\theta_k\|^2 \geq \operatorname{arctanh}(u_k) - \eta B^2 u_k \geq \operatorname{arctanh}(u_k) - \frac{u_k}{2}.$$

The map $x \mapsto \operatorname{arctanh} x - x/2$ has derivative $(1 + x^2)/[2(1 - x^2)] > 0$ on $(0, 1)$. The subtraction formula gives $\tanh(a - b) \geq \tanh a - \tanh b$ for $a \geq b \geq 0$. We also use $\tanh x \leq x$ for $x \geq 0$. Hence

$$u_{k+1} \geq \tanh\left(\operatorname{arctanh}(u_k) - \frac{u_k}{2}\right) \geq \frac{C^2 w_{\min,r}}{B^2} - \tanh\left(\frac{C^2 w_{\min,r}}{2B^2}\right) \geq \frac{C^2 w_{\min,r}}{2B^2} \geq u_{\min,r}.$$

This completes the joint inductive step and yields

$$w_k \geq w_{\min,r}, \quad u_k \geq u_{\min,r}, \quad k \in \mathbb{Z}_{\geq r}.$$

In particular, the matching-sign region is forward invariant. For $k \geq r$, the gradient satisfies

$$\|\nabla \mathcal{L}(\vartheta_k)\|^2 \geq \|\nabla_{\theta} \mathcal{L}(\vartheta_k)\|^2 = u_k^2 \|q_k\|^2 = 2u_k^2 \mathcal{L}(\vartheta_k) \geq 2u_{\min,r}^2 \mathcal{L}(\vartheta_k).$$

Combining this inequality with the descent inequality from Step 1 and using $1 - x \leq e^{-x}$ yields

$$\mathcal{L}(\vartheta_{r+j}) \leq (1 - \eta u_{\min,r}^2)^j \mathcal{L}(\vartheta_r) \leq e^{-\eta u_{\min,r}^2 j} \mathcal{L}(\vartheta_r), \quad j \in \mathbb{Z}_{\geq 0}.$$

Step 5. Mixed signs, finite exit or convergence to $\mathcal{S}$. Suppose $\langle \theta_0, c^* \rangle \nu_0 < 0$, and define the exit time $T := \inf\{k \in \mathbb{Z}_{\geq 0} : \langle \theta_k, c^* \rangle \nu_k \geq 0\}$, with $\inf \emptyset := \infty$. Thus $T \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$. The estimates below apply at indices $k < T$.

Put $\theta_k^\perp := \theta_k - w_k c^*$ and $G_k := \mathcal{L}(\vartheta_k) - C^2/2$. Then $\langle \theta_k^\perp, c^* \rangle = 0$, $w_k u_k < 0$, and

$$G_k = \frac{1}{2} \|\theta_k^\perp\|^2 u_k^2 + \frac{C^2}{2} (w_k u_k - 2) w_k u_k > 0.$$

The update has the form

$$\theta_{k+1}^\perp = (1 - \eta u_k^2) \theta_k^\perp, \quad w_{k+1} = (1 - \eta u_k^2) w_k + \eta u_k, \quad \nu_{k+1} - \nu_k = \eta C^2 (1 - u_k^2) \left[w_k - u_k \left(w_k^2 + \frac{\|\theta_k^\perp\|^2}{C^2} \right) \right].$$

The first two mixed-sign recurrences give $\|\theta_k^\perp\| \leq \|\theta_0^\perp\|$ and $|w_k| \leq \max\{|w_0|, \eta\}$. These bounds hold at $0 \leq k \leq T$ when $T < \infty$, and at every finite index when $T = \infty$. The final mixed-sign recurrence has sign opposite to ν_k , while its magnitude is at most $\eta[B^2 + C^2 \max\{|w_0|, \eta\}]$. Hence the same index ranges satisfy $|\nu_k| \leq \max\{|\nu_0|, \eta[B^2 + C^2 \max\{|w_0|, \eta\}]\}$.

Set $c_\nu := \operatorname{sech}^2(\max\{|\nu_0|, \eta[B^2 + C^2 \max\{|w_0|, \eta\}]\}) > 0$ and $c_{\text{mix}} := \min\{2C^2/B^2, c_\nu C\} > 0$. Orthogonality of $\theta_k^\perp$ and c^* gives $\|\nabla_{\theta} \mathcal{L}(\vartheta_k)\|^2 = C^2 u_k^2 + 2u_k^2 G_k$ and $|\nabla_{\nu} \mathcal{L}(\vartheta_k)| \geq c_\nu C^2 |w_k|$. Also, the confinement bound from Step 1 gives $G_k \leq 3B^2 u_k^2/8 + C^2 |w_k u_k|$. Since $\frac{1}{4} C^2 u_k^2 + c_\nu^2 C^4 w_k^2 \geq c_\nu C^3 |w_k u_k|$,

$$\|\nabla \mathcal{L}(\vartheta_k)\|^2 \geq \frac{3}{4} C^2 u_k^2 + c_\nu C^3 |w_k u_k| \geq c_{\text{mix}} \left(\frac{3}{8} B^2 u_k^2 + C^2 |w_k u_k| \right) \geq c_{\text{mix}} G_k.$$

Here $c_{\text{mix}} \leq 2C^2/B^2 \leq 1/8$, so $0 < \eta c_{\text{mix}}/2 < 1$. Combining the descent inequality from Step 1 with the preceding mixed-sign gradient lower bound yields

$$G_{k+1} \leq \left(1 - \frac{\eta c_{\text{mix}}}{2}\right) G_k \leq e^{-\eta c_{\text{mix}}/2} G_k, \quad k < T.$$

Suppose now that $T = \infty$. The preceding contraction estimate gives $G_k \rightarrow 0$, while $\|\nabla \mathcal{L}(\vartheta_k)\|^2 \geq C^2 u_k^2$ and the descent inequality from Step 1 gives the sharper bound below because $\mathcal{L}(\vartheta_k) > C^2/2$ throughout the mixed-sign region.

$$\sum_{k=0}^{\infty} u_k^2 \leq \frac{2G_0}{\eta C^2} < \infty.$$

This summability bound implies $u_k \rightarrow 0$, and hence $\nu_k \rightarrow 0$. Since $0 \leq \eta u_k^2 \leq 1/32$, $-\log(1 - \eta u_k^2) \leq (32/31)\eta u_k^2$. Thus the positive product $\prod_{k=0}^{\infty} (1 - \eta u_k^2)$ converges to a positive limit, and the first mixed-sign recurrence shows that $\theta_k^\perp$ converges to some $\theta_s \perp c^*$.

Because $\nu_k \rightarrow 0$, its increments vanish. In the mixed-sign region, the final mixed-sign recurrence satisfies $\eta C^2(1 - u_k^2)|w_k| \leq |\nu_{k+1} - \nu_k|$, so $w_k \rightarrow 0$. We conclude that

$$\vartheta_k \rightarrow (\theta_s, 0) \in \mathcal{S}.$$

Thus $T = \infty$ implies $\vartheta_0 \in \mathcal{W}^s(\mathcal{S})$.

Now suppose $\vartheta_0 \notin \mathcal{W}^s(\mathcal{S})$. If the initialization has matching signs, take $\tau = 0$. A boundary initialization outside $\mathcal{S}$ is handled by the boundary-entry calculation in Step 3. For a mixed-sign initialization, the implication $T = \infty \Rightarrow \vartheta_0 \in \mathcal{W}^s(\mathcal{S})$ proved above rules out $T = \infty$. After the finite exit, a landing in $\mathcal{S}$ would make ϑ_0 belong to $\mathcal{W}^s(\mathcal{S})$. Thus the exit either has matching signs or lands on the boundary outside $\mathcal{S}$, where the boundary-entry calculation applies. Therefore $\tau := \inf\{k \in \mathbb{Z}_{\geq 0} : \langle \theta_k, c^* \rangle \nu_k > 0\} < \infty$.

Set

$$\underline{u} := \min\left\{|\tanh \nu_\tau|, \frac{\min\{|\langle \theta_\tau, c^* \rangle|, C^2\}}{2B^2}\right\} > 0.$$

Applying the matching-sign rate from Step 4 at time τ gives

$$\mathcal{L}(\vartheta_{\tau+j}) \leq e^{-\eta \underline{u}^2 j} \mathcal{L}(\vartheta_\tau), \quad j \in \mathbb{Z}_{\geq 0}.$$

If $\mathcal{L}(\vartheta_\tau) = 0$, then $K_\epsilon \leq \tau$. Otherwise,

$$K_\epsilon \leq \tau + \left\lceil \frac{[\log(\mathcal{L}(\vartheta_\tau)/\epsilon)]_+}{\eta \underline{u}^2} \right\rceil.$$

This proves the pointwise logarithmic complexity. We do not claim a uniform bound on τ or $\underline{u}$.

If $\vartheta_0 \in \mathcal{W}^s(\mathcal{S})$, continuity gives $\mathcal{L}(\vartheta_k) \rightarrow C^2/2$.

Step 6. Null saddle basin. We adapt the standard center-stable-manifold argument for avoiding saddle points under gradient descent (Lee et al., 2016). The argument below also treats points at which the update map is singular. It remains to prove $\text{Vol}_{d+1}(\mathcal{D}_\eta \cap \mathcal{W}^s(\mathcal{S})) = 0$. The claim is immediate if $\mathcal{D}_\eta$ is empty. Otherwise, its definition implies

$$\eta \leq \frac{1}{32 \max\{C^2, 1\}}, \quad \eta C \leq \frac{1}{32}.$$

Reachable saddle limits. Let $\vartheta_0 \in \mathcal{D}_\eta \cap \mathcal{W}^s(\mathcal{S})$ and $\Phi^k(\vartheta_0) \rightarrow \vartheta_s = (\theta_s, 0) \in \mathcal{S}$. Applying the confinement argument above with this initialization and using the definition of $\mathcal{D}_\eta$ gives $\|\theta_s\|^2 \leq 3/(8\eta)$. Thus all such limits lie in the compact set $\mathcal{K}_\eta := \{(\theta, 0) \in \mathcal{S} : \|\theta\|^2 \leq 3/(8\eta)\}$.

Local center-stable disks. At $\vartheta_s = (\theta_s, 0) \in \mathcal{S}$, the Hessian kernel is $\{(v, 0) : v \in \mathbb{R}^d, \langle v, c^* \rangle = 0\}$. Direct diagonalization on its two-dimensional complement gives

$$\lambda_\pm(\vartheta_s) = \frac{1}{2} \left(\|\theta_s\|^2 \pm \sqrt{\|\theta_s\|^4 + 4C^2} \right), \quad \lambda_- < 0 < \lambda_+.$$

The derivative $D\Phi(\vartheta_s)$ has eigenvalue 1 with multiplicity $d-1$, together with $1-\eta\lambda_+(\vartheta_s)$ and $1-\eta\lambda_-(\vartheta_s)$. The inequality $\sqrt{a^2+4b^2} \leq a+2b$, the global step-size bounds above, and $\eta\|\theta_s\|^2 \leq 3/8$ give $0 < \eta\lambda_+(\vartheta_s) \leq 13/32 < 1$. Hence the eigenvalues of $D\Phi(\vartheta_s)$ form center, stable, and unstable groups whose dimensions are $d-1$, 1, and 1. In particular, $D\Phi(\vartheta_s)$ is nonsingular. The Inverse Function Theorem makes Φ a C^1 local diffeomorphism near each point of $\mathcal{K}_\eta$.

The local discrete-time Center-Stable Manifold Theorem (Shub, 1987, Theorem III.7, pp. 65–66) therefore supplies an open neighborhood U_{ϑ_s} and a d -dimensional embedded C^1 disk $W_{\text{loc}}^{cs}(\vartheta_s)$ satisfying

$$\{\vartheta \in U_{\vartheta_s} : \forall n \in \mathbb{Z}_{\geq 0}, \quad \Phi^n(\vartheta) \in U_{\vartheta_s}\} \subseteq W_{\text{loc}}^{cs}(\vartheta_s).$$

Each such disk is Lebesgue-null in $\mathbb{R}^{d+1}$. Compactness gives points $\vartheta_{s,1}, \dots, \vartheta_{s,N} \in \mathcal{K}_\eta$ such that $\mathcal{K}_\eta \subseteq \bigcup_{i=1}^N U_{\vartheta_{s,i}}$. Write $U_i := U_{\vartheta_{s,i}}$ and $W_i^{cs} := W_{\text{loc}}^{cs}(\vartheta_{s,i})$.

If an admissible trajectory converges to a point of $\mathcal{K}_\eta$, it eventually remains in some U_i . The local trapping property then places a finite iterate in W_i^{cs} . Thus

$$\mathcal{D}_\eta \cap \mathcal{W}^s(\mathcal{S}) \subseteq \bigcup_{i=1}^N \bigcup_{\ell=0}^{\infty} \Phi^{-\ell}(W_i^{cs}).$$

Null preimages. We next show that inverse images of null sets remain null. The determinant of $D\Phi$ is real analytic. Its value at the origin is $\det D\Phi(0,0) = 1 - \eta^2 C^2 > 0$, so the determinant is not the zero function. Its critical set $\text{Crit}(\Phi) := \{\vartheta : \det D\Phi(\vartheta) = 0\}$ is Lebesgue-null (Krantz & Parks, 2002, Section 4.1, p. 83).

Let $Z \subseteq \mathbb{R}^{d+1}$ be Lebesgue-null. At every point outside $\text{Crit}(\Phi)$, the Inverse Function Theorem gives a neighborhood U on which Φ has a C^1 inverse. We choose an open ball B whose compact closure lies in $\Phi(U)$, and set $O = (\Phi|_U)^{-1}(B)$. The inverse derivative is bounded on $\bar{B}$. Since B is convex, the inverse is Lipschitz on B . Second countability gives a collection $\{O_j\}_{j \geq 1}$ of these neighborhoods that covers the regular points. Lipschitz maps preserve null sets (Evans & Gariepy, 2015, Sections 2.2 and 2.4), so

$$\Phi^{-1}(Z) \setminus \text{Crit}(\Phi) \subseteq \bigcup_{j=1}^{\infty} (\Phi|_{O_j})^{-1}(Z \cap \Phi(O_j)), \quad \text{Vol}_{d+1}(\Phi^{-1}(Z) \setminus \text{Crit}(\Phi)) = 0.$$

The part inside $\text{Crit}(\Phi)$ is also null. Therefore

$$\text{Vol}_{d+1}(Z) = 0 \implies \text{Vol}_{d+1}(\Phi^{-1}(Z)) = 0.$$

Repeating the argument shows that each $\Phi^{-\ell}(W_i^{cs})$ is null. The countable union on the right of the stable-basin cover above is measurable and null. Its subset $\mathcal{D}_\eta \cap \mathcal{W}^s(\mathcal{S})$ is measurable and null by completeness of Lebesgue measure. $\square$

Proposition D.2 (Proposition 5.2). *Assume $\mathbb{E}[\|\mathbf{x}\|^2] < \infty$. Fix $\bar{\alpha}_t \in (0,1]$, a finite model pair $(\theta, \nu) \in \mathbb{R}^d \times \mathbb{R}$. Let $\theta^* \neq \mathbf{0}$ vary with $\|\theta^*\| \rightarrow \infty$ along a fixed ray, while the ground-truth mixing weights and all other model and data-generating quantities remain fixed. If $\theta \neq \mathbf{0}$, assume*

$$\mathbb{P}(\{\langle \theta, \mathbf{x} \rangle = 0\} \cup \{\langle \theta^*, \mathbf{x} \rangle = 0\}) = 0.$$

The expectations defining $\mathcal{L}_t$ and V_t are taken under the corresponding varying ground truth. Then, pointwise at the fixed model pair and fixed scale of the diffusion noise level,

$$\lim_{\|\theta^*\| \rightarrow \infty} |\nabla_\nu \mathcal{L}_t(\theta, \nu)| = 0, \quad \lim_{\|\theta^*\| \rightarrow \infty} V_t(\theta, \nu) = 0.$$

Proof of Proposition 5.2. We use the established scaled projections and forward observation

$$\mu_t := \sqrt{\bar{\alpha}_t} \langle \theta, \mathbf{x} \rangle, \quad \mu_t^* := \sqrt{\bar{\alpha}_t} \langle \theta^*, \mathbf{x} \rangle, \quad y_t = (-1)^{z+1} \mu_t^* + \varepsilon_t,$$

where $\varepsilon_t \sim \mathcal{N}(0, 1)$ is independent of $(\mathbf{x}, z)$. Proposition 4.6 and Corollary 4.5 give the score-matching and latent-variance identities

$$V_t(\theta, \nu) = \frac{1}{2\bar{\alpha}_t} \mathbb{E}[\mu_t^2 \operatorname{sech}^2(\mu_t y_t + \nu)], \quad \nabla_\nu \mathcal{L}_t(\theta, \nu) = -\frac{1}{\bar{\alpha}_t} \mathbb{E}[(\mu_t y_t + \mu_t^2 \tanh(\mu_t y_t + \nu)) \operatorname{sech}^2(\mu_t y_t + \nu)].$$

If $\theta = \mathbf{0}$, then $\mu_t = 0$ almost surely, so both quantities vanish for every signal norm. Hence assume $\theta \neq \mathbf{0}$.

For every $x \in \mathbb{R}$,

$$|x| \operatorname{sech}^2 x \leq \frac{2}{e}, \quad \operatorname{sech}^2 x \leq 1, \quad |\tanh x| \operatorname{sech}^2 x \leq 1.$$

Substituting $\mu_t y_t = (\mu_t y_t + \nu) - \nu$ therefore gives the ground-truth-independent majorants

$$|(\mu_t y_t + \mu_t^2 \tanh(\mu_t y_t + \nu)) \operatorname{sech}^2(\mu_t y_t + \nu)| \leq \frac{2}{e} + |\nu| + \mu_t^2.$$

The variance integrand is bounded in absolute value by μ_t^2 . Moreover,

$$\mathbb{E}[\mu_t^2] = \bar{\alpha}_t \mathbb{E}[\langle \theta, \mathbf{x} \rangle^2] \leq \bar{\alpha}_t \|\theta\|^2 \mathbb{E}[\|\mathbf{x}\|^2] < \infty.$$

Thus both absolute integrands have integrable majorants independent of $\|\theta^*\|$. By the projection assumption, almost surely

$$\mu_t y_t + \nu = (-1)^{z+1} \bar{\alpha}_t \langle \theta, \mathbf{x} \rangle \langle \theta^*, \mathbf{x} \rangle + \sqrt{\bar{\alpha}_t} \langle \theta, \mathbf{x} \rangle \varepsilon_t + \nu,$$

whose absolute value tends to infinity. Each integrand in the exact identities therefore tends to zero almost surely. The dominated convergence theorem, followed by division by the fixed positive $\bar{\alpha}_t$, proves both limits. $\square$

Lemma D.3 (Hard-Assignment Limit of the EM Imbalance Operator). Under the assumptions of Proposition 5.2, suppose $\theta \neq \mathbf{0}$, and define $q(\theta, \theta^*) := \mathbb{E}_{\mathbf{x}}[\operatorname{sgn}(\langle \theta, \mathbf{x} \rangle \langle \theta^*, \mathbf{x} \rangle)]$, which is constant along the fixed ray. Let N_{θ^*} and $\mathcal{H}_{\theta^*}$ denote the EM imbalance operator and cross-entropy when their ground-truth expectations are taken under (θ^*, ν^*) . Then, as $\|\theta^*\| \rightarrow \infty$ along the fixed ray,

$$N_{\theta^*}(\theta_t, \nu) \rightarrow q(\theta, \theta^*) \tanh \nu^*, \quad \partial_\nu N_{\theta^*}(\theta_t, \nu) \rightarrow 0,$$

and

$$\nabla_\nu \mathcal{H}_{\theta^*}(\theta_t, \nu) \rightarrow \tanh \nu - q(\theta, \theta^*) \tanh \nu^*, \quad \nabla_\nu^2 \mathcal{H}_{\theta^*}(\theta_t, \nu) \rightarrow \operatorname{sech}^2 \nu > 0.$$

If $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, I_d)$ and $\rho := \langle \theta, \theta^* \rangle / (\|\theta\| \|\theta^*\|)$, then $q(\theta, \theta^*) = (2/\pi) \arcsin \rho$.

Proof. Under the ground truth (θ^*, ν^*) , the candidate activation is $\mu_t y_t = (-1)^{z+1} \bar{\alpha}_t \langle \theta, \mathbf{x} \rangle \langle \theta^*, \mathbf{x} \rangle + \sqrt{\bar{\alpha}_t} \langle \theta, \mathbf{x} \rangle \varepsilon_t$. The projection condition gives $\tanh(\mu_t y_t + \nu) \rightarrow (-1)^{z+1} \operatorname{sgn}(\langle \theta, \mathbf{x} \rangle \langle \theta^*, \mathbf{x} \rangle)$ and $\operatorname{sech}^2(\mu_t y_t + \nu) \rightarrow 0$ almost surely. Bounded convergence, independence of z and $\mathbf{x}$, and $\mathbb{E}[(-1)^{z+1}] = \tanh \nu^*$ then give

$$N_{\theta^*}(\theta_t, \nu) \rightarrow q(\theta, \theta^*) \tanh \nu^*, \quad \partial_\nu N_{\theta^*}(\theta_t, \nu) = \mathbb{E}_*[\operatorname{sech}^2(\mu_t y_t + \nu)] \rightarrow 0.$$

Proposition 4.2 and its derivative give the two cross-entropy limits. For isotropic Gaussian covariates, Lemma A.12 applied to the standardized projections gives $q(\theta, \theta^*) = (2/\pi) \arcsin \rho$. $\square$

E Experimental Details and Additional Numerical Results

This appendix develops the covariance interpretation for the auxiliary local spectral diagnostic and gives the numerical settings for the figures placed alongside the results in Sections 3, 4, and 5, together with their synthesis in Section 6. Unless stated otherwise, $\mathbf{x} \sim \mathcal{N}(0, I_d)$. All computations use double precision, unit response-noise variance, and the convention for the mixing weights in Section 2.1.

Experimental notation. To avoid a second notation system, Table 1 cross-references quantities already introduced in the theorem statements or their preceding formal appendix. Numerical diagnostics such as fitted slopes, quantiles, and coverage are described in words rather than assigned new symbols.

Table 1: **Experimental notation.** Main-paper quantities reused in the numerical experiments.

Quantity	Use in the experiments
(θ, ν) and (θ^*, ν^*)	Model and ground-truth parameters in every study.
$\bar{\alpha}_t$ and $\bar{\alpha}_{T_n}$	Scale of the diffusion noise level in the endpoint and blindness studies, and terminal scale in the finite-sample study.
$(\hat{\theta}_{n,T_n}^{\text{SM}}, \hat{\nu}_{n,T_n}^{\text{SM}})$ and s_n	Integrated score matching estimator and the joint sign alignment in Theorem 3.2.
$I(\theta^*, \nu^*)$	Fisher-information benchmark for the standardized aligned estimator.
M, N, V , and V_t	Terms in the complete low-noise reference and the blindness study.
Σ	Covariate covariance in the endpoint and limiting-loss calculations.
$\mathcal{C}^*$	Fourth-order signal covariance in the auxiliary spectral diagnostic.
$c^*, \mathcal{L}(\vartheta), J(\vartheta), \Phi$, and $\mathcal{D}_\eta$	Limiting loss, update map, and admissible set used in the dynamics study.
$\eta, \mathcal{L}_0$, and B_0	Step size, initial loss, and explicit-envelope constant in the dynamics study.
$\mathcal{S}, \mathcal{W}^s(\mathcal{S})$, and K_ϵ	Exceptional saddle set, its stable set, and the first iteration K for which $\mathcal{L}(\vartheta_K) \leq \epsilon$.

Numerical settings. Tables 2 and 3 collect the settings without introducing experiment-specific mathematical notation. A separate entry lists the spectral study as an auxiliary local-mechanism diagnostic.

Table 2: **Finite-sample, endpoint, and blindness settings.**

Study	Model and numerical setting
Finite sample	$d = 1$, $(\theta^*, \nu^*) = (1.10, 0.55)$, $n = 250 \cdot 2^j$ for $j = 0, \dots, 4$, with 200 independent repetitions at each n , $\bar{\alpha}_0 = 1$, and $\bar{\alpha}_{T_n} = n^{-3/2}$. The diffusion integral uses 24-point Gauss–Legendre quadrature, the expectation over ξ uses 11-point Gauss–Hermite quadrature, and each fit uses four starts. The Fisher information uses an 80-point Gauss–Hermite rule in each Gaussian variable and an exact sum over the two latent classes.
Endpoints	$d = 1$, $(\theta^*, \nu^*) = (1.10, 0.55)$, and $(\theta, \nu) = (0.45, -0.35)$. Both the Gaussian covariate and response-noise expectations use 140-point Gauss–Hermite quadrature. For $j = 0, \dots, 8$, the low-noise grid uses $\bar{\alpha}_t = 1 - 10^{-6+j/2}$, while the high-noise grid uses $\bar{\alpha}_t = 10^{-6+j/2}$.
Blindness	$d = 1$, $\bar{\alpha}_t = \frac{1}{2}$, $(\theta, \nu) = (0.80, -0.35)$, $\nu^* = 0.55$, and $\theta^* = 3 \times 10^{j/5} > 0$ for $j = 0, \dots, 25$. Expectations use deterministic quadrature as detailed below.

Table 3: **Limiting-dynamics and spectral-diagnostic settings.**

Study	Model and numerical setting
Degenerate dynamics	$d = 2$, $c^* = \mathbf{0}$, $\eta = 1/50 = 2 \times 10^{-2}$, $\theta_0 = (\sinh(0.70), 0)$, and $\nu_0 = 0.70$, so that $J(\vartheta_0) = 0$. The exact map Φ is iterated for 30,000 steps and the theorem’s explicit envelope is evaluated at each step.
Nondegenerate dynamics	$d = 2$, $c^* = (0.80, 0)$, and $\eta = 1/50 = 2 \times 10^{-2}$. Every displayed start is checked to lie in $\mathcal{D}_\eta$ before the exact map Φ is iterated for 1,600 steps. The four starts are $((0.45, 0.15), 0.40)$, $((-0.45, 0.25), 0.40)$, $((5 \times 10^{-3}, 0.65), 0)$, and the exact saddle $((0, 0.65), 0)$.
Spectral diagnostic	$d = 20$, $\Sigma = I_d$, $\theta^* = (1, 0, \dots, 0)$, $\nu^* = \nu = 0$, $\bar{\alpha}_t = 5 \times 10^{-3}$, raw θ -update step size $\eta_\theta = 20$ (whose product with $\bar{\alpha}_t$ is 10^{-1}), and $\ \theta_0\ = 2 \times 10^{-2}$. The initial absolute cosine with θ^* is 10^{-1} . For the Gaussian and covariance-normalized spherical Student t designs with ten degrees of freedom, the experiment uses 1.6×10^5 common draws and 14 normalized updates.

Finite-sample estimation and calibration. For a data point $(y_0, \mathbf{x})$, define $\mu_0 = \langle \theta, \mathbf{x} \rangle$, $\mu_t = \sqrt{\bar{\alpha}_t} \mu_0$, and $y_t = y_0 \sqrt{\bar{\alpha}_t} + \xi \sqrt{1 - \bar{\alpha}_t}$ as in Section 2.1. After terms independent of (θ, ν) are removed, the implemented conditional Hyvärinen integrand is the sum of $-y_t \mu_0 \tanh(\mu_t y_t + \nu) / \sqrt{\bar{\alpha}_t}$ and $\mu_0^2 \{2 - \tanh^2(\mu_t y_t + \nu)\} / 2$. It is integrated over $\bar{\alpha}_t \in [\bar{\alpha}_{T_n}, \bar{\alpha}_0]$ and over $\xi \sim \mathcal{N}(0, 1)$. Since $n \bar{\alpha}_{T_n} / (1 - \bar{\alpha}_{T_n}) = n^{-1/2} / (1 - n^{-3/2}) \rightarrow 0$, the selected terminal scale satisfies the terminal-scale condition in Theorem 3.2.

Each fit uses L-BFGS-B with $\theta \in [-3, 3]$, $\nu \in [-2.5, 2.5]$, at most 140 iterations, function tolerance 10^{-11} , and projected-gradient tolerance 2×10^{-7} . Two starts are a data-moment estimate and its joint negative. The first coordinate of the data-moment estimate is $\sqrt{\max\{n^{-1} \sum_i (y_0^{(i)})^2 - 1, 8 \times 10^{-2}\}}$, and its second coordinate is obtained by applying $\operatorname{arctanh}$ to the ratio of $(n^{-1} \sum_i \mathbf{x}^{(i)} y_0^{(i)}) / \max\{n^{-1} \sum_i \|\mathbf{x}^{(i)}\|^2, 10^{-10}\}$ to this first coordinate, clipped to remain 8×10^{-2} away from $\{\pm 1\}$ by restricting it to $[-1 + 8 \times 10^{-2}, 1 - 8 \times 10^{-2}]$. The other two starts are $(0.55, 0.15)$ and $(-0.55, -0.15)$. Among the four starts, we retain the fit with the smallest objective and then apply the joint sign s_n to both estimated coordinates. The aligned error is this estimate minus the ground-truth pair, and Figure 1(a) uses its norm induced by $I(\theta^*, \nu^*)$, called the Fisher norm. Theorem 3.2 gives convergence of n times its square to the two-degree-of-freedom chi-square law χ_2^2 . Hence the median and 90% references are the square roots of the corresponding χ_2^2 quantiles divided by n . The empirical quantile bars are 95% percentile-bootstrap intervals from 2,000 resamples with seed 17031. Panel (b) compares all ordered values of n times the squared Fisher error with χ_2^2 quantiles at $n = 250$ and $n = 4000$.

For each sample size n , Table 4 reports the fraction of aligned estimates for which n times the squared Fisher error does not exceed the 95% χ_2^2 quantile. Its covariance discrepancy is the Frobenius distance between the empirical covariance of the $\sqrt{n}$ -scaled aligned errors and $I(\theta^*, \nu^*)^{-1}$, divided by the Frobenius norm of $I(\theta^*, \nu^*)^{-1}$. All 1,000 optimization runs terminated without solver failure.

Table 4: **Finite-sample calibration.** Joint 95% Fisher-ellipse coverage and relative covariance discrepancy.

Sample size n	Joint 95% Fisher-ellipse coverage	Relative covariance discrepancy
250	93.5%	0.220
500	96.5%	0.093
1000	94.0%	0.127
2000	94.5%	0.089
4000	95.0%	0.070

Endpoint-gradient checks. We use an exact sum over the latent class and 140-point Gauss–Hermite quadrature for both the Gaussian covariate and response-noise expectations at every scale of the diffusion

noise level. At low noise, the θ component of the complete reference in Proposition 4.7 is the sum of $\Sigma(\theta - M(\theta, \nu))$, $-(\nabla_\theta M(\theta, \nu))^\top \Sigma \theta$, and $\nabla_\theta V(\theta, \nu)$. The ν component is $-\langle \nabla_\theta N(\theta, \nu), \theta \rangle + \nabla_\nu V(\theta, \nu)$. The plotted remainder uses the direct $\bar{\alpha}_t = 1$ gradient to stabilize the subtraction at the smallest scales. As a separate check, that endpoint differs from the complete operator expression in Proposition 4.7 by 2.24964×10^{-9} in Euclidean norm. Its θ - and ν -component residuals are 7.606×10^{-10} and 2.117×10^{-9} . This separate identity check confirms that the endpoint reference includes the correction terms and is not the bare EM residual. At high noise, the reference is the gradient of the effective-product limiting loss in Proposition 4.8.

Table 5 records the componentwise log-log slope deviations from the first-order benchmark. Each fit uses the five smallest of the nine endpoint distances.

Table 5: **Endpoint-remainder slopes.** Entries are written relative to the unit-slope benchmark.

Gradient quantity	Low-noise slope (panel a)	High-noise slope (panel b)
θ component	$1 + 2.3191 \times 10^{-5}$	$1 - 3.8124 \times 10^{-5}$
ν component	$1 + 1.0247 \times 10^{-5}$	$1 - 3.00596 \times 10^{-4}$

Remark (Spiked fourth-order covariance and the local spectral diagnostic). Proposition 4.8 identifies $\mathcal{L}(\theta, \nu) = \frac{1}{2} \|\theta \tanh \nu - c^*\|_\Sigma^2$, where $c^* := \theta^* \tanh \nu^*$, as the leading high-noise objective. For Gaussian covariates and for elliptical covariates with zero mean, covariance $\Sigma \succ 0$, and finite fourth moments, define the fourth-order signal covariance using the covariate marginal by

$$\mathcal{C}^* := \mathbb{E}_{\mathbf{x}} [\mathbf{x} \mathbf{x}^\top (\mathbf{x}^\top \theta^*)^2].$$

Lemmas A.18 and A.19 give

$$\mathcal{C}^* = \begin{cases} \sigma_\mu^2 \Sigma + 2\Sigma \theta^* (\Sigma \theta^*)^\top, & \mathbf{x} \sim \mathcal{N}(0, \Sigma), \\ (1 + \kappa) [\sigma_\mu^2 \Sigma + 2\Sigma \theta^* (\Sigma \theta^*)^\top], & \mathbf{x} \text{ elliptical}, \end{cases} \quad \sigma_\mu^2 := \|\theta^*\|_\Sigma^2,$$

where $\kappa := \mathbb{E}_{\mathbf{x}} [\|\Sigma^{-1/2} \mathbf{x}\|^4] / (d(d+2)) - 1 > -1$. If $\theta^* \neq \mathbf{0}$ and $\nu^* := \Sigma^{1/2} \theta^*$, whitening gives $\sigma_\mu^2 I_d + 2\nu^* (\nu^*)^\top$ in the Gaussian case and $(1 + \kappa) [\sigma_\mu^2 I_d + 2\nu^* (\nu^*)^\top]$ in the elliptical case, so $\text{span}\{\nu^*\}$ is the unique leading eigenspace after whitening. This also makes $\text{span}\{\theta^*\}$ the leading generalized eigenspace of $\mathcal{C}^* \nu = \lambda \Sigma \nu$. The corresponding generalized spectral gap is $2\sigma_\mu^2$ in the Gaussian case and $2(1 + \kappa)\sigma_\mu^2$ in the elliptical case. When $\Sigma = I_d$, the Gaussian covariance reduces to $\|\theta^*\|^2 I_d + 2\theta^* (\theta^*)^\top$, whereas the elliptical covariance is $(1 + \kappa)$ times this matrix. These covariance identities define the auxiliary directional comparator used below.

On the balanced slice $\nu^* = \nu = 0$, the leading high-noise objective is flat in θ , so Proposition 4.8 supplies no leading-order direction. The $\mathcal{O}(\bar{\alpha}_t)$ remainder test of that proposition is Figure 2(b), not Figure 5. The latter is only an auxiliary next-order directional comparison at small positive $\bar{\alpha}_t$. Using n covariate draws $\{\mathbf{x}^{(i)}\}_{i=1}^n$ with $\Sigma = I_d$, form the sample spiked covariance

$$\hat{\mathcal{C}}^* := \frac{1}{n} \sum_{i=1}^n \mathbf{x}^{(i)} (\mathbf{x}^{(i)})^\top ((\mathbf{x}^{(i)})^\top \theta^*)^2$$

and the normalized power step

$$\theta \mapsto \frac{(I_d + 2\gamma \hat{\mathcal{C}}^*) \theta}{\|(I_d + 2\gamma \hat{\mathcal{C}}^*) \theta\|}, \quad \gamma := \eta_\theta \bar{\alpha}_t.$$

In parallel, with ν clamped at zero, apply the finite-noise score-matching update $\theta \leftarrow \theta - \eta_\theta \nabla_\theta \mathcal{L}_t(\theta, 0)$ and renormalize to a fixed radius. The two trajectories use respective normalization radii 1 and 2×10^{-2} , which do not affect the absolute cosine $|\cos \angle(\theta_k, \theta^*)|$ plotted in Figure 5. The label t_{10} denotes spherical Student- t covariates with ten degrees of freedom (covariance-normalized), contrasted with isotropic Gaussian

covariates. The experiment compares directions only: it asserts neither equality of the unnormalized updates nor that the comparison follows from the proposition’s $\mathcal{O}(\bar{\alpha}_t)$ bound. Isotropic $\Sigma = I_d$ gives the power step an ordinary-eigenvector interpretation; anisotropic covariance would require the corresponding generalized-eigenvector map.

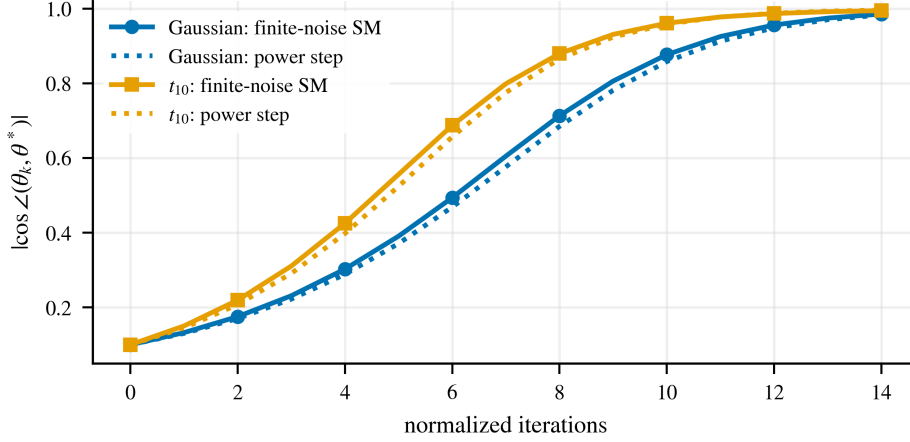


Figure 5: **Auxiliary next-order directional comparison on the balanced slice** $\nu^* = \nu = 0$. Absolute cosine $|\cos \angle(\theta_k, \theta^*)|$ for normalized finite-noise score-matching updates $\theta \leftarrow \theta - \eta_\theta \nabla_\theta \mathcal{L}_t(\theta, 0)$ (solid) and normalized power steps $\theta \mapsto (I_d + 2\gamma \hat{\mathcal{C}}^*)\theta / \|(I_d + 2\gamma \hat{\mathcal{C}}^*)\theta\|$ with $\gamma = \eta_\theta \bar{\alpha}_t$ (dotted). Curves labeled t_{10} use spherical Student- t covariates with ten degrees of freedom; the remaining curves use isotropic Gaussians.

Limiting-dynamics diagnostics. The implementation iterates $\vartheta_{k+1} = \Phi(\vartheta_k)$. For the degenerate target, $\mathcal{L}_0 = \frac{1}{2} \sinh^2(0.70) \tanh^2(0.70) \approx 0.1050944$, $J(\vartheta_0) = 0$, and $B_0 = 4$. Since $\eta = 1/50$, the left-hand side in the definition of $\mathcal{D}_\eta$ is $32\eta = 16/25 = 0.64 < 1$. The reference curve in Figure 3(a) is the theorem’s bound $(\mathcal{L}_0^{-1/2} + \eta K / (\sqrt{6} B_0))^{-2}$. We fit a linear regression of $\log \mathcal{L}(\vartheta_K)$ on $\log K$ for every integer K from 3×10^3 through 3×10^4 . The fitted slope is $-2 - 1.5035 \times 10^{-2}$. It is 1.5035×10^{-2} steeper than the K^{-2} reference. The coefficient of determination is $1 - 4.6533 \times 10^{-7}$. This fit is a finite-window diagnostic. The pointwise comparison with the explicit bound checks the theorem’s $\mathcal{O}(K^{-2})$ upper bound.

For $c^* \neq \mathbf{0}$, the three displayed generic starts in Table 6 are selected to show different entrance times while remaining in $\mathcal{D}_\eta$. The exact saddle start satisfies $\nu_0 = 0$ and $\langle \theta_0, c^* \rangle = 0$ and verifies the predicted plateau $\|c^*\|^2/2 = 0.32$. Across the one degenerate and four nondegenerate paths, the maximum admissibility left-hand side is 0.64 and the largest one-step loss increase is zero (at the stationary saddle), consistent along these trajectories with the theorem’s forward-invariance and monotonicity conclusions.

Table 6: **Nondegenerate-dynamics initializations.**

Start	θ_0	ν_0	Role
Same-sign	(0.45, 0.15)	0.40	Attraction to the matching-sign branch.
Opposite-sign	(−0.45, 0.25)	0.40	Crosses toward the target branch.
Near saddle	$(5 \times 10^{-3}, 0.65)$	0	Perturbation from the saddle manifold.
Exact saddle	(0, 0.65)	0	Stationary at loss $\ c^*\ ^2/2$.

For each displayed generic start, the analysis fits a linear regression of $\log \mathcal{L}(\vartheta_K)$ on K while the loss lies between 10^{-3} and 10^{-12} . Here $K_{10^{-2}}$, for example, is the first iteration K with $\mathcal{L}(\vartheta_K) \leq 10^{-2}$. Table 7 reports the four hitting times, the fitted slope, and one minus its coefficient of determination.

We omit values below 10^{-14} from the semilogarithmic panel as a roundoff display cutoff.

Table 7: **Hitting times and geometric tail fits.**

Start	$K_{10^{-2}}$	$K_{10^{-4}}$	$K_{10^{-6}}$	$K_{10^{-8}}$	Tail slope	Tail lack of fit
Same-sign	136	296	475	680	-0.022705	1.71×10^{-3}
Opposite-sign	328	498	694	907	-0.021873	6.67×10^{-4}
Near saddle	614	844	1082	1322	-0.019248	5.68×10^{-6}

Pointwise high-SNR blindness. The experiment in Figure 4 uses the scalar covariate $x \sim \mathcal{N}(0, 1)$ and a fixed positive ground-truth direction. This numerical derivation uses the boundary-layer coordinate $u = \sqrt{\|\theta^*\|} x$. The diffused ground-truth signal norm is $\sqrt{\bar{\alpha}_t} \|\theta^*\|$. Because $\bar{\alpha}_t$ is fixed in each sweep covered by Proposition 5.2, it diverges as $\|\theta^*\| \rightarrow \infty$. With the established latent class $z \in \{1, 2\}$ and $\varepsilon_t \sim \mathcal{N}(0, 1)$, the candidate activation is the sum of $(-1)^{z+1} \bar{\alpha}_t \theta u^2$, $\sqrt{\bar{\alpha}_t} \theta u \varepsilon_t / \sqrt{\|\theta^*\|}$, and ν , so its nonsaturated region has fixed width in u . We evaluate $\nabla_\nu \mathcal{L}_t$ and V_t by quadrature, multiplying the former by $\|\theta^*\|^{1/2}$ and the latter by $\|\theta^*\|^{3/2}$ before integrating over $(-\infty, 0] \cup [0, \infty)$, and undo the scaling afterward. This prevents the adaptive quadrature routine from having to resolve a shrinking neighborhood of $x = 0$ and keeps the reported components comparable in magnitude. These factors are numerical preconditioners tailored to this one-dimensional Gaussian design and are undone after integration. The same adaptive call also evaluates two diagnostic components. The first is a second copy of the preconditioned score matching gradient obtained by direct differentiation. The second is the scaled boundary correction used for the cross-entropy gradient.

We use an exact sum over the two latent classes. We evaluate the response-noise expectation with paired 80- and 160-node Gauss–Hermite rules. We compare their results without conflating this check with the adaptive integration in u . The latter uses vector quadrature with absolute tolerance 10^{-11} , relative tolerance 10^{-9} , and subdivision limit 500. Its returned error estimate covers the adaptive u -integration, not Gauss–Hermite discretization. A production run is rejected if any reported scaled component has paired-order difference exceeding $10^{-8} + 10^{-6}$ times the larger absolute result from the 80- and 160-node rules. At the publication settings, the direct-score formula, which is derived without the integration-by-parts identity, differs from that identity by at most 2.85×10^{-10} in absolute value over the displayed sweep. The largest 80-versus-160-node difference in the score gradient obtained by direct differentiation is also 2.85×10^{-10} . The largest adaptive error estimate for the scaled five-component vector is 1.60×10^{-10} . We use the direct formula for the plotted score gradient because it preserves exact-model cancellation, and retain the integration-by-parts formula as an independent identity check. We take absolute values after integrating the signed quantities.

Panels (a)–(b) of Figure 4 use 1,200 common-random-number draws at $\|\theta^*\| = 3$ and $\|\theta^*\| = 100$. Their background is the fixed candidate’s response-score sensitivity $|\partial_\nu s_{\theta_t, \nu}(y_t, x)|$ on a common logarithmic scale. Panel (c) contains the imbalance gradient and posterior-variance term stated in Proposition 5.2. At the largest displayed $\|\theta^*\|$, the computed values of $|\nabla_\nu \mathcal{L}_t|$ and V_t are 1.6688×10^{-3} and 2.9399×10^{-9} .

The cross-entropy control appears in Appendix Figure 6(c). It evaluates $\nabla_\nu \mathcal{H}(\theta_t, \nu) = \tanh \nu - \mathbb{E}_*[\tanh(\mu_t y_t + \nu)]$ along the same fixed-candidate ray. For the displayed aligned one-dimensional specialization, the candidate activation is the sum of $(-1)^{z+1} \bar{\alpha}_t \theta \|\theta^*\| x^2$, $\sqrt{\bar{\alpha}_t} \theta x \varepsilon_t$, and ν . Its hyperbolic tangent therefore converges almost surely to $(-1)^{z+1}$, and bounded convergence gives $|\nabla_\nu \mathcal{H}(\theta_t, \nu)| \rightarrow |\tanh(-0.35) - \tanh(0.55)| \approx 0.836896$. Its nonzero limit therefore agrees with Lemma D.3 and provides a contrast. The score matching diagnostics lose their local imbalance signal, whereas this cross-entropy gradient remains informative. This is a fixed-candidate comparison of population gradients, not a comparison of optimized estimators.

Figure 6(a)–(b) provides two scope checks. In the fixed-scale comparisons, each value of $\bar{\alpha}_t$ is held constant while $\|\theta^*\|$ varies, so each curve is a separate specialization covered by Proposition 5.2. The moving-candidate comparison instead sets $\theta = 0.8/\|\theta^*\|$ while retaining the main experiment’s scale of the diffusion noise level, ground-truth ray, and imbalance parameters. Because the candidate changes with $\|\theta^*\|$, this curve illustrates the proposition’s pointwise-in-candidate scope.

Along the moving candidate used in the control, differentiation and dominated convergence give $\nabla_\nu \mathcal{L}_t(0.8/\|\theta^*\|, \nu) \rightarrow -0.8 \mathbb{E}_*[(-1)^{z+1} x^2 \operatorname{sech}^2(0.8 \bar{\alpha}_t (-1)^{z+1} x^2 + \nu)]$. A separate integration gives the signed limiting value -0.3023024610 to ten decimal places, with magnitude 0.3023024610. The plotted absolute

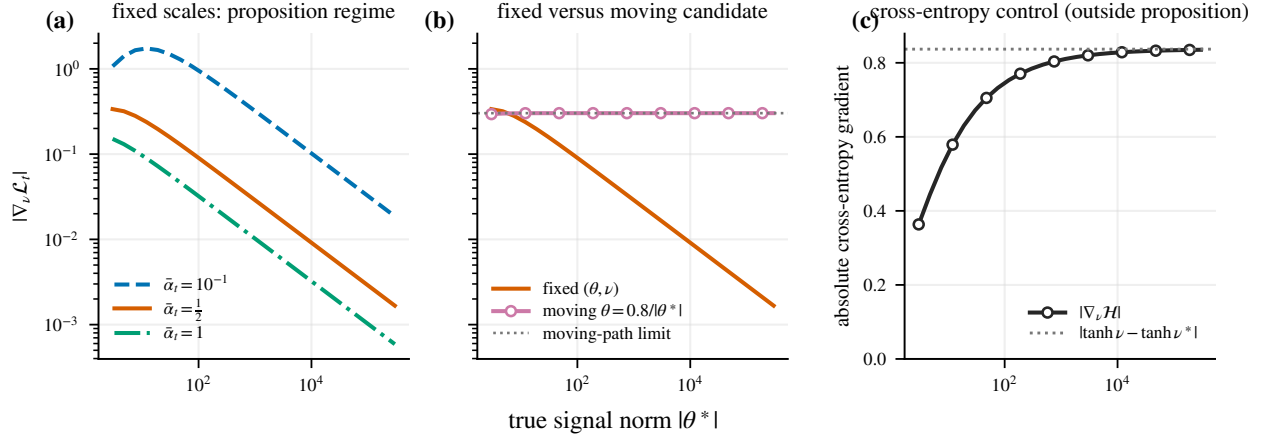


Figure 6: **Scope and objective controls for pointwise high-SNR blindness.** (a) Fixed- $\bar{\alpha}_t$ curves provide separate checks covered by the proposition as the true signal $\|\theta^*\|$ grows, for $\bar{\alpha}_t \in \{10^{-1}, \frac{1}{2}, 1\}$. (b) The moving-candidate curve uses $\theta = 0.8/\|\theta^*\|$, for which the candidate activation need not diverge and the moving-candidate score matching gradient $|\nabla_{\nu} \mathcal{L}_t(0.8/\|\theta^*\|, \nu)|$ approaches 0.302302. That curve is outside the fixed-candidate hypothesis of Proposition 5.2 and illustrates its pointwise-in-candidate scope. (c) The separate population cross-entropy control is evaluated along the same fixed-candidate ray as Figure 4. It approaches $|\tanh \nu - \tanh \nu^*| \approx 0.836896$, showing that this selected cross-entropy diagnostic retains imbalance signal.

gradient at $\|\theta^*\| = 3 \times 10^5$ differs from that magnitude by 3.2×10^{-12} . This is the discrepancy at the largest displayed signal norm. The largest 80-versus-160-node difference over the control sweep is 8.72×10^{-8} .